Notes to EGMO

by: ShadowTheDP

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CHAPTER 0
Preliminaries

This PDF give all section solution of the book *Eucildean Geometry in Mathematical Olympiads* written by **Evan Chen**.

Here's all of the typos or mistakes in *EGMO* : https://web.evanchen.cc/textbooks/geombook-errata.pdf

Here's the solution provided by **Evan Chen** : https://web.evanchen.cc/textbooks/AGEST.pdf

0.1. The description of mathenv

Take an example:

Definition 0.1 (something). the main body

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At the very first of this example, is the type name of this example, followed by the serial number of the example, the string inside the brackets is usually the source in the book *EGMO*, the string after the ":" is the main body of this mathenv.

0.2. Center of a Triangle

- The **Orthocenter** of $\triangle ABC$, usually denoted by H, is the intersection of the perpendiculars (or altitudes) from A to BC, B to CA, and C to AB. The triangle formed by the feet of these altitudes is called the **orthic triangle**.

- The **Centroid**, usually denoted by G, is the intersection the medians, which are the lines joining each vertex to the midpoint of the opposite side. The triangle formed by the midpoints is called the **medial triangle**.

- The **Incenter**, usually denoted by I , is the intersection of the angle bisectors of the angles of $\triangle ABC$. It is also the center of a circle (the **incircle**) tangent to all three sides. The radius of the incircle is called the inradius r. The triangle formed by the points of tangency of the incircle is called **contact triangle**

- The **Circumcenter**, usually denoted by O, is the center of the unique circle (the **circumcircle**) passing through the vertices of $\triangle ABC$. The radius of this circumcircle is called the circumradius R.

0.3. Notation

Consider a triangle ABC. Throughout this text, let $a = BC, b = CA, c = AB$, and abbreviate $A = \angle BAC, B = \angle CBA, C = \angle ACB$.

Here's some common used notation in this pdf:

- $\angle$: a standard angle
- $\measuredangle$: a directed angle from $\angle$
- $s := \frac{a+b+c}{2}$: the semiperimeter of $\triangle ABC$
- $[P_1P_2P_3...P_n]$: the area of polygon $P_1P_2P_3...P_n$,
- $(P_1P_2P_3...P_n)$: a circle with a sequence of points $P_1, P_2, P_3, ...P_n$ all lying on it
- $AB \cap CD$: the intersection of the two lines AB and CD
- For algebraic computations, we may use sigma notation as follows:

$$\sum_{\text{cyc}} f(a,b,c) = f(a,b,c) + f(b,c,a) + f(c,a,b)$$

$$\sum_{\text{sym}} f(a,b,c) = f(a,b,c) + f(a,c,b) + f(b,a,c) + f(b,c,a) + f(c,a,b) + f(c,b,a)$$

CHAPTER 1
Angle Chasing

Mathematics is the language in which God has written the universe.

— *Galileo Galilei*

Problem 1.1 (Problem 1.36.). Let $ABCDE$ be a convex pentagon such that $BCDE$ is a square with center O and $\angle A = 90°$. Prove that AO bisects $\angle BAE$.

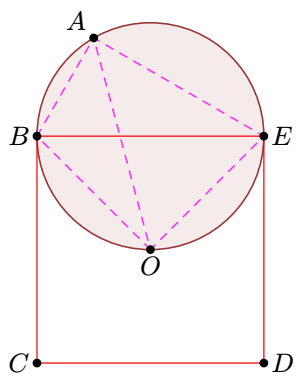

Figure 2A | Problem 1.36.

Proof. Since O is the center of $BCDE$, we have $\angle BOE = 90° = \angle BAE \Rightarrow A, B, O, E$ concyclic. Also, $BO = EO$. The result follows. □

Problem 1.2 (Problem 1.37 (BAMO 1999/2).). Let $O = (0, 0), A = (0, a),$ and $B = (0, b)$, where $0 < a < b$ are reals. Let Γ be a circle with diameter AB and let P be any other point on Γ. Line PA meets the x-axis again at Q. Prove that $\angle BQP = \angle BOP$.

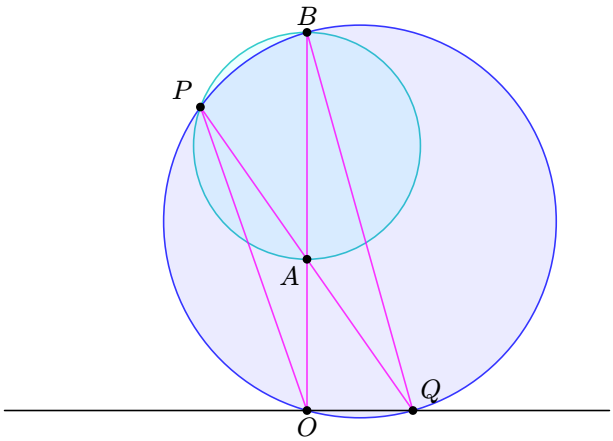

Figure 2B | Problem 1.37 (BAMO 1999/2).

Proof. Obviously we have $\angle BPQ = \angle BPA = 90°$ and $\angle BOQ = 90°$, therefore P, B, Q, O are concyclic. The result follows. □

Problem 1.3 (Problem 1.38.). In cyclic quadrilateral $ABCD$, let I_1 and I_2 denote the incenters of $\triangle ABC$ and $\triangle DBC$, respectively. Prove that I_1I_2BC is cyclic.

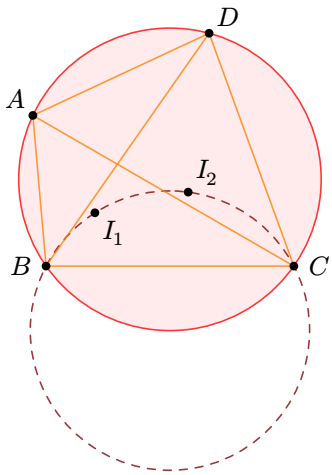

Figure 2C | Problem 1.38.

Proof. In fact, we have

$$\begin{aligned}\angle BI_1C &= \frac{\pi}{2} + \frac{\angle BAC}{2} \\ &= \frac{\pi}{2} + \frac{\angle BDC}{2} \\ &= \angle BI_2C.\end{aligned}$$

As result. □

Problem 1.4 (Problem 1.39 (CGMO 2012/5).). Let ABC be a triangle. The incircle of $\triangle ABC$ is tangent to AB and AC at D and E respectively. Let O denote the circumcenter of $\triangle BCI$ .Prove that $\angle ODB = \angle OEC$.

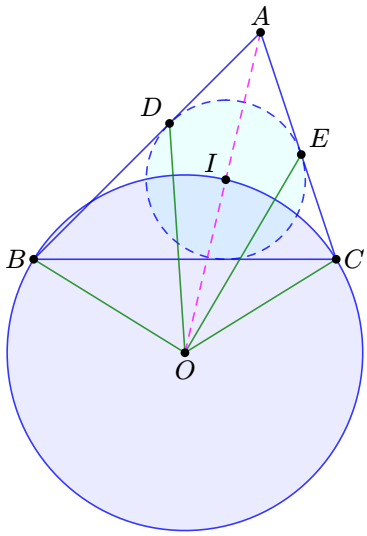

Figure 2D | Problem 1.39 (CGMO 2012/5).

Proof. Well-known A, I, O colinear. Since $AD = AE, AI$ is an angle bisector, we have $\triangle ADO \cong \triangle AEO$, so $\angle AOD = \angle AOE$, the result follows. □

Problem 1.5 (Problem 1.40 (Canada 1991/3).). Let P be a point inside circle ω. Consider the set of chords of ω that contain P. Prove that their midpoints all lie on a circle.

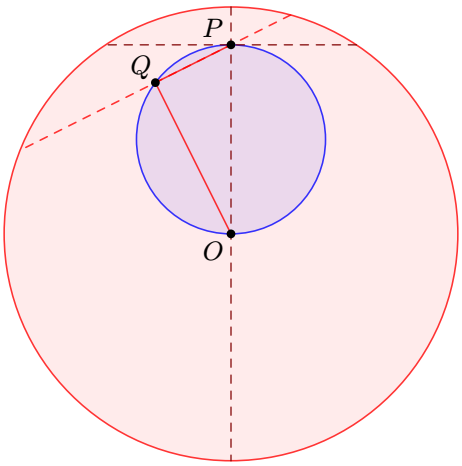

Figure 2E | Problem 1.40 (Canada 1991/3).

Proof. It's sufficient to proof all Q lie on the circle with PO is diameter. In fact we have $OQ \perp QP$ at all the time. □

Problem 1.6 (Problem 1.41 (Russian Olympiad 1996).). Points E and F are on side BC of convex quadrilateral $ABCD$ (with E closer than F to B). It is known that $\angle BAE = \angle CDF$ and $\angle EAF = \angle FDE$. Prove that $\angle FAC = \angle EDB$.

Proof. Since $\angle EAF = \angle FDE$, we have A, E, F, D concyclic. Also it's sufficient to show A, B, C, D are concyclic. In fact,

$$\begin{aligned}\angle BAD + \angle DCB &= \angle BAE + \angle EAF + \angle FAD + \angle DCF \\ &= \angle CDF + \angle FDE + \angle FED + \angle DCF \\ &= \pi.\end{aligned}$$

As result. □

Problem 1.7 (Lemma 1.42.). Let ABC be an acute triangle inscribed in circle Ω . Let X be the midpoint of the arc BC not containing A and define Y, Z similarly. Show that the orthocenter of XYZ is the incenter I of ABC.

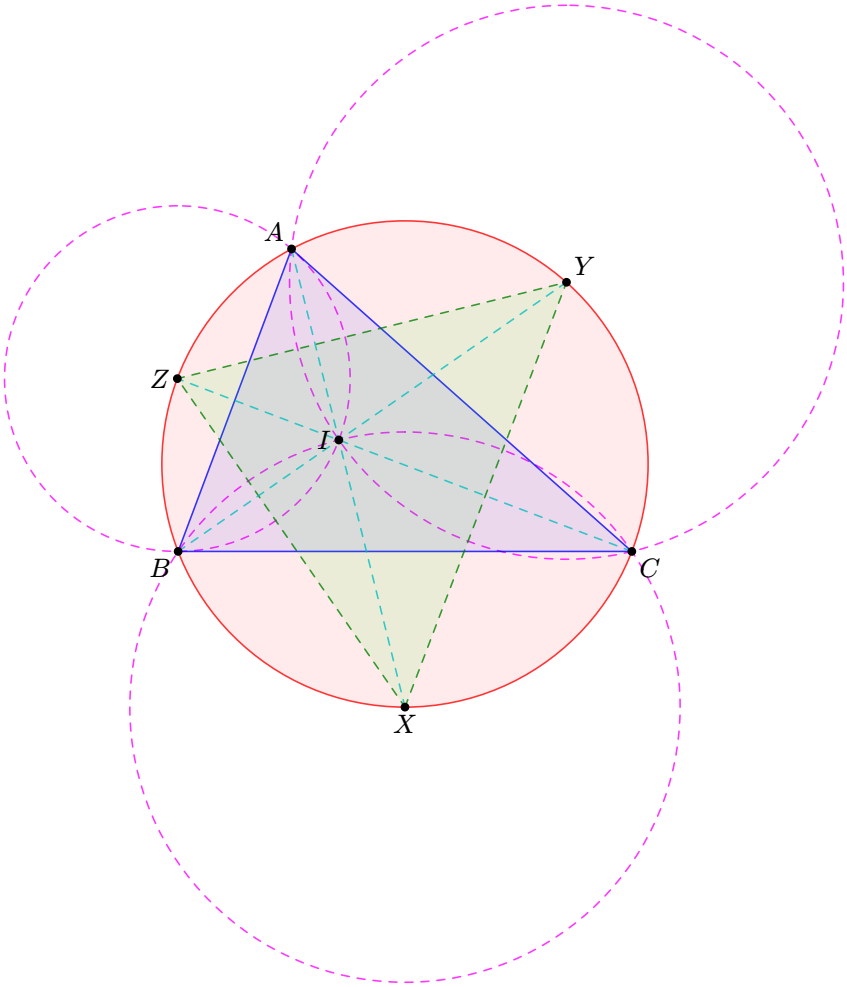

Figure 2F | Lemma 1.42

Proof. Notice that it's sufficient to show $AX \perp ZY$, since A, I, X colinear and AI is radical axis of (AIB) and (AIC), same applies on BY and CZ, the result follows. □

Problem 1.8 (Problem 1.43 (JMO 2011/5).). Points A, B, C, D, E lie on a circle ω and point P lies outside the circle. The given points are such that:

i) lines PB and PD are tangent to ω,
ii) P, A, C are collinear,
iii) $DE \parallel AC$.

Prove that BE bisects AC.

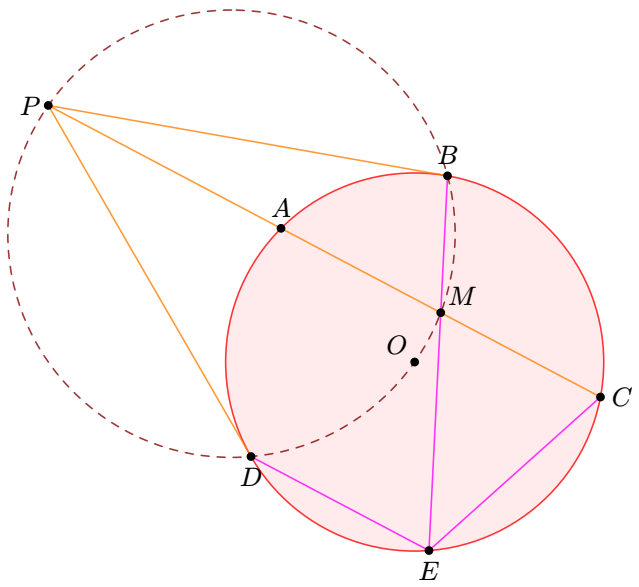

Figure 2G | Problem 1.43 (JMO 2011/5).

Proof (Angle Chasing). Since $\angle PBO = \angle PDO = 90°$, we have P, B, O, D concyclic with PO is the diameter. Now we have to claim M lies on the circle. Since $DE \parallel PC$

$$\angle PMB = \angle DEB = \angle PDB.$$

Then $\angle PMO = 90°$, which is obvious. □

Proof (Projective Geometry). Notice the $ABCD$ is a harmonic quadrilateral, Since $DE \parallel AC$

$$-1 = (A, C; B, D) \overset{E}{=} (A, C; BE \cap AC, D) = (A, C; BE \cap AC, P_\infty).$$

which shows $M = BE \cap AC$ is midpoint. □

Problem 1.9 (Lemma 1.44 (Three Tangents).). Let ABC be an acute triangle. Let BE and CF be altitudes of triangle ABC, and denote by M the midpoint of BC. Prove that ME, MF, and the line through A parallel to BC are all tangents to (AEF).

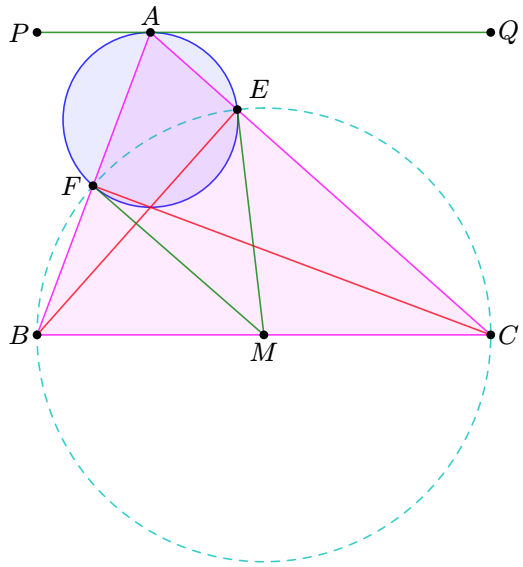

Figure 2H | Lemma 1.44 (Three Tangents).

Proof. Notice that B, F, E, C concyclic, with M is the center of the circle. To show ME, MF is tangent to $\odot AEF$. It's sufficient to show $\angle MFE = \angle MEF = \angle A$. In fact,

$$\begin{aligned}\angle MFE = \angle MEF &= \frac{1}{2}(\pi - \angle EMF)\\ &= \frac{1}{2}(\angle BMF + \angle CME)\\ &= \frac{1}{2}(\pi - 2\angle A + \pi - 2\angle B)\\ &= \angle A.\end{aligned}$$

Next it's to show the line through A parallel to BC is tangents to $\odot AEF$.

$$\angle QAC = \angle C = \angle AFE.$$

which is obvious. □

Problem 1.10 (Lemma 1.45 (Right Angles on Incircle Chord).). The incircle of ABC is tangent to BC, CA, AB at D, E, F respectively. Let M and N be the midpoints of BC and AC, respectively. Ray BI meets line EF at K. Show that $BK \perp CK$. Then show K lies on line MN.

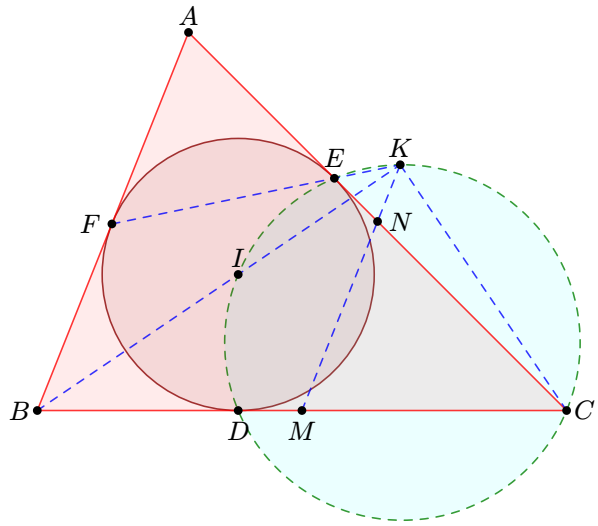

Figure 2I | Lemma 1.45 (Right Angles on Incircle Chord).

Proof.

Claim 1 - I, E, K, C, D are concyclic, with diameter IC.

Proof of Claim. It's sufficient to show $\angle IKE = \angle ILE = \frac{1}{2}\angle C$. Since

$$\begin{aligned}\angle IKE &= \angle BKF \\ &= \pi - \angle BFE - \angle FBK \\ &= \pi - \left(\frac{\pi}{2} + \frac{1}{2}\angle A\right) - \frac{1}{2}\angle B \\ &= \frac{1}{2}\angle C.\end{aligned}$$

as result. ■

Therefore, $\angle BKC = 90° \Rightarrow BK \perp CK$, also we have M is the circumcenter of BKC. Now it suffices to show $MK \parallel AB$, since

$$\angle MKB = \angle MBK = \angle ABK.$$

we are done □

Problem 1.11 (Problem 1.46 (Canada 1997/4).). The point O is situated inside the parallelogram $ABCD$ such that $\angle AOB + \angle COD = 180°$. Prove that $\angle OBC = \angle ODC$.

Proof. Let O'be a point outside the parallelogram $ABCD$ such that $\triangle O'DC \cong \triangle OAB$, we can easily have $O'OAD, O'OBC$ are parallelogram, also $O'DOC$ concyclic.
We are done since $\angle ODC = \angle OO'C = \angle OBC$. □

Problem 1.12 (Problem 1.47 (IMO 2006/1).). Let ABC be triangle with incenter I . A point P in the interior of the triangle satisfies

$$\angle PBA + \angle PCA = \angle PBC + \angle PCB.$$

Show that $AP \geq AI$ and that equality holds if and only if $P = I$.

Proof. Since

$$\angle PBA + \angle PCA = \angle PBC + \angle PCB = \frac{B+C}{2}.$$

Notice that B, P, I, C concyclic. Let O be circumcenter of the circle. Well-known A, I, O colinear, then we are done. □

Problem 1.13 (Lemma 1.48 (Simson Line).). Let ABC be a triangle and P be any point on (ABC). Let X, Y, Z be the feet of the perpendiculars from P onto lines BC, CA, and AB. Prove that points X, Y, Z are collinear.

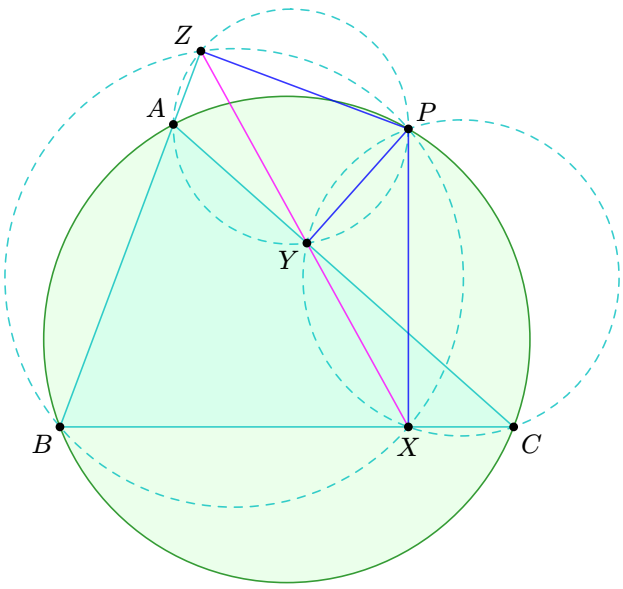

Figure 2J | Lemma 1.48 (Simson Line).

Proof. It's sufficient to show $\angle ZXP = \angle YXP$. Obviously we have the following:

- P, Z, A, Y concyclic, with diameter AP.
- P, Z, B, X concyclic, with diameter BP.
- P, Y, X, C concyclic, with diameter CP.

In fact, we have

$$\angle YXP = \angle YCP = \angle ACP = \angle ABP = \angle ZBP = \angle ZXP.$$

Hence the result. □

Problem 1.14 (Problem 1.49 (USAMO 2010/1).). Let $AXYZB$ be a convex pentagon inscribed in a semicircle of diameter AB. Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ, respectively. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle XOZ$, where O is the midpoint of segment AB.

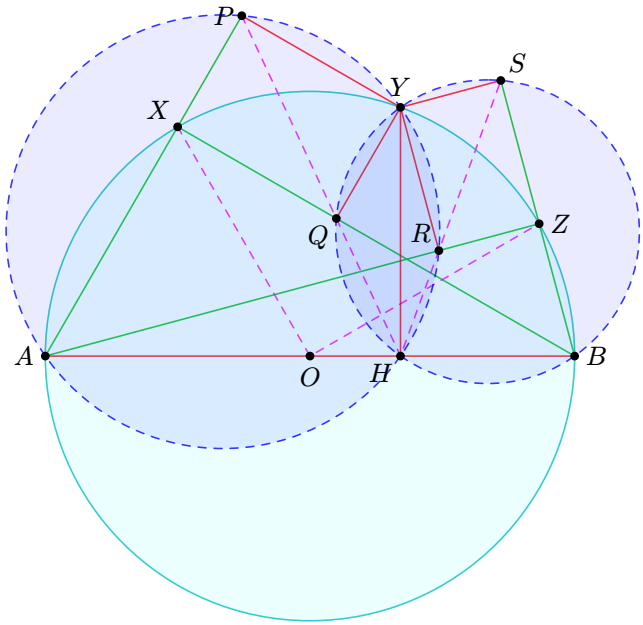

Figure 2K | Problem 1.49 (USAMO 2010/1).

Proof. During the process of drawing the graph, we have our first claim:

Claim 1 - PQ, RS, AB are concurrent at point H, also $YH \perp AB$.

Proof of Claim. Define $H :=$ The foot of the perpendicular from point Y to line AB. According to the property of Simson Line, we can have P, Q, H and S, R, H collinear, the result follows. ■

Back onto this problem, it suffices to show $\angle PHS = \angle XBZ = \frac{1}{2}\angle XOZ$, therefore, we have another claim:

Claim 2 - Q, Y, S, B, H are concyclic, with diameter YB.

Proof of Claim. Obviously we have $\angle YSB = \angle YHB = \angle YQB = \frac{\pi}{2}$. We get Y, Q, H, B and Y, H, B, S concyclic, as result. ■

done. □

Problem 1.15 (Problem 1.50 (IMO 2013/4).). Let ABC be an acute triangle with orthocenter H, and let W be a point on the side BC, between B and C. The points M and N are the feet of the altitudes drawn from B and C, respectively. ω_1 is the circumcircle of triangle BWN and X is a point such that WX is a diameter of ω_1. Similarly, ω_2 is the circumcircle of triangle CWM and Y is a point such that WY is a diameter of ω_2. Show that the points X, Y, and H are collinear.

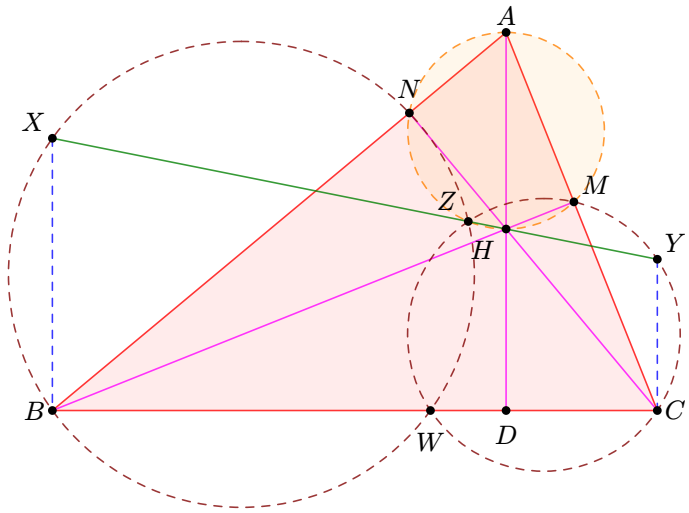

Figure 2L | Problem 1.50 (IMO 2013/4).

Proof (Angle Chasing). Obviously A, N, Z, H, M are concyclic with diameter AH.

Claim 1 - Z, H, Y are collinear.

Proof of Claim. We have

$$\measuredangle MZH = \measuredangle MAH = \frac{\pi}{2} - \measuredangle WCM = \frac{\pi}{2} - \measuredangle WYM = \measuredangle MWY = \measuredangle MZY$$

which is obvious. ■

Same applies on Z, H, X. In conclusion, X, Z, H, Y are collinear □

Proof (Computational Geometry). First we define $D := AH \cap BC, BW = ta, CW = (1-t)a$.

Since WX, WY are the diameter, $\angle XBW = \angle YCW = \frac{\pi}{2}$. It's sufficient to show

$$HD = \frac{CD}{a} BX + \frac{BD}{a} CY$$

First calculate BX and CY,

$$\begin{aligned}
BX &= \frac{BW}{\tan BXW} \\
&= BW\frac{\cos BNW}{\sin BNW} \\
&= BW\frac{BN^2+NW^2-BW^2}{2BN\cdot NW}\frac{NW}{BW\sin B} \\
&= \frac{BN^2+NW^2-BW^2}{2BN\sin B} \\
CY &= \frac{CW}{\tan WYC} \\
&= CW\frac{\cos CMW}{\sin CMW} \\
&= CW\frac{CM^2+MW^2-CW^2}{2CM\cdot MW}\frac{MW}{CW\sin C} \\
&= \frac{CM^2+MW^2-CW^2}{2CM\sin C} \\
HD &= \frac{c\cos B}{2\tan C}+\frac{b\cos C}{2\tan B}
\end{aligned}$$

From Stewart's theorem we get the followings:

$$\begin{aligned}
NW^2 &= \frac{NC^2BW+NB^2CW}{BC}-BW\cdot CW \\
MW^2 &= \frac{MC^2BW+MB^2CW}{BC}-BW\cdot CW
\end{aligned}$$

substitute those variable and simplify, we have:

$$\begin{aligned}
BX &= \frac{a(1-t)}{\tan B} \\
CY &= \frac{at}{\tan C}
\end{aligned}$$

It's sufficient to show,

$$\begin{aligned}
\frac{c\cos B}{2\tan C}+\frac{b\cos C}{2\tan B} &= \frac{a(1-t)}{\tan B}\frac{CD}{a}+\frac{at}{\tan C}\frac{BD}{a} \\
\frac{c}{2}\sin B+\frac{b}{2}\sin C &= (1-t)CD\tan C+tBD\tan B \\
\frac{1}{2}AD+\frac{1}{2}AD &= AD
\end{aligned}$$

which is obvious. □

Problem 1.16 (Problem 1.51 (IMO 1985/1).). A circle has center on the side AB of the cyclic quadrilateral $ABCD$. The other three sides are tangent to the circle. Prove that $AD+BC=AB$.

Proof. Consider $T := \odot(COD)\cap AB$, we have

$$\angle DTA = \angle DCO = \frac{1}{2}\angle DCB = \frac{1}{2}(\pi-\angle BAD) = \frac{\pi}{2}-\frac{1}{2}\angle TAD$$

That means triangle DAT is isosceles. Same as triangle CBT, the result follows. □

CHAPTER 2
Circles

Mathematics, rightly viewed, possesses not only truth but supreme beauty.

— *Bertrand Russell*

Problem 2.1 (Lemma 2.24.). Let ABC be a triangle with I_A, I_B, and I_C as excenters. Prove that triangle $I_AI_BI_C$ has orthocenter I and that triangle ABC is its orthic triangle.

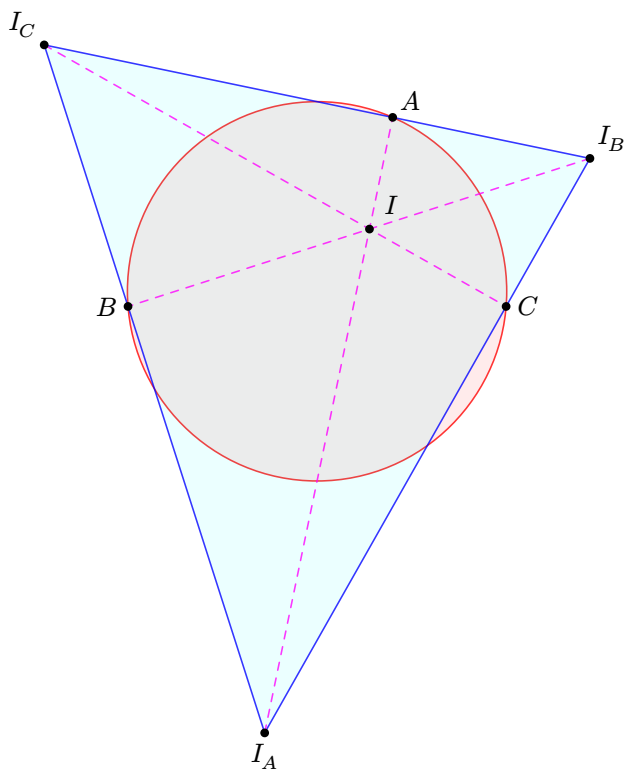

Figure 3A | Lemma 2.24.

Proof. It's obvious that I_A, I, A collinear, it suffices to show $I_AA \perp I_CAI_B$, which is trivial. The same applies to the rest. Then we're done. □

Problem 2.2 (Theorem 2.25 (The Pitot Theorem).). Let $ABCD$ be a quadrilateral. If a circle can be inscribed in it, prove that $AB + CD = BC + DA$.

Proof. Let AB, BC, CD, DA be tangent to the circle at points $E, F, G,$ and H respectively. We easily have,

$$AB + CD = AE + EB + CG + GD = AH + FB + CF + HD = BC + DA$$

done. □

Problem 2.3 (Problem 2.26 (USAMO 1990/5).). An acute-angled triangle ABC is given in the plane. The circle with diameter AB intersects altitude CC' and its extension at points M and N, and the circle with diameter AC intersects altitude BB' and its extensions at P and Q. Prove that the points M, N, P, Q lie on a common circle.

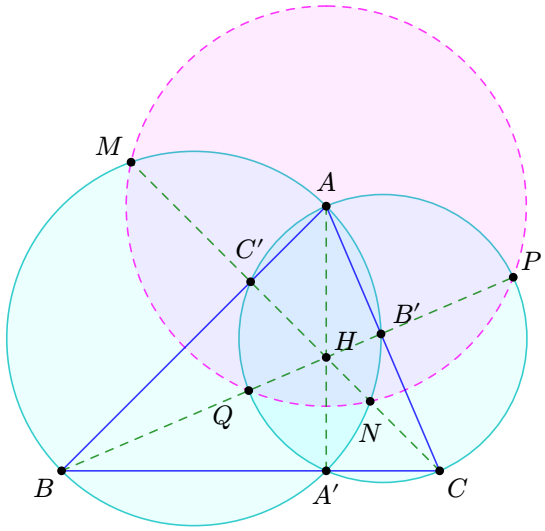

Figure 3B | Problem 2.26 (USAMO 1990/5).

Proof. Obviously, $A' = (ABB') \cap (ACC') \cap BC$ and $AA' \perp BC$, since AA', BB', CC' collinear at orthocenter H, the result is trivially true. □

Problem 2.4 (Problem 2.27 (BAMO 2012/4).). Given a segment AB in the plane, choose on it a point M different from A and B. Two equilateral triangles AMC and BMD in the plane are constructed on the same side of segment AB. The circumcircles of the two triangles intersect in point M and another point N.

i) Prove that AD and BC pass through point N.
ii) Prove that no matter where one chooses the point M along segment AB, all lines MN will pass through some fixed point K in the plane.

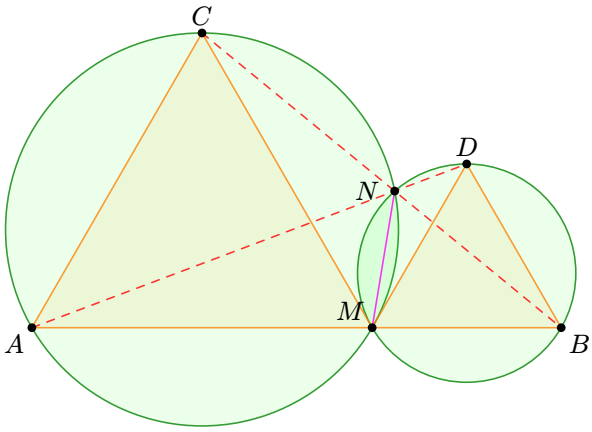

Figure 3C | Problem 2.27 (BAMO 2012/4).

Proof (Part 1). With $\angle CNA = \angle ANM = \angle MNB = \angle BND = 60°$, the result is now obvious. □

Proof (Part 2). Define K s.t. A, K, B, N concyclic and AKB is a equilateral triangle. It suffices to show

$$\frac{\sin AKN}{\sin BKN} = \frac{\sin AKM}{\sin BKM}$$

In fact, we have

$$\frac{\sin AKN}{\sin BKN} = \frac{AN}{BN}, \quad \frac{\sin AKM}{\sin BKM} = \frac{AM}{BM}\frac{BK}{AK} = \frac{AM}{BM}$$

with MN bisects $\angle ANB$. We are done. □

Problem 2.5 (Problem 2.28 (JMO 2012/1).). Given a triangle ABC, let P and Q be points on segments AB and AC, respectively, such that $AP = AQ$. Let S and R be distinct points on segment BC such that S lies between B and R, $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P, Q, R, S are concyclic.

Proof. Since $\angle BPS = \angle PRS$, that means BPA is tangent to $\odot PSR$ at point P. Same applies on the line CQA, we notice that

$$AP^2 = \text{Pow}(A, (PSR)) = \text{Pow}(A, (QSR)) = AQ^2$$

but $A \notin BC$ which is the radical axis of these two circle, that implies $(PSR) = (QSR)$, we are done. □

Problem 2.6 (Problem 2.29 (IMO 2008/1).). Let H be the orthocenter of an acute-angled triangle ABC. The circle Γ_A centered at the midpoint of BC and passing through H intersects the sideline BC at points A_1 and A_2. Similarly, define the points B_1, B_2, C_1, and C_2. Prove that six points A_1, A_2, B_1, B_2, C_1, and C_2 are concyclic.

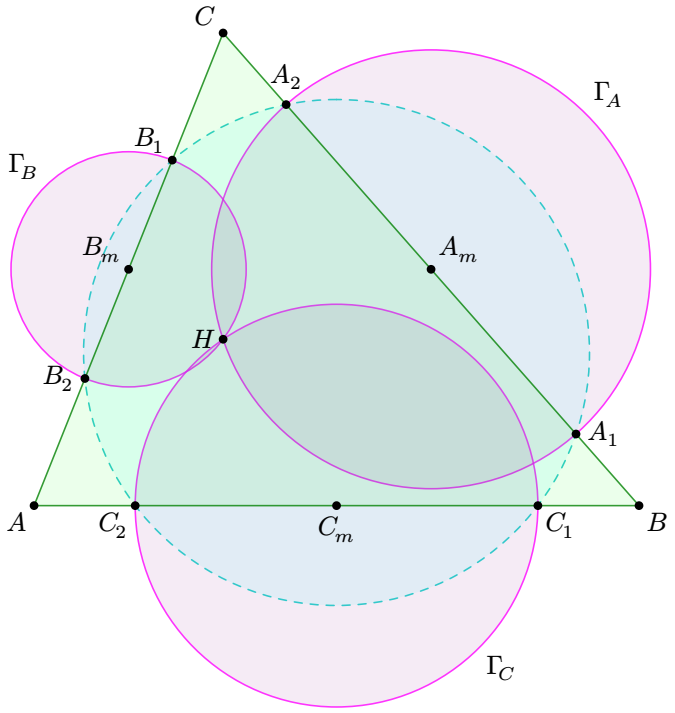

Figure 3D | Problem 2.29 (IMO 2008/1).

Proof. According to the cyclic of those six points, it's sufficient to show B_1, B_2, C_1, C_2 concyclic.

First we observe that $AH \perp BC$. Since B_mC_m is the median of the triangle ABC, which $B_mC_m \parallel BC$, we get $AH \perp B_mC_m$. This implies that AH is the root axis of $\odot B_m$ and $\odot C_m$. The result follows from the property of the radical axis. □

Problem 2.7 (Problem 2.30 (USAMO 1997/2).). Let ABC be a triangle. Take points D, E, F on the perpendicular bisectors of BC, CA, AB respectively. Show that the lines through A, B, C perpendicular to EF, FD, DE respectively are concurrent.

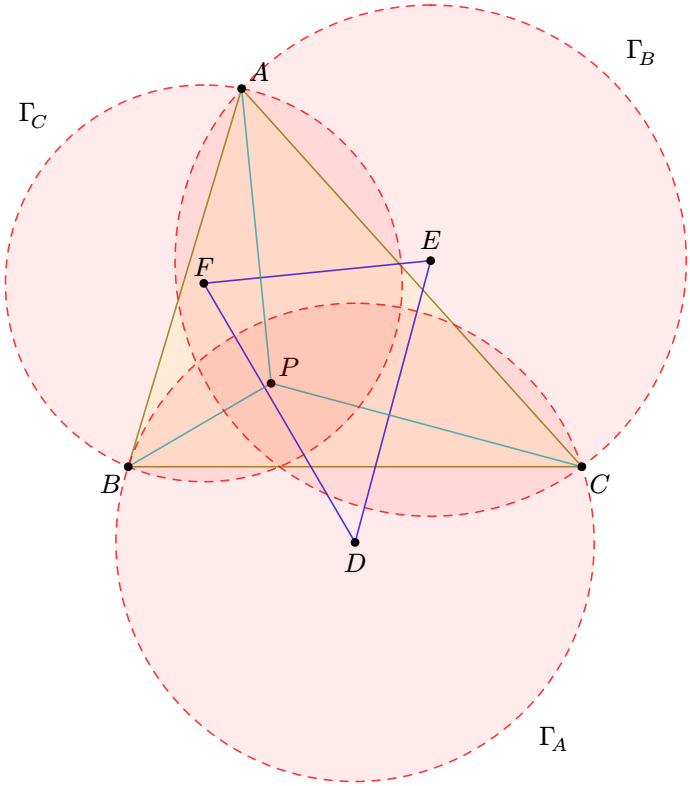

Figure 3E | Problem 2.30 (USAMO 1997/2).

Proof. Define Γ_A as a circle with center D and radius DB, define Γ_B, Γ_C similarly. For the radical axis of Γ_A and Γ_B must be perpendicular to DE and passes through C, same applies on other situation, the result is now obvious. □

Problem 2.8 (Problem 2.31 (IMO 1995/1).). Let A, B, C, D be four distinct points on a line, in that order. The circles with diameters AC and BD intersect at X and Y . The line XY meets BC at Z. Let P be a point on the line XY other than Z. The line CP intersects the circle with diameter AC at C and M, and the line BP intersects the circle with diameter BD at B and N. Prove that the lines AM, DN, XY are concurrent.

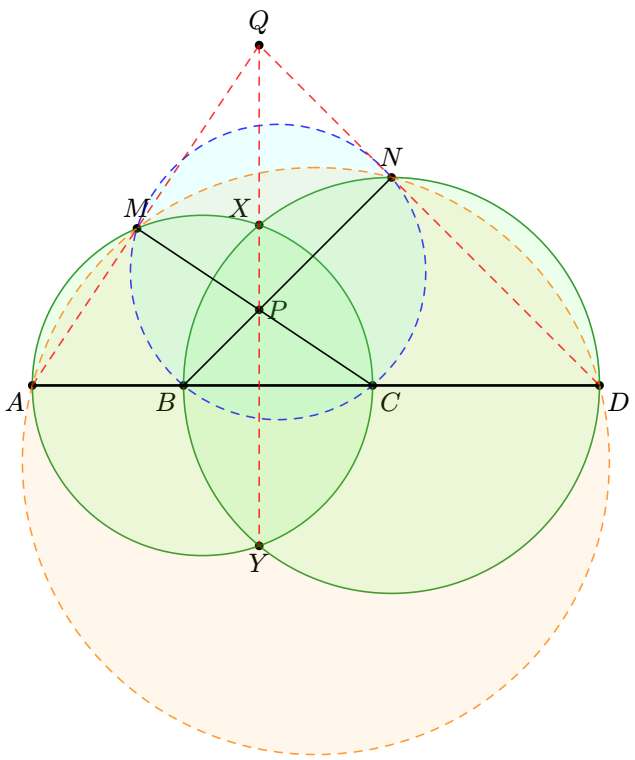

Figure 3F | Problem 2.31 (IMO 1995/1).

Proof.

Claim 1 - N, M, C, B are concyclic.

Proof of Claim. Since $NP \cdot PC = XP \cdot PY = BP \cdot PM$, we have the result easily. ■

It suffices to show A, N, M, D concyclic. In fact, we have

$$\measuredangle DAM = \measuredangle CAM = \frac{\pi}{2} - \measuredangle MCB = \frac{\pi}{2} - \measuredangle MNB = \frac{\pi}{2} + \measuredangle BNM = \measuredangle DNM$$

as desired. □

Problem 2.9 (Problem 2.32 (USAMO 1998/2).). Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be concentric circles, with $\mathcal{C}_2$ in the interior of $\mathcal{C}_1$. From a point A on $\mathcal{C}_1$ one draws the tangent AB to $\mathcal{C}_2$ ($B \in \mathcal{C}_2$). Let C be the second point of intersection of ray AB and $\mathcal{C}_1$, and let D be the midpoint of AB. A line passing through A intersects $\mathcal{C}_2$ at E and F in such a way that the perpendicular bisectors of DE and CF intersect at a point M on AB. Find, with proof, the ratio $\frac{AM}{MC}$.

Proof. Since we have $AE \cdot AF = AB^2 = AD \cdot AC \Rightarrow D, C, F, E$ concyclic. Therefore, M is the midpoint of DC. Then we have

$$\frac{AM}{AC} = \frac{5}{3}$$

□

Problem 2.10 (Problem 2.33 (IMO 2000/1).). Two circles G_1 and G_2 intersect at two points M and N. Let AB be the line tangent to these circles at A and B, respectively, so that M lies closer to AB than N. Let CD be the line parallel to AB and passing through the point M, with C on G_1 and D on G_2. Lines AC and BD meet at E; lines AN and CD meet at P; lines BN and CD meet at Q. Show that $EP = EQ$.

Proof. Define $O_2 :=$ the center of G_2, since $AB \perp BO_2$ and $AB \parallel CD$ we have $MB = DB$, on the other hand, we have $\angle EBA = \angle BDM = \angle ABM$, similarly, $\angle EAB = \angle ACM = \angle BAM$, we get $\triangle EAB \cong \triangle MAB \Longrightarrow EB = MB$.

That implies $\angle EMQ = \frac{\pi}{2}$, it's well-known that AB's midpoint lies on the radical axis of these two circle. Since $AB \parallel PQ$, we deduce M is the midpoint of PQ. It follows that $EP = EQ$ as well. □

Problem 2.11 (Problem 2.34 (Canada 1990/3).). Let $ABCD$ be a cyclic quadrilateral whose diagonals meet at P. Let W, X, Y, Z be the feet of P onto AB, BC, CD, DA, respectively. Show that $WX + YZ = XY + WZ$.

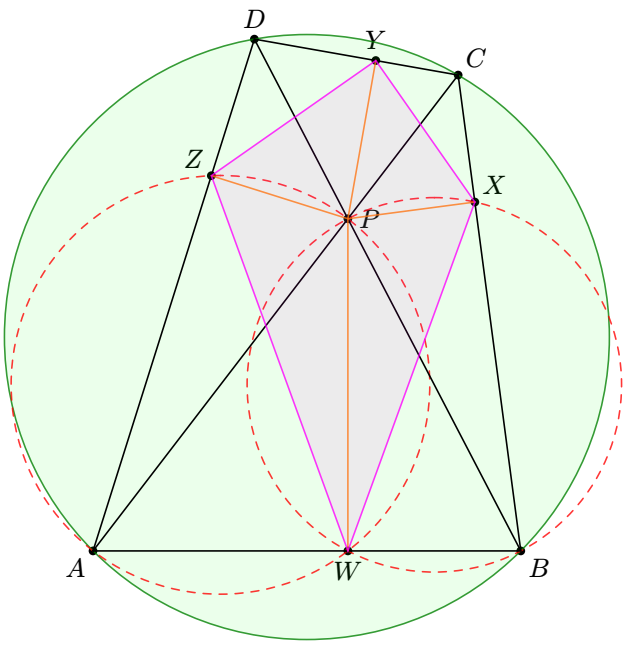

Figure 3G | Problem 2.34 (Canada 1990/3).

Proof. Obviously, P, X, B, W and P, Z, A, W are concyclic. Since we have

$$\angle PWX = \angle PBX = \angle DBC = \angle DAC = \angle ZAP = \angle ZWP$$

that implies P lies on the angle bisector of $\angle ZWX$, similarly, P also lies on the angle bisector of $\angle WXY, \angle XYZ, \angle YWZ$. Hence, the distance from P to each side were the same, the result is now trivial. □

Problem 2.12 (Problem 2.35 (IMO 2009/2).). Let ABC be a triangle with circumcenter O. The points P and Q are interior points of the sides CA and AB, respectively. Let K, L, and M be the midpoints of the segments BP, CQ, and PQ, respectively, and let Γ be the circle passing through K, L, and M. Suppose that the line P Q is tangent to the circle Γ . Prove that $OP = OQ$.

Proof. Notice that it's sufficient to show

$$Pow(P,(ABC)) = Pow(Q,(ABC)) \iff AQ \cdot QB = AP \cdot PC。$$

In fact, we have,

$$\begin{aligned}
\frac{AQ}{AP} &= \frac{PC}{QB} \\
\frac{\sin\angle APQ}{\sin\angle AQP} &= \frac{ML}{MK} \\
\frac{\sin\angle QPC}{\sin\angle PQB} &= \frac{\sin\angle MKL}{\sin\angle MLK} \\
\frac{\sin\angle MPC}{\sin\angle MQB} &= \frac{\sin\angle PML}{\sin\angle QMK}
\end{aligned}$$

which is obvious. □

Problem 2.13 (Problem 2.36.). Let AD, BE, CF be the altitudes of a scalene triangle ABC with circumcenter O. Prove that $(AOD), (BOE)$, and (COF) intersect at point X other than O.

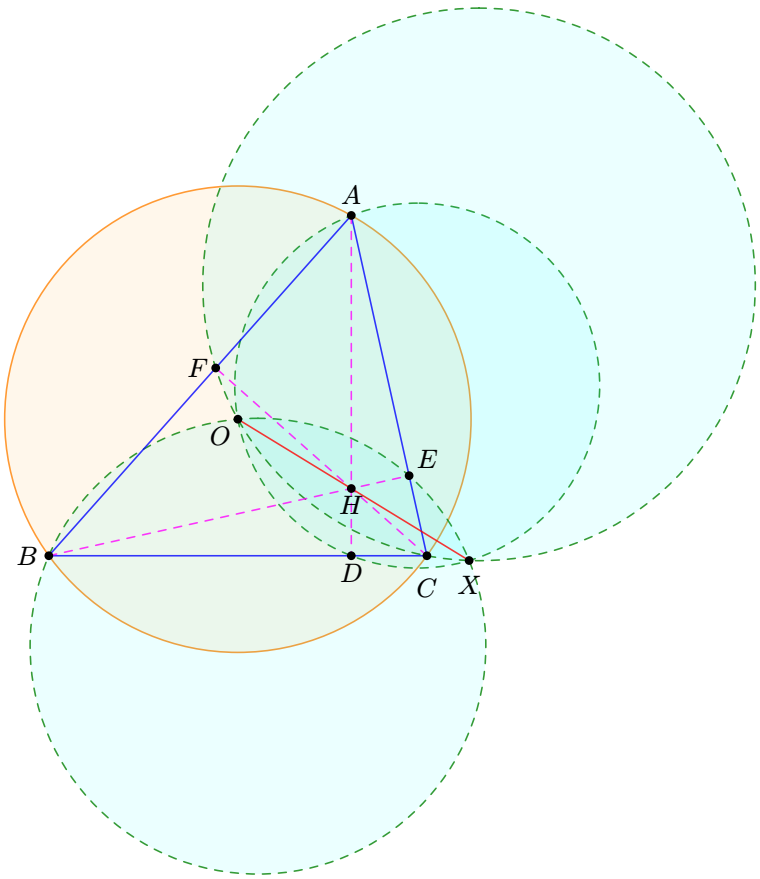

Figure 3H | Problem 2.36.

Proof. Notice that $Pow(H,(AOD)) = Pow(H,(BOE)) = Pow(H,(COF))$, the result follows. □

Problem 2.14 (Problem 2.37 (Canada 2007/5).). Let the incircle of triangle ABC touch sides BC, CA, and AB at D, E, and F, respectively. Let $\omega, \omega_1, \omega_2$, and ω_3 denote the circumcircles of triangles ABC, AEF, BDF, and CDE respectively. Let ω and ω_1 intersect at A and P, ω and ω_2 intersect at B and Q, ω and ω_3 intersect at C and R.

i) Prove that ω_1, ω_2, and ω_3 intersect in a common point.

ii) Show that lines PD, QE, and RF are concurrent.

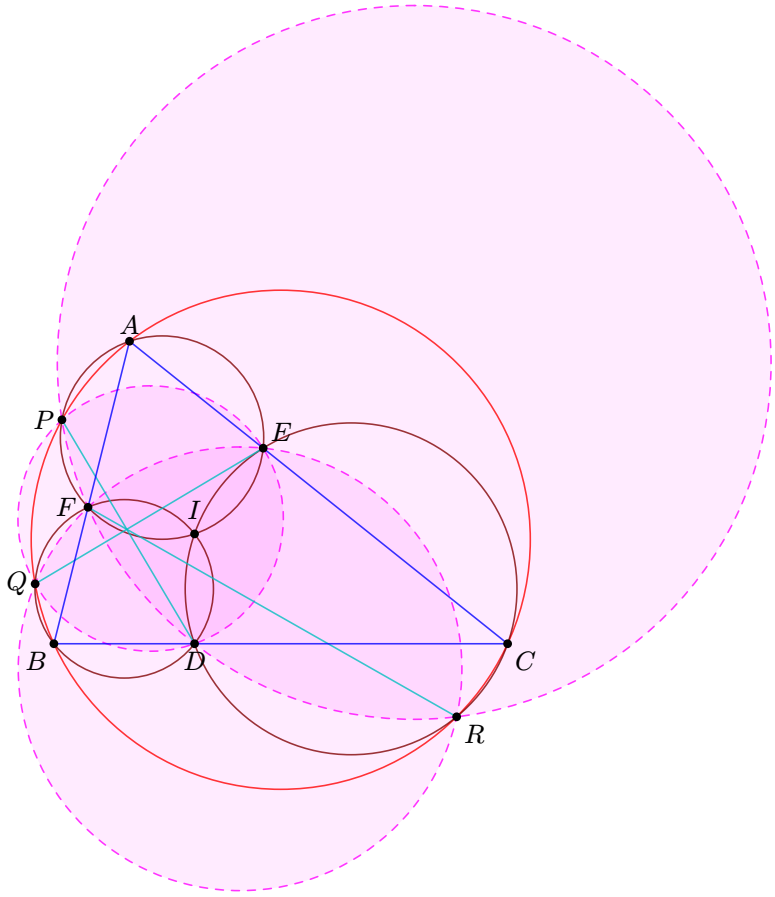

Figure 3I | Problem 2.37. (Canada 2007/5).

Proof (Part 1). Obvious. □

Proof (Part 2). Notice that it suffices to show E, D, Q, P concyclic.

Lemma 1 - Define $M := PD \cap \odot(ABC)$ other than P, M is the midpoint of arc BC

Proof of Lemma. It suffices to show $\angle BPD = \angle CPD$. In fact, we have

$$\frac{PB}{PC} = \frac{BF}{CE} = \frac{BD}{CD}$$

■

Since we have

$$\begin{aligned}\angle EPD &= \angle APD - \angle APE \\ &= B + \frac{A}{2} - \left(\frac{\pi}{2} - \frac{A}{2}\right) \\ &= A + B - \frac{\pi}{2} \\ \angle EQD &= \angle BQE - \angle BQD \\ &= A + \frac{B}{2} - \left(\frac{\pi}{2} - \frac{B}{2}\right) \\ &= A + B - \frac{\pi}{2}\end{aligned}$$

Hence the result. □

Problem 2.15 (Problem 2.38 (Iran TST 2011/1).). In acute triangle ABC, $\angle B$ is greater than $\angle C$. Let M be the midpoint of BC and let E and F be the feet of the altitudes from B and C, respectively. Let K and L be the midpoints of ME and MF, respectively, and let T be on line KL such that $TA \parallel BC$. Prove that $TA = TM$.

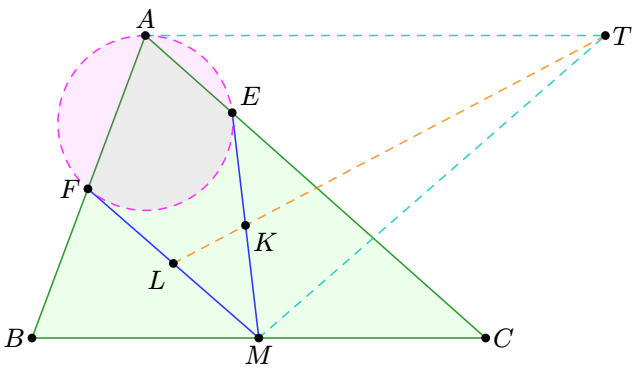

Figure 3J | Problem 2.38 (Iran TST 2011/1).

Proof.

Lemma 1 - TA, ME, MF are both tangent to $\odot(AEF)$

Proof of Lemma. See Problem 1.9 ■

Notice that $LM^2 = LF^2$ and $KM^2 = KE^2$, that implies KL is the radical axis of $\odot(AEF)$ and $\odot(M)$. The result is now trivial. □

CHAPTER 3
Lengths and Ratios

Twenty years from now you will be more disappointed by the things that you didn't do than by the ones you did do. So throw off the bowlines. Sail away from the safe harbor. Catch the trade winds in your sails. Explore. Dream. Discover.

— *Often attributed to Mark Twain*

Problem 3.1 (Problem 3.16.). Let ABC be a triangle with contact triangle DEF. Prove that AD, BE, CF concur. The point of concurrency is the Gergonne point of triangle ABC.

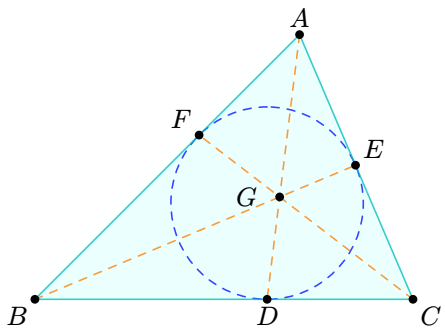

Figure 4A | Problem 3.16.

Proof. In fact, we have

$$\frac{AE}{EC}\frac{CD}{DB}\frac{BF}{FA} = \frac{s-a}{s-c}\frac{s-c}{s-b}\frac{s-b}{s-a} = 1, \quad s = \frac{a+b+c}{2}.$$

□

Problem 3.2 (Lemma 3.17.). In cyclic quadrilateral $ABCD$, points X and Y are the orthocenters of $\triangle ABC$ and $\triangle BCD$. Show that $AXYD$ is a parallelogram.

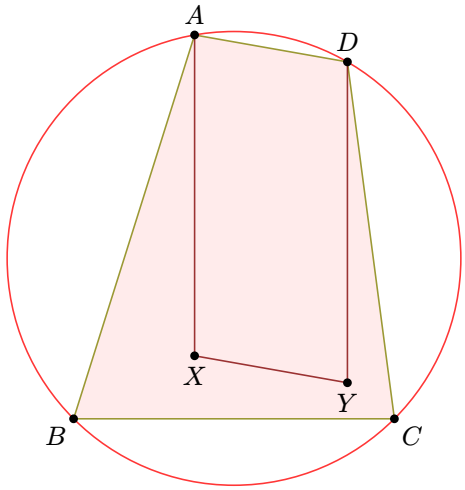

Figure 4B | Lemma 3.17.

Proof. Obviously $AX \parallel DY$. It suffices to show $AX = DY$. WLOG Assume $R = 1$.

$$AX = 2\cos\angle BAC = 2\cos\angle BDC = DY$$

which is clear. □

Problem 3.3 (Problem 3.18.). Let AD, BE, CF be concurrent cevians in a triangle, meeting at P. Prove that

$$\frac{PD}{AD} + \frac{PE}{BE} + \frac{PF}{CF} = 1$$

Proof. It's trivial since

$$\frac{PD}{AD} + \frac{PE}{BE} + \frac{PF}{CF} = \frac{[BPC]}{[BAC]} + \frac{[CPA]}{[CBA]} + \frac{[APB]}{[ACB]} = \frac{[ABC]}{[ABC]} = 1$$

□

Problem 3.4 (Problem 3.19 (Shortlist 2006/G3).). Let $ABCDE$ be a convex pentagon such that

$$\angle BAC = \angle CAD = \angle DAE \text{ and } \angle ABC = \angle ACD = \angle ADE.$$

Diagonals BD and CE meet at P. Prove that ray AP bisects CD.

Proof. Define $X := BD \cap AC, Y := CE \cap AD$

Obviously we have $\triangle ABC \sim \triangle ACD \sim \triangle ADE \implies ABCD \sim ACDE$, that implies $\frac{AX}{XC} = \frac{AY}{YD}$. From Ceva's theorem, the result is trivial. □

Problem 3.5 (Problem 3.20 (BAMO 2013/3).). Let H be the orthocenter of an acute triangle ABC. Consider the circumcenters of triangles ABH, BCH, and CAH. Prove that they are the vertices of a triangle that is congruent to ABC.

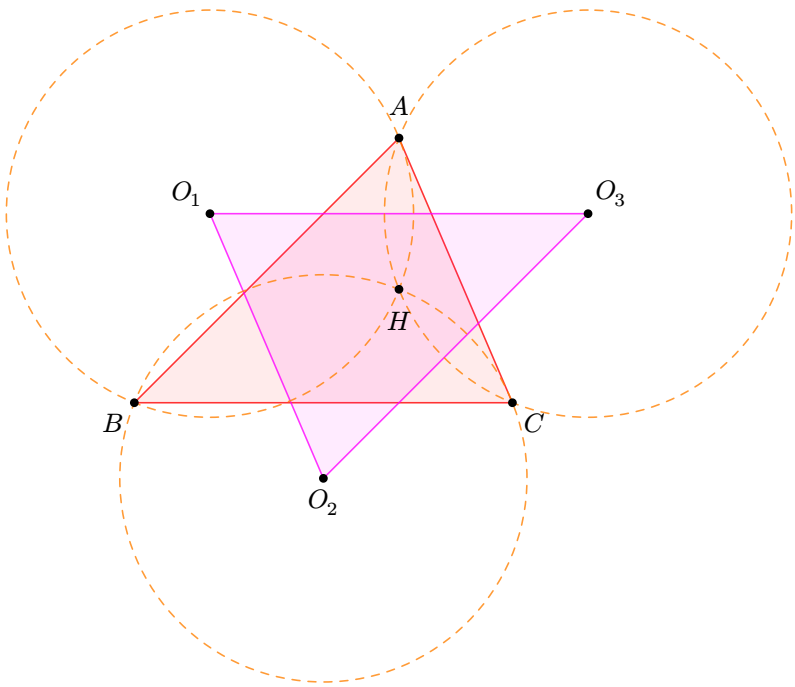

Figure 4C | Problem 3.20 (BAMO 2013/3).

Proof. Notice that

$$\begin{aligned}\angle O_1HO_3 &= \angle O_1HA + \angle O_3HA \\ &= \left(\frac{\pi}{2} - \frac{\angle AO_1H}{2}\right) + \left(\frac{\pi}{2} - \frac{\angle AO_3H}{2}\right) \\ &= \pi - \angle ABH - \angle ACH \\ &= 2A\end{aligned}$$

Same applies on $\angle O_1HO_2, \angle O_2HO_3$, thus H is the circumcenter of $\triangle O_1O_2O_3$. On the other hand,

$$\begin{aligned}\angle O_1BC + \angle O_3CB &= B + C + \angle O_1BA + \angle O_3CA \\ &= B + C + \left(\frac{\pi}{2} - \frac{\angle BO_1A}{2}\right) + \left(\frac{\pi}{2} - \frac{\angle CO_1A}{2}\right) \\ &= \pi\end{aligned}$$

conclude the things above we get O_1BCO_3 is a parallelogram, same for O_1ACO_2 and O_2BAO_3. Therefore, $\triangle O_1O_2O_3 \cong \triangle ABC$ (SSS) □

Problem 3.6 (Problem 3.21 (USAMO 2003/4).). Let ABC be a triangle. A circle passing through A and B intersects segments AC and BC at D and E, respectively. Lines AB and DE intersect at F, while lines BD and CF intersect at M. Prove that $MF = MC$ if and only if $MB \cdot MD = MC^2$.

Proof. Notice that

$$MB \cdot MD = MC^2 \iff \angle BCM = \angle CDM = \angle ADB = \angle AEB \overset{\text{Ceva}}{\iff} MF = MC$$

□

Problem 3.7 (Theorem 3.22 (Monge's Theorem).). Consider disjoint circles $\omega_1, \omega_2, \omega_3$ in the plane, no two congruent. For each pair of circles, we construct the intersection of their common external tangents. Prove that these three intersections are collinear.

Proof. Let O_1, O_2, O_3 be the center of these three circle respectively. Applying Menelaus' Theorem solve this problem. □

Problem 3.8 (Theorem 3.23 (Cevian Nest).). Let AX, BY, CZ be concurrent cevians of ABC. Let XD, YE, ZF be concurrent cevians in triangle XYZ. Prove that rays AD, BE, CF concur.

Proof. Define $P := AD \cap BC,\ Q := BE \cap CA,\ R := CF \cap AB$

Notice that

$$\begin{aligned}\frac{BP}{PC} &= \frac{AB \sin ZAD}{AC \sin YAD} \\ &= \frac{AB}{AC}\frac{ZD \cdot YA}{YD \cdot ZA}\end{aligned}$$

same applies on $\frac{CQ}{QA}$ and $\frac{AR}{RB}$. The result follows since Ceva. □

Problem 3.9 (Problem 3.24.). Let ABC be an acute triangle and suppose X is a point on (ABC) with $AX \parallel BC$ and $X \neq A$. Denote by G the centroid of triangle ABC, and by K the foot of the altitude from A to BC. Prove that K, G, X are collinear.

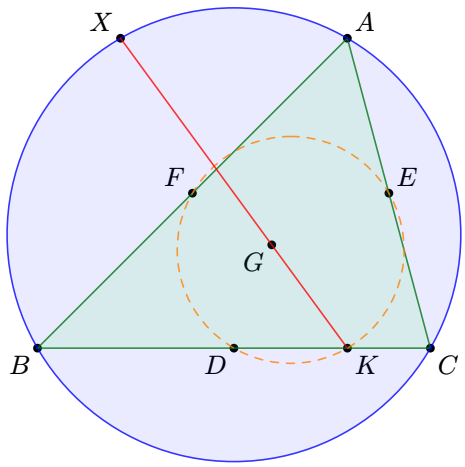

Figure 4D | Problem 3.24.

Proof (Homothety). It's well-known that G is still the centroid of the medial triangle. The problem is solved by using homothety at point G with Nine-point circle and circumcircle. □

Proof (Complex number). WLOG Assume circumcircle be the unit circle. We define

- $A = a$
- $C = c$
- $B = b = \overline{c} = \frac{1}{c}$

- $X = x = \overline{a} = \frac{1}{a}$

After some calculation

- $G = g = \frac{1}{3}(a+b+c) = \frac{1}{3c}(ac+1+c^2)$
- $K = \frac{1}{2}(a+b+c-bc\overline{a}) = \frac{1}{2ac}(a^2c+a+ac^2-c)$

It suffices to show

$$\begin{aligned}\frac{K-X}{G-X} &= \frac{\frac{1}{2ac}(a^2c+a+ac^2-c)-\frac{1}{a}}{\frac{1}{3c}(ac+1+c^2)-\frac{1}{a}} \\ &= \frac{3a(a^2c+a+ac^2-c)-6ac}{2a^2(ac+1+c^2)-6ac} \\ &= \frac{3a^3c+3a^2+3a^2c^2-9ac}{2a^3c+2a^2+2a^2c^2-6ac} \\ &= \frac{3}{2} \in \mathbb{R}\end{aligned}$$

done. □

Problem 3.10 (Problem 3.25 (USAMO 1993/2).). Let $ABCD$ be a quadrilateral whose diagonals AC and BD are perpendicular and intersect at E. Prove that the reflections of E across AB, BC, CD, DA are concyclic.

Proof. Define $W := EW \perp AB,\ X := EX \perp BC,\ Y := EY \perp CD,\ Z := EZ \perp DA$, Obviously $A, W, E, Z;\ B, X, E, W;\ C, Y, E, X;\ D, Y, E, Z$ are concyclic. Further more,

$$\begin{aligned}\angle WXY + \angle WZY &= \angle WXE + \angle EXY + \angle WAE + \angle EZY \\ &= \angle WBE + \angle ECY + \angle WAE + \angle EDY \\ &= \pi\end{aligned}$$

$\Longrightarrow W, X, Y, Z$ are concyclic, the result follows by homothety. □

Problem 3.11 (Problem 3.26 (EGMO 2013/1).). The side BC of the triangle ABC is extended beyond C to D so that $CD = BC$. The side CA is extended beyond A to E so that $AE = 2CA$. Prove that if $AD = BE$ then the triangle ABC is right-angled.

Proof. Define $F := BE \cap AD$, notice that A is the centroid of $\triangle EBD$, also $FA = \frac{1}{2}AD = \frac{1}{2}BE \Longrightarrow FA = FB = FE$, which implies $AB \perp CE$ hence the result. □

Problem 3.12 (Problem 3.27 (APMO 2004/2).). Let O be the circumcenter and H the orthocenter of an acute triangle ABC. Prove that the area of one of the triangles AOH, BOH, and COH is equal to the sum of the areas of the other two.

Proof. WLOG Assume $\angle A < \angle B < \angle C,\ R = 1$, We need to show $[BOH] = [AOH] + [COH]$

$$\begin{aligned}[AOH] = \frac{1}{2}AHd(O, AH) &= \cos A \sin\Big(A - 2\Big(\frac{\pi}{2} - C\Big)\Big) \\ &= \frac{1}{2}(\sin(2B) - \sin(2C))\end{aligned}$$

$$[BOH] = \frac{1}{2}BHd(O, BH) = \cos B \sin\left(B - 2\left(\frac{\pi}{2} - C\right)\right)$$
$$= \frac{1}{2}(\sin(2A) - \sin(2C))$$
$$[COH] = \frac{1}{2}CHd(O, CH) = \cos C \sin\left(C - 2\left(\frac{\pi}{2} - B\right)\right)$$
$$= \frac{1}{2}(\sin(2A) - \sin(2B))$$

The result is now obvious. □

Problem 3.13 (Problem 3.28 (Shortlist 2001/G1).). Let A_1 be the center of the square inscribed in acute triangle ABC with two vertices of the square on side BC. Thus one of the two remaining vertices of the square is on side AB and the other is on AC. Points B_1 and C_1 are defined in a similar way for inscribed squares with two vertices on sides AC and AB, respectively. Prove that lines AA_1, BB_1, CC_1 are concurrent.

Proof. Notice that

$$\frac{\sin \angle A_1AB}{\sin \angle A_1AC} = \frac{\sin(B + \frac{\pi}{4})}{\sin(C + \frac{\pi}{4})}$$

Same applies on other two ratios, hence the result by Ceva. □

Problem 3.14 (Problem 3.29 (USA TSTST 2011/4).). Acute triangle ABC is inscribed in circle ω. Let H and O denote its orthocenter and circumcenter, respectively. Let M and N be the midpoints of sides AB and AC, respectively. Rays MH and NH meet ω at P and Q, respectively. Lines MN and PQ meet at R. Prove that $OA \perp RA$.

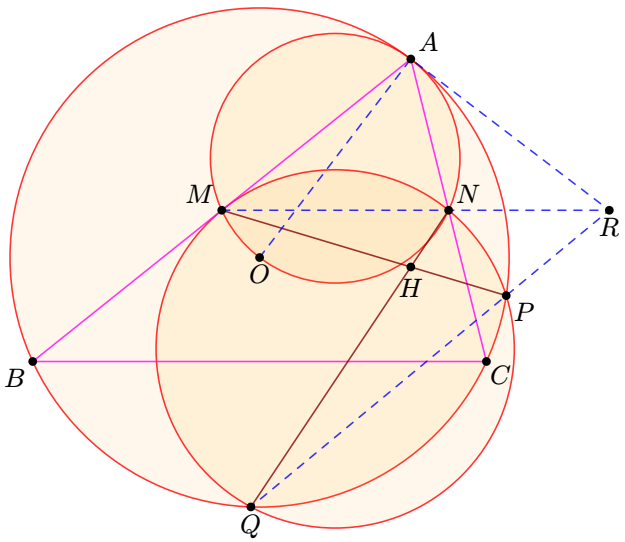

Figure 4E | Problem 3.29 (USA TSTST 2011/4).

Proof. Notice that $HP \cdot HM = \frac{1}{2}Pow(H, \odot(ABC)) = HQ \cdot HN \implies M, N, P, Q$ concyclic. Also since R is the radical center of $(ABC), (AMN), (MNPQ)$, the result follows. □

Problem 3.15 (Problem 3.30 (USAMO 2015/2).). Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90°$ and $AP = AQ < BP$. Let X be a variable point on segment

PQ. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that XT is perpendicular to AX. Let M denote the midpoint of chord ST. As X varies on segment PQ, show that M moves along a circle.

Proof. Notice that $Pow(A, \odot(N_9)) = \frac{1}{2}AP^2$ which is fixed, so is AN_9, since the homothety from the centroid G, the result follows. □

CHAPTER 4
Assorted Configurations

Exploration is really the essence of the human spirit.

— Frank Borman, Apollo 8 astronaut

Problem 4.1 (Problem 4.41 (Hong Kong 1998).). Let $PQRS$ be a cyclic quadrilateral with $\angle PSR = 90°$ and let H and K be the feet of the altitudes from Q to lines PR and PS. Prove that HK bisects QS.

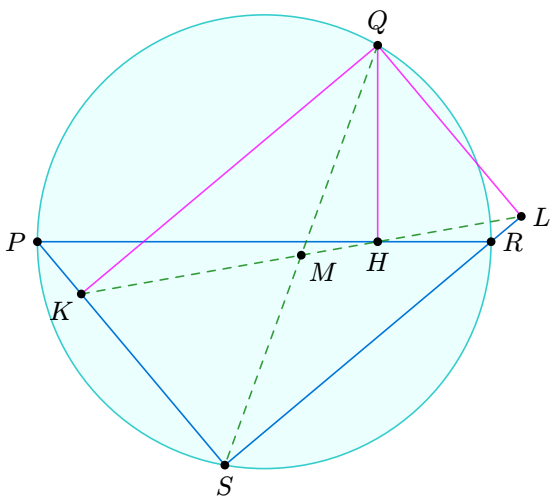

Figure 5A | Problem 4.41 (Hong Kong 1998).

Proof. According to Simson line, define $L :=$ The foot of the perpendicular from Q to SR. It suffices to show LK bisects QS, which is obvious since S is the orthocenter of $\triangle PRS$. □

Problem 4.2 (Problem 4.42 (USAMO 1988/4).). Suppose $\triangle ABC$ is a triangle with incenter I . Show that the circumcenters of $\triangle IAB, \triangle IBC,$ and $\triangle ICA$ lie on a circle whose center is the circumcenter of $\triangle ABC$.

Proof. Well-known that the circumcenter of $\triangle IAB, \triangle IBC,$ and $\triangle ICA$ lie on the midpoint of arc $AB,$ arc $BC,$ arc CA, respectively, the result follows. □

Problem 4.3 (Problem 4.43 (USAMO 1995/3).). Given a non-isosceles, non-right triangle ABC, let O denote its circumcenter, and let $A_1, B_1,$ and C_1 be the midpoints of sides $BC, CA,$ and AB, respectively. Point A_2 is located on the ray OA_1 so that $\triangle OAA_1$ is similar to $\triangle OA_2A$. Points B_2 and C_2 on rays OB_1 and OC_1, respectively, are defined similarly. Prove that lines $AA_2, BB_2,$ and CC_2 are concurrent.

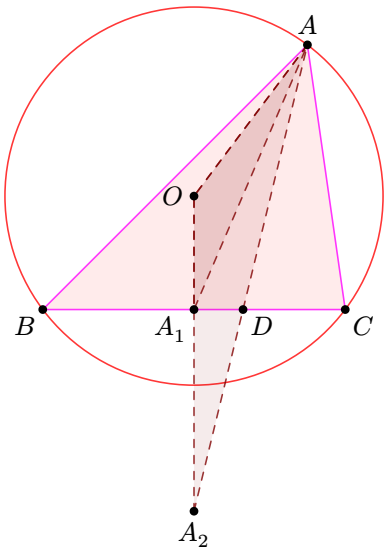

Figure 5B | Problem 4.43 (USAMO 1995/3).

Proof. Define $D := AA_2 \cap BC,\ E := BB_2 \cap CA,\ F := CC_2 \cap AB$. Since we have

$$\begin{gathered}
\measuredangle AA_1O + 90^\circ + \measuredangle A_1BA + \angle BAA_1 = 0 \\
\measuredangle OAA_2 + \measuredangle A_1BA + \measuredangle BAA_1 = 90^\circ \\
(\measuredangle BAC - \measuredangle BAO - \measuredangle A_2AC) + \measuredangle A_1BA + \measuredangle BAA_1 = 90^\circ \\
-\measuredangle ACB - 90^\circ + \measuredangle ACB - \measuredangle A_2AC + \measuredangle BAA_1 = 90^\circ \\
\measuredangle BAA_1 = \measuredangle A_2AC
\end{gathered}$$

which implies AA_2 is the A-symmedian of $\triangle ABC$, according to Ceva

$$\frac{BD}{DC}\frac{CE}{EA}\frac{AF}{FB} = \left(\frac{AB}{BC}\right)^2\left(\frac{BC}{CA}\right)^2\left(\frac{CA}{AB}\right)^2 = 1$$

as desired. □

Problem 4.4 (Problem 4.44 (USA TST 2014).). Let ABC be an acute triangle and let X be a variable interior point on the minor arc BC. Let P and Q be the feet of the perpendiculars from X to lines CA and CB, respectively. Let R be the intersection of line PQ and the perpendicular from B to AC. Let ℓ be the line through P parallel to XR. Prove that as X varies along minor arc BC, the line always passes through a fixed point.

Proof. According to the property of Simson line, it's easy to show $PXRH$ is a parallelogram, and H is the fixed point which satisfies the requirements of the question. □

Problem 4.5 (Problem 4.45 (USA TST 2011/1).). In an acute scalene triangle ABC, points D, E, F lie on sides BC, CA, AB, respectively, such that $AD \perp BC, BE \perp CA, CF \perp$

AB. Altitudes AD, BE, CF meet at orthocenter H. Points P and Q lie on segment EF such that $AP \perp EF$ and $HQ \perp EF$. Lines DP and QH intersect at point R. Compute $\frac{HQ}{HR}$.

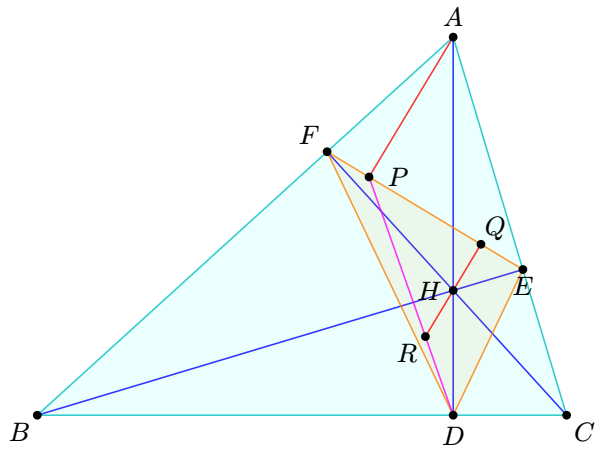

Figure 5C | Problem 4.45 (USA TST 2011/1).

Solution (Projective Geometry). Notice that

$$(R, Q; H, P_\infty) \overset{P}{=} (D, EF \cap AD; H, A) = -1.$$

Since $AP \parallel QR$, which implies H is the midpoint of $QR \Rightarrow \frac{HQ}{HR} = 1$. □

Solution (Pure Geometry). Notice that for $\triangle DEF$, H is incenter, A is D-excenter, P is the tangent point of the excircle and EF, well-known R is the opposite of Q for the incircle. Hence $\frac{HQ}{HR} = 1$ □

Problem 4.6 (Problem 4.46 (ELMO Shortlist 2012).). Circles Ω and ω are internally tangent at point C. Chord AB of Ω is tangent to ω at E, where E is the midpoint of AB. Another circle, ω_1 is tangent to Ω, ω, and AB at D, Z, and F respectively. Rays CD and AB meet at P. If $M \neq C$ is the midpoint of major arc AB, show that

$$\tan \angle ZEP = \frac{PE}{CM}$$

Proof. Notice that $\tan \angle ZEP = \frac{EZ}{CE}$, it suffices to show

$$\frac{EZ}{PE} = \frac{CE}{MC}$$

which is equivalent to $\triangle ZCM \sim \triangle ZEP$ since $\angle ZCM = \angle ZEP$. It suffices to show

$$\angle ZPE = \angle ZME \iff P, Z, E, M \text{ concyclic} \iff PZ \perp ZM$$

In fact, we have

$$\angle CEF = \angle CDF = 90^\circ \Rightarrow C, D, F, E \text{ concyclic}$$

this implies M lies on the radical axis of ω and ω_1, on the other hand, P is the homothetic center of ω and ω_1, the result follows. □

Problem 4.7 (Problem 4.47 (USAMO 2011/5).). Let P be a point inside convex quadrilateral $ABCD$. Points Q_1 and Q_2 are located within $ABCD$ such that

$$\angle Q_1BC = \angle ABP,\ \angle Q_1CB = \angle DCP,$$

$$\angle Q_2AD = \angle BAP,\ \angle Q_2DA = \angle CDP.$$

Prove that $Q_1Q_2 \parallel AB$ if and only if $Q_1Q_2 \parallel CD$.

Proof. Define $E := AD \cap BC$, notice that Q_1 is the isogonal conjugate of P with respect to $\triangle CDE$, Q_2 is the isogonal conjugate of P with respect to $\triangle ABE$, that implies E, Q_1, Q_2 are collinear, the result is now obvious. □

Problem 4.8 (Problem 4.48 (Japanese Olympiad 2009).). Triangle ABC is inscribed in circle. A circle with center O is drawn, tangent to side BC at a point P, and internally tangent to the arc BC of not containing A at a point Q. Show that if $\angle BAO = \angle CAO$ then $\angle PAO = \angle QAO$.

Proof. Define $M := AO \cap (ABC)$, $N := PQ \cap (ABC)$, obviously MN is the perpendicular bisector of the chord BC. According to the Reim's theorem, Q, O, P, A are concyclic. Therefore,

$$\angle PAO = \angle PQO = \angle QPO = \angle QAO$$

□

Problem 4.9 (Problem 4.49.). Let ABC be a triangle and let its incircle touch BC at D. Let T be the tangency point of the A-mixtilinear incircle with (ABC). Prove that $\angle BTA = \angle CTD$.

Proof. Notice that it suffices to show A, E, S collinear, where E, S is the reflection of D, T over the perpendicular bisector of BC. Well-known AT and AE is isogonal, redefine $S := AE \cap (ABC)$, we have S is the reflection of T easily, the result follow. □

Problem 4.10 (Problem 4.50 (Vietnam TST 2003/2).). Let ABC be a scalene triangle with circumcenter O and incenter I . Let H, K, L be the feet of the altitudes of triangle ABC from the vertices A, B, C, respectively. Denote by A_0, B_0, C_0 the midpoints of these altitudes AH, BK, CL, respectively. The incircle of triangle ABC touches the sides BC, CA, AB at the points D, E, F, respectively. Prove that the four lines $A_0D, B_0E, C_0F,$ and OI are concurrent.

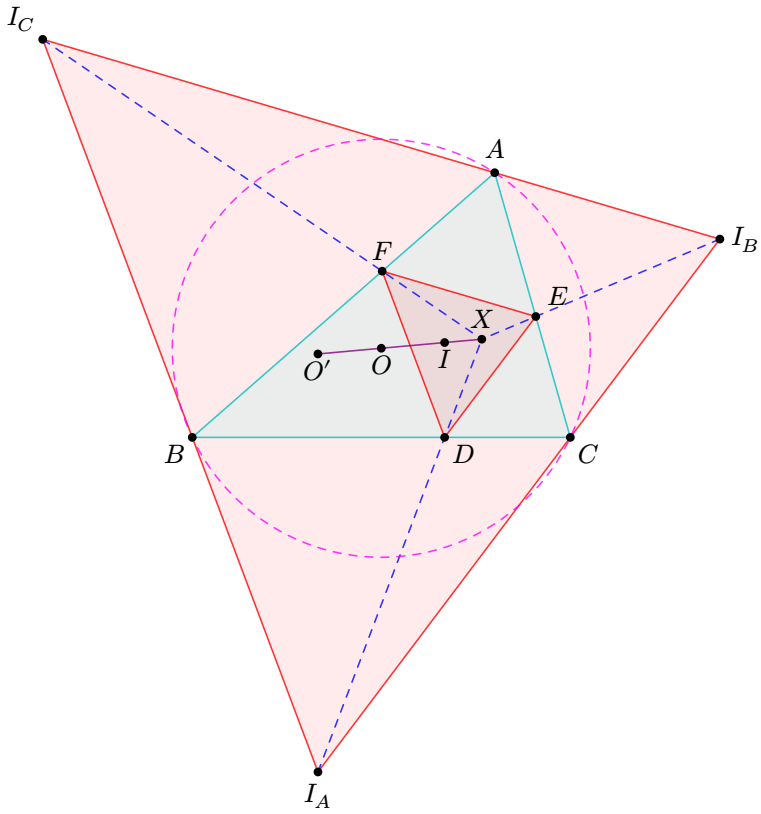

Figure 5D | Problem 4.50 (Vietnam TST 2003/2).

Proof. Let I_A, I_B, I_C denote the excenters. Well-known A_0, D, I_A collinear, we take I_AD instead, same for the others, also with the homothety taking $\triangle I_AI_BI_C \mapsto \triangle DEF$, we already have DI_A, EI_B, FI_C concur at the same point X. Define O' be the circumcenter of $\triangle I_AI_BI_C$, then we have $O' \mapsto I$ under h, and since O became the nine-point center of $\triangle I_AI_BI_C$, which implies I, O, X, O' collinear. Hence the result. □

Problem 4.11 (Problem 4.51 (Sharygin 2013).). The incircle of $\triangle ABC$ touches BC, CA, AB at points A', B' and C' respectively. The perpendicular from the incenter I to the C-median meets the line $A'B'$ in point K. Prove that $CK \parallel AB$.

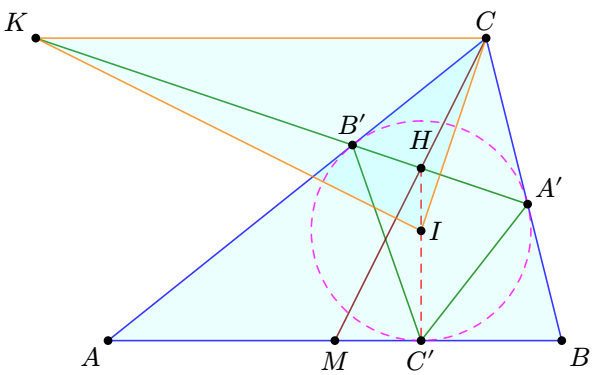

Figure 5E | Problem 4.51 (Sharygin 2013).

Proof. Notice that $H := CM \cap A'B'$ is the orthocenter of $\triangle KCI$, it suffices to show H, C', I collinear, which is obvious. □

Problem 4.12 (Problem 4.52 (APMO 2012/4).). Let ABC be an acute triangle. Denote by D the foot of the perpendicular line drawn from the point A to the side BC, by M the midpoint of BC, and by H the orthocenter of ABC. Let E be the point of intersection of the circumcircle Γ of the triangle ABC and the ray MH, and F be the point of intersection (other than E) of the line ED and the circle Γ. Prove that $\frac{BF}{CF} = \frac{AB}{AC}$ must hold.

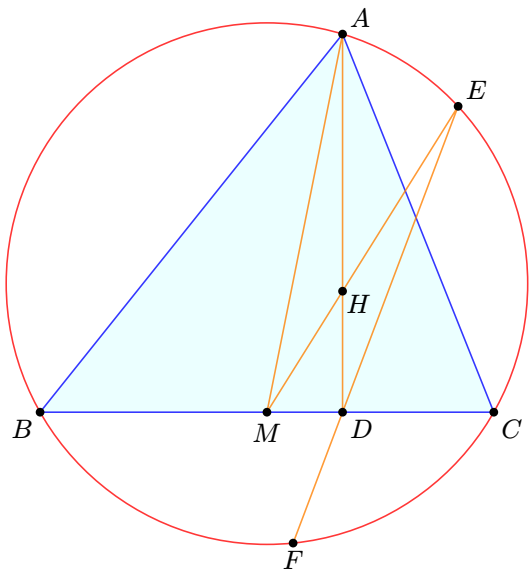

Figure 5F | Problem 4.52 (APMO 2012/4).

Proof. Well-known A, E, D, M concyclic. In fact we have

$$\angle BAF = \angle BEF = \angle BED = \pi - \angle EBC - \angle EDB$$
$$= \pi - \angle EBC - (\pi - \angle MAE) = \angle MAE - \angle EAC = \angle MAC$$

which show AF is A-symmedian, hence the result. □

Problem 4.13 (Problem 4.53 (Shortlist 2002/G7).). The incircle Ω of the acute triangle ABC is tangent to BC at a point K. Let AD be an altitude of triangle ABC, and let M be the midpoint of the segment AD. If N is the common point of the circle Ω and the line KM (distinct from K), then prove that the incircle Ω and the circumcircle of triangle BCN are tangent to each other at the point N.

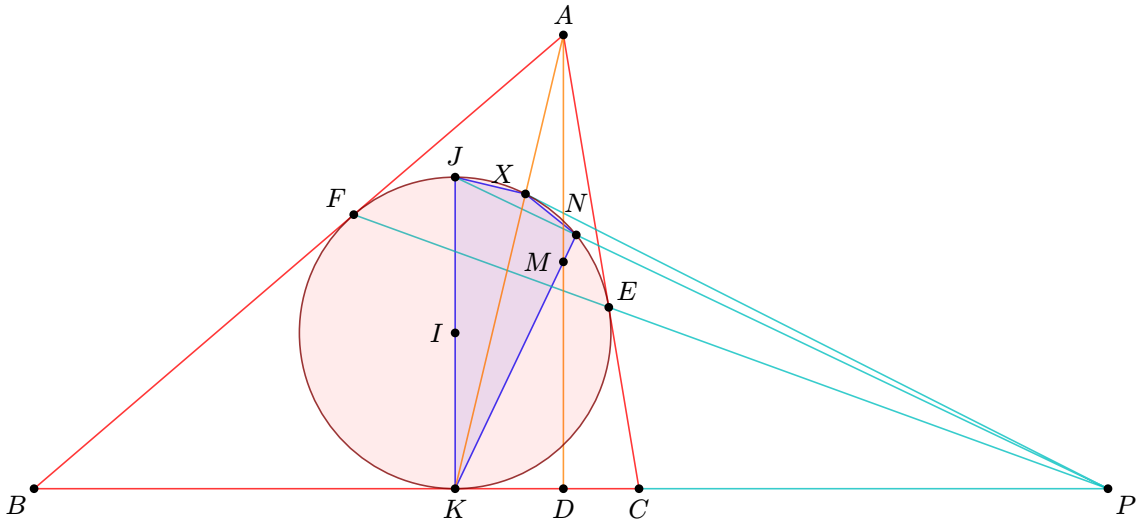

Figure 5G | Problem 4.53 (Shortlist 2002/G7).

Proof. It's sufficient to show KN bisects $\angle BNC$.

Define J, X, P, s.t. $EF \cap BC =: P$, $AK \cap (KEF) =: X \neq K$, JK is the diameter of (KEF). Well-known $(B, C; X, P) = -1$, it suffices to show J, N, P collinear, since $\{KD, KM, KA, KI\}$ is harmonic bundles. We have $JXNK$ is harmonic quadrilateral. Well-known $P = XX \cap KK$, the result follows. $\square$

CHAPTER 5
Computational Geometry

The gladdest moment in human life is a departure into unknown lands.

— *Sir Richard Burton*

Problem 5.1 (Problem 5.16 (Star Theorem).). Let $A_1A_2A_3A_4A_5$ be a convex pentagon. Suppose rays A_2A_3 and A_5A_4 meet at the point X_1. Define X_2, X_3, X_4, X_5 similarly. Prove that

$$\prod_{i=1}^{5} X_iA_{i+2} = \prod_{i=1}^{5} X_iA_{i+3}$$

where the indices are taken modulo 5.

Proof. It suffices to show

$$\begin{aligned}
\prod_{i=1}^{5} X_iA_{i+2} : \prod_{i=1}^{5} X_iA_{i+3} &= 1 \\
\prod_{i=1}^{5} \frac{X_iA_{i+2}}{X_iA_{i+3}} &= 1 \\
\prod_{i=1}^{5} \frac{\sin\angle X_iA_{i+3}A_{i+2}}{\sin\angle X_iA_{i+2}A_{i+3}} &= 1 \\
\prod_{i=1}^{5} \frac{\sin\angle X_{i-1}A_{i+2}A_{i+1}}{\sin\angle X_iA_{i+2}A_{i+3}} &= 1
\end{aligned}$$

which is obvious. □

Problem 5.2 (Problem 5.17.). Let ABC be a triangle with inradius r. If the exradii of ABC are r_A, r_B, r_C, show that the triangle has area $\sqrt{r \cdot r_A \cdot r_B \cdot r_C}$.

Proof. Well-known $r_A = \frac{sr}{s-a},\ r_B = \frac{sr}{s-b},\ r_C = \frac{sr}{s-c}$, it suffices to show

$$\sqrt{\frac{s^3r^4}{(s-a)(s-b)(s-c)}} = [ABC]$$
$$\sqrt{\frac{s^4r^4}{s(s-a)(s-b)(s-c)}} = [ABC]$$
$$(sr)^2 = [ABC]^2$$

which is obvious. □

Problem 5.3 (Problem 5.18 (APMO 2013/1).). Let ABC be an acute triangle with altitudes AD, BE and CF, and let O be the center of its circumcircle. Show that the segments OA, OF, OB, OD, OC, OE dissect the triangle ABC into three pairs of triangles that have equal areas.

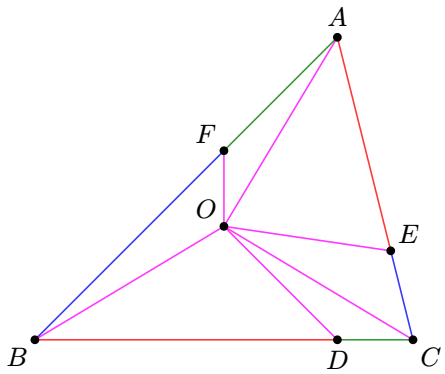

Figure 6A | Problem 5.18 (APMO 2013/1).

Proof. WLOG Assume $R = 1$, calculate the area for each triangle:

$$[AOE] = \frac{1}{2}c\cos A\cos B$$
$$[EOC] = \frac{1}{2}a\cos C\cos B$$
$$[COD] = \frac{1}{2}b\cos C\cos A$$
$$[DOB] = \frac{1}{2}c\cos B\cos A$$
$$[BOF] = \frac{1}{2}a\cos B\cos C$$
$$[FOA] = \frac{1}{2}b\cos A\cos C$$

Hence the result. □

Problem 5.4 (Problem 5.19 (EGMO 2013/1).). The side BC of the triangle ABC is extended beyond C to D so that $CD = BC$. The side CA is extended beyond A to E so that $AE = 2CA$. Prove that if $AD = BE$ then triangle ABC is right-angled.

Proof. We start from $AD = BE$:

$$\begin{aligned} AD^2 &= BE^2 \\ AE^2 + AB^2 - 2AE \cdot AB \cos\angle BAE &= AC^2 + CD^2 - 2AC \cdot CD \cos\angle ACD \\ (2b)^2 + c^2 + 4bc\cos A &= b^2 + a^2 + 2ba\cos C \\ (2b)^2 + c^2 + 2(b^2 + c^2 - a^2) &= b^2 + a^2 + (a^2 + b^2 - c^2) \\ b^2 + c^2 &= a^2 \end{aligned}$$

and we are done. □

Problem 5.5 (Problem 5.20 (Harvard-MIT Math Tournament 2013).). Let triangle ABC satisfy $2BC = AB + AC$ and have incenter I and circumcircle ω. Let D be the intersection of AI and ω (with A, D distinct). Prove that I is the midpoint of AD.

Proof. Let DEF be the contact triangle, with D, E, F lie on BC, CA, AB respectively, and M lies on BC, s.t. $DM \perp BC$. Since $2BC = AB + AC$, we have $AE = AF = \frac{a}{2}$. Notice that

$$\begin{aligned} AI &= \frac{AE}{\tan\frac{A}{2}} = \frac{a}{2\tan\frac{A}{2}} \\ ID = IC &= \frac{IM}{\tan\frac{A}{2}} = \frac{a}{2\tan\frac{A}{2}} \end{aligned}$$

the result is now obvious. □

Problem 5.6 (Problem 5.21 (USAMO 2010/4).). Let ABC be a triangle with $\angle A = 90°$. Points D and E lie on sides AC and AB, respectively, such that $\angle ABD = \angle DBC$ and $\angle ACE = \angle ECB$. Segments BD and CE meet at I. Determine whether or not it is possible for segments AB, AC, BI, ID, CI, IE to all have integer lengths.

Proof. Actually, not even AB, AC, BI, CI be all integer. Since

$$\begin{aligned} BC^2 &= BI^2 + CI^2 - 2BI \cdot CI \cos\angle BIC \\ AB^2 + AC^2 - BI^2 - CI^2 &= \sqrt{2} BI \cdot CI \\ \sqrt{2} &= \frac{AB^2 + AC^2 - BI^2 - CI^2}{BI \cdot CI} \end{aligned}$$

therefore, there's no way for AB, AC, BI, CI be all integer. □

Problem 5.7 (Problem 5.22 (Iran Olympiad 1999).). Let I be the incenter of triangle ABC and let ray AI meet the circumcircle of ABC at D. Denote the feet of the perpendiculars from I to lines BD and CD by E and F, respectively. If $IE + IF = \frac{1}{2}AD$, calculate $\angle BAC$.

Solution. WLOD Assume $R = 1$, get start from the condition $IE + IF = \frac{1}{2}AD$:

$$\begin{aligned}
IE + IF &= \frac{1}{2}AD \\
ID(\sin C + \sin B) &= \frac{1}{2}\left(\sin\left(B + \frac{A}{2}\right) + \sin\left(C + \frac{A}{2}\right)\right) \\
\sin\frac{A}{2}\left(\sin\frac{C+B}{2}\sin\frac{C-B}{2}\right) &= \frac{1}{4}\sin\frac{C-B}{2} \\
2\sin\frac{A}{2}\cos\frac{A}{2} &= \frac{1}{2} \\
\sin A &= \frac{1}{2} \\
\Rightarrow \angle A &= 30^\circ \text{ or } 150^\circ
\end{aligned}$$

Hence the result. □

Problem 5.8 (Problem 5.23 (CGMO 2002/4).). Circles Γ_1 and Γ_2 intersect at two points B and C, and BC is the diameter of Γ_1. Construct a tangent line to circle Γ_1 at C intersecting Γ_2 at another point A. Line AB meets Γ_1 again at E and line CE meets Γ_2 again at F. Let H be an arbitrary point on segment AF. Line HE meets Γ_1 again at G, and BG meets AC at D. Prove that

$$\frac{AH}{HF} = \frac{AC}{CD}$$

Proof. In fact, we have

$$\begin{aligned}
\frac{AC}{CD} &= \frac{AB\sin CBE}{DB\sin CBG} \\
&= \frac{\sin CDG\sin CBE}{\sin CAE\sin CBG} \\
&= \frac{\tan CBE}{\tan CBG} \\
&= \frac{CE\cdot BG}{GC\cdot BE}
\end{aligned}$$

$$\begin{aligned}
\frac{AH}{HF} &= \frac{EA\sin BEG}{EF\sin CEG} \\
&= \frac{CE\sin BEG}{BE\sin CEG} \\
&= \frac{CE\cdot BG}{BE\cdot CG}
\end{aligned}$$

we are done. □

Problem 5.9 (Problem 5.24 (IMO 2007/4).). In triangle ABC the bisector of angle BCA intersects the circumcircle again at R, the perpendicular bisector of BC at P, and the perpendicular bisector of AC at Q. The midpoint of BC is K and the midpoint of AC is L. Prove that the triangles RPK and RQL have the same area.

Proof. We will prove this question by using computational geometry, WLOG Assume $R = 1$ and start at RC:

$$\begin{aligned}
RC &= \frac{RB \cdot AC + RA \cdot BC}{AB} \\
&= \frac{2\sin\frac{C}{2}(a+b)}{c} \\
&= 2\sin\frac{C}{2}\frac{a+b}{c} \\
[RPK] &= \frac{1}{2}RP \cdot \frac{a}{2}\sin\frac{C}{2} \\
&= \frac{1}{2}\left(2\sin\frac{C}{2}\frac{a+b}{c} - \frac{a}{2\cos\frac{C}{2}}\right)\frac{a}{2}\sin\frac{C}{2} \\
&= \frac{a}{8}\tan\frac{C}{2}(a+b-a) \\
&= \frac{1}{8}ab\tan\frac{C}{2} \\
[RQL] &= \frac{1}{2}RQ \cdot \frac{b}{2}\sin\frac{C}{2} \\
&= \frac{1}{2}\left(2\sin\frac{C}{2}\frac{a+b}{c} - \frac{b}{2\cos\frac{C}{2}}\right)\frac{b}{2}\sin\frac{C}{2} \\
&= \frac{b}{8}\tan\frac{C}{2}(a+b-b) \\
&= \frac{1}{8}ab\tan\frac{C}{2}
\end{aligned}$$

Hence the result. □

Problem 5.10 (Problem 5.25 (JMO 2013/5).). Quadrilateral $XABY$ is inscribed in the semicircle ω with diameter XY. Segments AY and BX meet at P. Point Z is the foot of the perpendicular from P to line XY. Point C lies on ω such that line XC is perpendicular to line AZ. Let Q be the intersection of segments AY and XC. Prove that

$$\frac{BY}{XP} + \frac{CY}{XQ} = \frac{AY}{AX}$$

Proof. Notice that $AZ \parallel CY$, then

$$\angle BXY = \angle PXZ = \angle PAZ = \angle CYA = \angle CXA$$

Let $\angle BXY = \angle CXA = \alpha$, $\angle BXA = \angle CXY = \beta$, it suffices to show

$$\frac{BY}{XP} + \frac{CY}{XQ} = \frac{1}{\tan(\frac{\pi}{2} - \alpha - \beta)} = \tan(\alpha + \beta)$$

WLOG Assume $R = 1$, in fact, we have

$$\begin{aligned}
BY &= 2\sin\alpha \\
XP &= \frac{AX}{\cos\beta} = \frac{2\cos(\alpha+\beta)}{\cos\beta} \\
CY &= 2\sin\beta \\
XQ &= \frac{AX}{\cos\alpha} = \frac{2\cos(\alpha+\beta)}{\cos\alpha}
\end{aligned}$$

$$\text{LHS} = \frac{\sin\alpha\cos\beta + \sin\beta\cos\alpha}{\cos(\alpha+\beta)} = \frac{\sin(\alpha+\beta)}{\cos(\alpha+\beta)} = \tan(\alpha+\beta) = \text{RHS}$$

Hence the result. □

Problem 5.11 (Problem 5.26 (CGMO 2007/5).). Point D lies inside triangle ABC such that $\angle DAC = \angle DCA = 30°$ and $\angle DBA = 60°$. Point E is the midpoint of segment BC. Point F lies on segment AC with $AF = 2FC$. Prove that $DE \perp EF$.

Proof. Define $M :=$ midpoint of AC, $N :=$ midpoint of DC, obviously $DM \perp AC$ and $FN \perp DC$, we have D, N, F, M concyclic, with diameter DF, also $ME \parallel AB$ and $EN \parallel BD \Longrightarrow \angle DBA = \angle NEM = 60° \Longrightarrow D, E, N, F, M$ concyclic, because of $\angle MDN = \angle MEN = 60°$. Hence the result. □

Problem 5.12 (Problem 5.27 (ISL 2011/G1).). Let ABC be an acute triangle. Let ω be a circle whose center L lies on the side BC. Suppose that ω is tangent to AB at B' and AC at C'. Suppose also that the circumcenter O of triangle ABC lies on the shorter arc $B'C'$ of ω. Prove that the circumcircle of ABC and ω meet at two points.

Proof. It suffices to show $r = OL \geq OM = \cos A > \frac{1}{2}R$, with $M =$ midpoint of BC, which mean $\angle A < 60°$, in fact, we have

$$\angle B'OC' = \pi - \frac{1}{2}\angle B'LC = \pi - \frac{1}{2}(\pi - A) = \frac{\pi}{2} - \frac{A}{2}$$

with $\angle B'OC' > \angle BOC = 2A$, we get $A < 60°$, as needed. □

Problem 5.13 (Problem 5.28 (IMO 2001/1).). Consider an acute-angled triangle ABC. Let P be the foot of the altitude of triangle ABC issuing from the vertex A, and let O be the circumcenter of triangle ABC. Assume that $\angle C \geq \angle B + 30°$. Prove that $\angle A + \angle COP < 90°$.

Proof. Notice that it suffices to show $PC < PO \Longleftrightarrow PC^2 < PO^2$

$$\begin{aligned}
PC^2 < PO^2 &\Longleftrightarrow (b\cos C)^2 < \cos^2 A + \left(\frac{a}{2} - b\cos C\right)^2 \\
&\Longleftrightarrow (b\cos C)^2 < 1 - \frac{a^2}{4} + \frac{a^2}{4} - ab\cos C + (b\cos C)^2 \\
&\Longleftrightarrow 4\sin A\sin B\cos C < 1 \\
&\Longleftrightarrow 2\sin A(\sin(B+C) - \sin(C-B)) < 1 \\
&\Longleftrightarrow 2\sin A\left(\sin A - \frac{1}{2}\right) < 2\left(1 - \frac{1}{2}\right) = 1
\end{aligned}$$

Hence the result. □

Problem 5.14 (Problem 5.29 (IMO 2001/5).). Let ABC be a triangle. Let AP bisect $\angle BAC$ and let BQ bisect $\angle ABC$, with P on BC and Q on AC. If $AB + BP = AQ + QB$ and $\angle BAC = 60°$, what are the angles of the triangle?

Proof. Let $\angle ABC = 2x$, since

$$\begin{aligned}
BP &= \frac{\sin 30°}{\sin(150° - 2x)}AB \\
AQ &= \frac{\sin x}{\sin(120° - x)}AB \\
BQ &= \frac{\sin 60°}{\sin(120° - x)}AB
\end{aligned}$$

we have

$$1+\frac{\sin 30^\circ}{\sin(150^\circ-2x)}=\frac{\sin x+\sin 60^\circ}{\sin(120^\circ-x)}$$

Let $y=\frac{1}{2}x$, notice that

$$\begin{aligned}\sin(120^\circ-2y)&=\sin(60^\circ+2y)=2\sin(30^\circ+y)\cos(30^\circ+y)\\ \sin 2y+\sin 60^\circ&=2\sin(30^\circ+y)\cos(30^\circ-y)\end{aligned}$$

therefore,

$$\begin{aligned}\frac{1}{2\sin(150^\circ-4y)}=\frac{\cos(30^\circ-y)}{\cos(30^\circ+y)}-1&=\frac{2\sin 30^\circ\sin y}{\cos(30^\circ+y)}\\ 2\sin y\sin(150^\circ-4y)&=\cos(30^\circ+y)\\ 2\left(-\frac{1}{2}\right)(\cos(5y+30^\circ)-\cos(3y+30^\circ))&=\cos(30^\circ+y)\\ \cos(y+30^\circ)+\cos(5y+30^\circ)&=\cos(3y+30^\circ)\\ 2\cos(3y+30^\circ)\cos 2y&=\cos(3y+30^\circ)\\ \cos(3y+30^\circ)(2\cos 2y-1)&=0\end{aligned}$$

Recall

$$y=\frac{1}{2}x=\frac{1}{4}\angle B<\frac{1}{4}(180^\circ-\angle A)=30^\circ$$

It's impossible for $\cos 2y=\frac{1}{2}$, then $\cos(3y+30^\circ)=0$, the only possible value for y is 20°, giving us $\angle B=40^\circ$ and $\angle C=80^\circ$. □

Problem 5.15 (Problem 5.30 (IMO 2001/6).). Let $a>b>c>d$ be positive integers and suppose that

$$ac+bd=(b+d+a-c)(b+d-a+c)$$

Prove that $ab+cd$ is not prime.

Proof. Construct a quadrilateral $WXYZ$, with $WX=a, XY=b, YZ=d, ZW=c$. Notice that the problem condition can be rewrite as $a^2-ac+c^2=b^2+bd+d^2$, which can be consider as the length of XZ, finally shows $WXYZ$ concyclic. Also since the strong form of Ptolemy's theorem

$$XZ^2=\frac{(ad+bc)(ab+cd)}{ac+bd}$$

Since $a>b>c>d$, we have $ab+cd>ac+bd>ad+bc$, assume $ab+cd$ isn't a prime then $\left(b^2+bd+d^2\right)^2=XZ^2=\frac{(ad+bc)(ab+cd)}{ac+bd}\notin\mathbb{Z}$, which is a contradiction. The result follows. □

CHAPTER 6
Complex number

Problem 6.1 (Problem 6.29.). Give a proof of the inscribed angle theorem using complex numbers.

Proof. Too easy, just do it yourself. □

Problem 6.2 (Lemma 6.30 (Complex Chord).). Show that a point P lies on a chord AB of the unit circle if and only if $p + ab\overline{p} = a + b$.

Proof. It's easy to spot that the foot from P to AB is itself, therefore

$$p = \frac{1}{2}(p + a + b - ab\overline{p})$$

we have the result after simplify. □

Problem 6.3 (Problem 6.31.). Let $ABCD$ be a cyclic quadrilateral. Let H_A, H_B, H_C, H_D denote the orthocenters of triangles $BCD, CDA, DAB,$ and ABC, respectively. Prove that $AH_A, BH_B, CH_C,$ and DH_D concur.

Proof. Let $A = a, B = b, C = c, D = d$ lies on the unit circle, then we have

$$\begin{aligned} H_A &= b + c + d \\ H_B &= a + c + d \\ H_C &= a + b + d \\ H_D &= a + b + c \end{aligned}$$

Notice that for $P := \frac{a+b+c+d}{2}$

$$\begin{aligned} P &= \frac{1}{2}A + \left(1 - \frac{1}{2}\right)H_A \\ &= \frac{1}{2}B + \left(1 - \frac{1}{2}\right)H_B \\ &= \frac{1}{2}C + \left(1 - \frac{1}{2}\right)H_C \\ &= \frac{1}{2}D + \left(1 - \frac{1}{2}\right)H_D \end{aligned}$$

which means P lies on $AH_A, BH_B, CH_C,$ and DH_D at the same time, hence the result. □

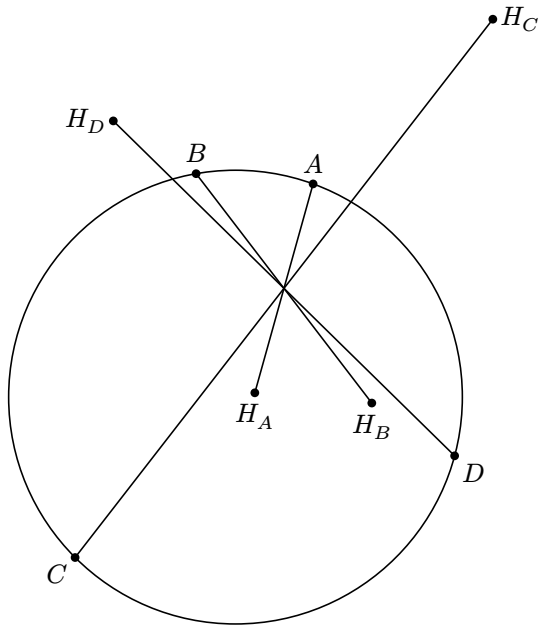

Figure 7A | Problem 6.31.

Problem 6.4 (Problem 6.32.). Let $ABCD$ be a quadrilateral circumscribed around a circle with center I. Prove that I lies on the line joining the midpoints of AC and BD.

Proof. Denote the circle (unit circle) tangent at W, X, Y, Z, which lie on AB, BC, CD, DA respectively. Therefore, we have

$$A = \frac{2zw}{z+w}$$
$$B = \frac{2wx}{w+x}$$
$$C = \frac{2xy}{x+y}$$
$$D = \frac{2yz}{y+z}$$
$$M = \frac{A+C}{2} = \frac{wxy+wxz+wyz+xyz}{(x+y)(z+w)}$$
$$N = \frac{B+D}{2} = \frac{wxy+wxz+wyz+xyz}{(w+x)(y+z)}$$

It suffices to show

$$\frac{M-I}{N-I} = \frac{M-0}{N-0} = \frac{M}{N} = \frac{(w+x)(y+z)}{(x+y)(z+w)} \in \mathbb{R}$$

Since

$$\begin{aligned}\overline{\left(\frac{M}{N}\right)} &= \frac{\left(\frac{1}{w}+\frac{1}{x}\right)\left(\frac{1}{y}+\frac{1}{z}\right)}{\left(\frac{1}{x}+\frac{1}{y}\right)\left(\frac{1}{z}+\frac{1}{w}\right)} \\ &= \frac{(x+w)(z+y)}{(y+x)(w+z)}\end{aligned}$$

Hence the result. □

Problem 6.5 (Problem 6.33 (Chinese TST 2011).). Let ABC be a triangle, and let A', B', C' be points on its circumcircle, diametrically opposite A, B, C, respectively. Let P be any point inside ABC and let D, E, F be the feet of the altitudes from P onto BC, CA, AB, respectively. Let X, Y, Z denote the reflections of A', B', C' over D, E, F, respectively. Show that triangles XYZ and ABC are similar to each other.

Proof. Assume (ABC) is unit circle. Let $A = a, B = b, C = c, P = p$, then we have

$$\begin{aligned}
A' = -a, B' = -b, C' = -c \\
D = \frac{1}{2}(p + b + c - bc\overline{p}) \\
E = \frac{1}{2}(p + c + a - ca\overline{p}) \\
F = \frac{1}{2}(p + a + b - ab\overline{p}) \\
X = p + a + b + c - bc\overline{p} \\
Y = p + a + b + c - ca\overline{p} \\
Z = p + a + b + c - ab\overline{p}
\end{aligned}$$

It suffices to show

$$\frac{C - A}{B - A} = \frac{Z - X}{Y - X}$$

Which is obvious. □

Problem 6.6 (Proposition 6.34 (Napoleon's Theorem).). Let ABC be a triangle and erect equilateral triangles on sides BC, CA, AB outside of ABC with centers O_A, O_B, O_C. Prove that $\triangle O_A O_B O_C$ is equilateral and that its center coincides with the centroid of triangle ABC.

CHAPTER 7
Inversion

Problem 7.1 (Problem 8.23.). Let ABC be a right triangle with $\angle C = 90°$ and let X and Y be points in the interiors of CA and CB, respectively. Construct four circles passing through C, centered at A, B, X, Y. Prove that the four points lying on at exactly two of these four circles are concyclic.

Proof. Perform an inversion around C with arbitrary radius.

After inversion, obviously $\odot A$ and $\odot X$ will be a line perpendicular to segment CB, same for $\odot B$ and $\odot Y$. With $\angle C = 90°$, we know that the quadrilateral from these four cline is a rectangle, which is concyclic, hence the result. □

Problem 7.2 (Problem 8.24.). Let $\omega_1, \omega_2, \omega_3, \omega_4$ be circles with consecutive pairs tangent at A, B, C, D, prove that quadrilateral $ABCD$ is cyclic.

Proof. Perform an inversion around A with arbitrary radius. Denote the center of ω_i^* as O_i, $i = 1, 2, 3, 4$.

Under inversion, ω_1 and ω_2 become two straight line. WLOG Assume they are vertical. ω_3 and ω_4 keep as circle which fall between them with ω_4 and ω_1 tangent at D, ω_3 and ω_4 tangent at C, ω_2 and ω_3 tangent at B. It suffice to show B, C, D collinear.

Notice that it's easy to show $\triangle DO_4C \sim \triangle BO_3C$ since O_4, C, O_3 collinear, the result follow. □

Problem 7.3 (Problem 8.25.). Let A, B, C be three collinear points and P be a point not on this line. Prove that the circumcenters of $\triangle PAB, \triangle PBC,$ and $\triangle PCA$ lie on a circle passing through P.

Proof (Pure Geometry). WLOG Assume A, B, C lies in order. Notice that let D be a arbitrary point lies on $\triangle PAB$, define $E := AD \cap (PAC)$, $F := BD \cap CE$, with the property of complete quadrilateral, F must lies on (PBC), the result follows by the Miquel's point property. □

Proof (Inversion). Perform an inversion around P with $r = 1$. Denote Z^* as the inversion of Z and the center of $(PAB), (PBC), (PAC)$ as O_1, O_2, O_3, respectively.

Under inversion, $(PAB), (PBC), (PAC)$ become line AB, BC, CA, respectively, with P, A, B, C concyclic. Well-known that AB perpendicular bisect PO_1^*, same holds for PO_2^* and PO_3^*, notice that $O_1^*O_2^*O_3^*$ is the Steiner line of point P with respect to $\triangle ABC$, hence the result. □

Problem 7.4 (Problem 8.26 (BAMO 2008/6).). A point D lies inside triangle ABC. Let A_1, B_1, C_1 be the second intersection points of the lines $AD, BD,$ and CD with the circumcircles of $BDC, CDA,$ and ADB, respectively. Prove that

$$\frac{AD}{AA_1} + \frac{BD}{BB_1} + \frac{CD}{CC_1} = 1.$$

Proof. Perform an inversion around D with $r = 1$. Denote Z^* as the inversion of Z.

We obtain a triangle $A^*B^*C^*$, with A_1^*, B_1^*, C_1^* lie on BC, CA, AB, respectively. Then we have

$$\frac{AD}{AA_1} = \frac{\frac{1}{A^*D}}{AD + DA_1} = \frac{1}{A^*D\left(\frac{1}{A^*D} + \frac{1}{DA_1^*}\right)} = \frac{DA_1^*}{DA_1^* + A^*D} = \frac{DA_1^*}{A^*A_1^*}$$

Same holds for $\frac{BD}{BB_1}, \frac{CD}{CC_1}$, notice that the result is obvious, since all ratio can be expressed as areas. □

Problem 7.5 (Problem 8.27 (Iran Olympiad 1996).). Consider a semicircle with center O and diameter AB. A line intersects line AB at M and the semicircle at C and D such that $MC > MD$ and $MB < MA$. Suppose (AOC) and (BOD) meet at a point K other than O. Prove that $\angle MKO = 90°$.

Proof. Perform an inversion around O with $(ABCD)$. Denote Z^* as the inversion of Z.

We obtain that $K^* = AC \cap BD$, $M^* = (OCD) \cap AB$. It suffices to show $K^*M^* \perp AB$. Notice that (OCD) is the nine-point circle of $\triangle K^*AB$, hence the result. □

Problem 7.6 (Problem 8.28 (Shortlist 2003/G4).). Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that Γ_1, Γ_3 are externally tangent at P, and Γ_2, Γ_4 are externally tangent at the same point P. Suppose that Γ_1 and Γ_2, Γ_2 and Γ_3, Γ_3 and Γ_4, Γ_4 and Γ_1 meet at A, B, C, D, respectively, and that all these points are different from P. Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}$$

.

Proof. Perform an inversion around P with $r = 1$. Denote Z^* as the inversion of Z.

From inversion, we obtain that Γ_1 and Γ_3 become vertical line, Γ_2 and Γ_4 become horizontal line, $ABCD$ form a rectangle.Back on to the question, in fact we have

$$\text{LHS} = \frac{(A^*B^* \cdot PA \cdot PB)(B^*C^* \cdot PB \cdot PC)}{(A^*D^* \cdot PA \cdot PD)(D^*C^* \cdot PD \cdot PC)} = \text{RHS}$$

, and we are done. □

Problem 7.7 (Problem 8.29.). Let ABC be a triangle with incenter I and circumcenter O. Prove that line IO passes through the centroid G_1 of the contact triangle.

Proof. Perform an inversion around I with respect to the incircle of $\triangle ABC$. Denote Z^* as the inversion of Z.

Notice that under inversion, $(ABC) \mapsto$ the nine-point circle (N_9) of contact triangle, which means O, I, N_9 collinear, also since I, G, N_9 colinear, we get the result. □

Problem 7.8 (Problem 8.30 (NIMO 2014).). Let ABC be a triangle and let Q be a point such that $AB \perp QB$ and $AC \perp QC$. A circle with center I is inscribed in $\triangle ABC$, and is tangent to $BC, CA,$ and AB at points $D, E,$ and F, respectively. If ray QI intersects EF at P, prove that $DP \perp EF$.

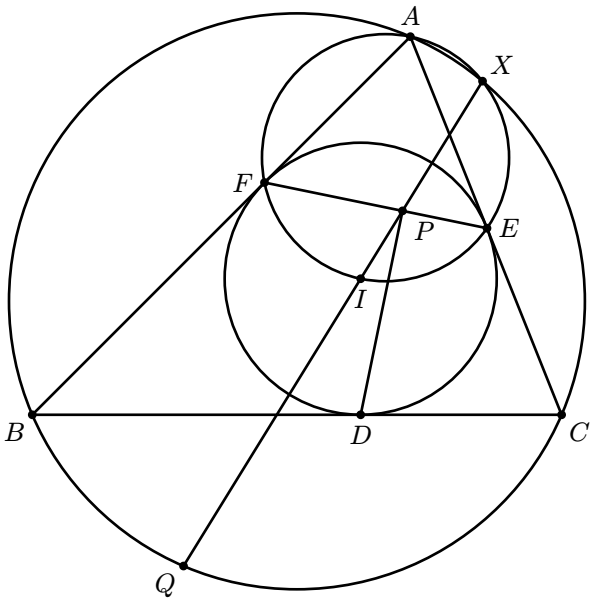

Figure 8A | Problem 8.30 (NIMO 2014).

Proof. Perform an inversion around I with respect to the incircle of $\triangle ABC$. Denote Z^* as the inversion of Z.

From observe, A, B, C, Q concyclic with diameter AQ. Expand QI intersect the circumcircle at X, since $\angle AXI = \angle AEI = \angle AFI = 90°$, A, F, I, E, X concyclic with diameter AI. Under inversion, $P = X^*$, since $(ABC) \mapsto$ the nine-point circle (N_9) of contact triangle, the result is obvious. □

Problem 7.9 (Problem 8.31 (EGMO 2013/5).). Let Ω be the circumcircle of the triangle ABC. The circle ω is tangent to the sides AC and BC, and it is internally tangent to the circle Ω at the point P. A line parallel to AB intersecting the interior of triangle ABC is tangent to ω at Q. Prove that $\angle ACP = \angle QCB$.

Proof. Perform an Force-Overlaid Inversion at A. Denote Z^* as the inversion of Z.

Define $Q' :=$ the A-excircle tangency point at segment AB, obviously A, Q, Q' collinear. Under Force-Overlaid Inversion, well-known P, Q switch their places, the result follows. □

Problem 7.10 (Problem 8.32 (Russian Olympiad 2009).). In triangle ABC with circumcircle Ω , the internal angle bisector of $\angle A$ intersects BC at D and Ω again at

E. The circle with diameter DE meets Ω again at F. Prove that AF is a symmedian of triangle ABC.

Proof. Define $M :=$ midpoint of BC, it suffices to show under Force-Overlaid Inversion, M, F are corresponding point.

Notice that AF, AM already isogonal, also $AF \cdot AM = AD \cdot AE$, hence the result. □

Problem 7.11 (Problem 8.33 (Shortlist 1997).). Let $A_1A_2A_3$ be a non-isosceles triangle with incenter I. Let $C_i, i = 1, 2, 3$, be the smaller circle through I tangent to A_iA_{i+1} and A_iA_{i+2} (indices taken mod 3). Let $B_i, i = 1, 2, 3$, be the second point of intersection of C_{i+1} and C_{i+2}. Prove that the circumcenters of the triangles $A_1B_1I, A_2B_2I, A_3B_3I$ are collinear.

Proof. It suffices to show $(A_1B_1I), (A_2B_2I), (A_3B_3I)$ got two common point. Perform an inversion around I with respect to the incircle of $\triangle A_1A_2A_3$. Denote Z^* as the inversion of Z. Let incircle of $\triangle A_1A_2A_3$ tangent to A_2A_3, A_3A_1, A_1A_2 at D, E, F respectively.

We got the following. A triangle $B_1^*B_2^*B_3^*$. Let $O_i, i = 1, 2, 3$, be the smaller circle through I tangent to $B_i^*B_{i+1}^*$ and $B_i^*B_{i+2}^*$, also they both internally tangent to (I) at D, E, F, respectively, with $I \neq A_i^* = O_{i+1} \cap O_{i+2}, i = 1, 2, 3$.

Well-known A_1^* is the midpoint of EF, same for A_2^*, A_3^*. From what we get, O_1, O_2, O_3 are congruent $\Longrightarrow EF \parallel B_2^*B_3^*$, same applies on DE, FD. Summarize them we get $\triangle B_1^*B_2^*B_3^*$, $\triangle A_1^*A_2^*A_3^*$ are homothetic, hence the result. □

Problem 7.12 (Problem 8.34 (IMO 1993/2).). Let A, B, C, D be four points in the plane, with C and D on the same side of the line AB, such that $AC \cdot BD = AD \cdot BC$ and $\angle ADB = 90^\circ + \angle ACB$. Find the ratio $\frac{AB \cdot CD}{AC \cdot BD}$, and prove that the circumcircles of the triangles ACD and BCD are orthogonal.

Proof. Perform an inversion around A with $r = 1$. Denote Z^* as the inversion of Z, O_1, O_2 be the circumcenter of $\triangle ADC, \triangle BDC$, respectively.

transforming the condition…

$$\begin{aligned} AC \cdot BD &= AD \cdot BC \\ \Longleftrightarrow \frac{1}{AC^*}\frac{B^*D^*}{AB^* \cdot AD^*} &= \frac{1}{AD^*}\frac{B^*C^*}{AB^* \cdot AC^*} \\ \Longleftrightarrow B^*D^* &= B^*C^* \end{aligned}$$

$$\begin{aligned} \angle ADB &= 90^\circ + \angle ACB \\ \Longleftrightarrow \angle AB^*D^* &= 90^\circ + \angle AB^*C^* \\ \Longleftrightarrow \angle D^*B^*C^* &= 90^\circ \end{aligned}$$

which means $\triangle D^*B^*C^*$ is isosceles right-triangle, the result is now obvious

$$\frac{AB \cdot CD}{AC \cdot BD} = \frac{AC^* \frac{C^*D^*}{AC^* \cdot AD^*}}{AB^* \frac{B^*D^*}{AB^* \cdot AD^*}} = \frac{C^*D^*}{B^*D^*} = \sqrt{2}$$

Notice that $\angle DAC + \angle DBC = \angle ADB - \angle ACB = 90^\circ$, which implies that $\angle DCO_1 + \angle DCO_2 = 180^\circ$, hence the result. □

Problem 7.13 (Problem 8.35 (IMO 1996/2).). Let P be a point inside a triangle ABC such that $\angle APB - \angle ACB = \angle APC - \angle ABC$. Let D, E be the incenters of triangles APB, APC, respectively. Show that the lines AP, BD, CE concur.

Proof. Perform an inversion around A with $r = 1$. Denote Z^* as the inversion of Z.

transforming the condition…

$$\begin{aligned} \angle APB - \angle ACB &= \angle APC - \angle ABC \\ \iff \angle AB^*P^* - \angle AB^*C^* &= \angle AC^*P^* - \angle AC^*B^* \\ \iff \angle P^*B^*C^* &= \angle P^*C^*B^* \\ \iff P^*C^* &= P^*B^* \end{aligned}$$

it suffice to show

$$\begin{aligned} \frac{AB}{BP} &= \frac{AC}{CP} \\ \left(AB^* \frac{B^*P^*}{AB^* \cdot AP^*}\right) &= \left(AC^* \frac{C^*P^*}{AC^* \cdot AP^*}\right) \\ B^*P^* &= C^*P^* \end{aligned}$$

which is obvious. □

Problem 7.14 (Problem 8.36 (IMO 2015/3).). Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocenter, and F the foot of the altitude from A. Let M be the midpoint of BC. Let Q be the point on Γ such that $\angle HQA = 90°$ and let K be the point on Γ such that $\angle HKQ = 90°$. Assume that the points $A, B, C, K,$ and Q are all different and lie on Γ in this order. Prove that the circumcircles of triangles KQH and FKM are tangent to each other.

Proof. Perform a negative inversion around H with $r = \sqrt{AH \cdot HF}$. Denote Z^* as the inversion of Z.

Notice that this inversion map $(ABC) \mapsto (N_9)$, well-known M, H, Q collinear, also, $(KQH) \mapsto \ell$, $(FKM) \mapsto (AK^*Q)$ s.t. $M \in \ell \perp MHQ$, $K^* = (N_9) \cap \ell$, it suffices to show ℓ tangent to (AK^*Q) at K^*. Denote O_1, O_2 as the circumcenter of $(AHQ), (AK^*Q)$, respectively. Obviously, $AQ \perp O_1O_2$. Since $AQ \parallel \ell$, $O_1O_2 \perp \ell$. Notice that $O_1K^* \perp \ell$ as O_1M is the diameter of (N_9), we have $O_2K^* \perp \ell$, hence the result. □

Problem 7.15 (Problem 8.37 (ELMO Shortlist 2013).). Let ω_1 and ω_2 be two orthogonal circles, and let the center of ω_1 be O. Diameter AB of ω_1 is selected so that B lies strictly inside ω_2. The two circles tangent to ω_2 through both O and A touch ω_2 at F and G. Prove that quadrilateral $FOGB$ is cyclic.

Proof. Perform the first inversion around O with respect to the ω_1. Denote Z^* as the inversion of Z, P be the circumcenter of ω_2.

Notice that AF^*, AG^* is tangent to ω_2^* at F^*, G^*, respectively. It suffices to show F^*BG^* collinear.

Perform the second inversion around P with respect to the ω_2. Denote Z^* as the inversion of Z.

It suffices to show $F^*PG^*B^*$ concyclic. Notice that they all lie on the circle with diameter AP, as result. $\square$

CHAPTER 8
Projective Geometry

Problem 8.1 (Lemma 9.40 (Incircle Polars).). Let ABC be a triangle with contact triangle DEF and incenter I. Lines EF and BC meet at K. Prove that $IK \perp AD$.

Proof. Obvious. □

Problem 8.2 (Theorem 9.41 (Desargues' Theorem).). Let ABC and XYZ be triangles in the projective plane. We say that the two triangles are perspective from a point if lines $AX, BY,$ and CZ concur (possibly at infinity), and we say they are perspective from a line if the points $AB \cap XY, BC \cap YZ, CA \cap ZX$ are collinear. Prove that these two conditions are equivalent.

Proof. Notice that, the problem is purely projective! It reduces the problem into a simple homothetic. □

Problem 8.3 (Problem 9.42 (USA TSTST 2012/4).). In scalene triangle ABC, let the feet of the perpendiculars from A to BC, B to CA, C to AB be A_1, B_1, C_1, respectively. Denote by A_2 the intersection of lines BC and B_1C_1. Define B_2 and C_2 analogously. Let D, E, F be the respective midpoints of sides BC, CA, AB. Show that the perpendiculars from D to AA_2, E to BB_2, and F to CC_2 are concurrent.

Proof. We claim that the concurrent point is the orthocenter of $\triangle ABC$. Denote as H.

Let $M := DH \cap (ABC)$, it suffices to show $A_2 = AM \cap B_1C_1 \cap BC$, in fact, A_2 is the radical center of $(BCC_1B_1), (ABC), (AB_1C_1)$. □

Problem 8.4 (Problem 9.43 (Singapore TST).). Let ω and O be the circumcircle and circumcenter of right triangle ABC with $\angle B = 90°$. Let P be any point on the tangent to ω at A other than A, and suppose ray PB intersects ω again at D. Point E lies on line CD such that $AE \parallel BC$. Prove that $P, O,$ and E are collinear.

Proof. Let $F := AE \cap ABC$, applying Pascal's theorem solve this problem. □

Problem 8.5 (Problem 9.44 (Canada 1994/5).). Let ABC be an acute triangle. Let AD be the altitude on BC, and let H be any interior point on AD. Lines BH and CH, when extended, intersect AC, AB at E and F respectively. Prove that $\angle EDH = \angle FDH$.

Proof. Notice that, it suffices to show $(F, FC \cap DE; H, C) = -1$, which is obvious. □

Problem 8.6 (Problem 9.45 (Bulgarian Olympiad 2001).). Let ABC be a triangle and let k be a circle through C tangent to AB at B. Side AC and the C-median of $\triangle ABC$ intersect k again at D and E, respectively. Prove that if the intersecting point of the tangents to k through C and E lies on the line BD then $\angle ABC = 90°$.

Proof. Notice that the condition means $DEBC$ is harmonic quadrilateral. Denote $M :=$ midpoint of $AB, X :=$ the intersecting point of the tangents to k through C and E. Since both CX and AB is tangent to k,

$$-1 = (B, D; E, C) \overset{C}{=} (B, A; M, X)$$

which means $CX \parallel AB$, implies that CB is the diameter of k, hence the result. □

Problem 8.7 (Problem 9.46 (ELMO Shortlist 2012).). Let ABC be a triangle with incenter I. The foot of the perpendicular from I to BC is D, and the foot of the perpendicular from I to AD is P. Prove that $\angle BPD = \angle DPC$.

Proof. Define circle (I) tangent to AC, AB at E, F, respectively. It suffices to show EF, IP, BC collinear. In fact, A, E, F, I, P lie on the circle with diameter AI, also B, I, P, C lie on the circle with diameter II_A, applying the radical axis theorem solve the problem. □

Problem 8.8 (Problem 9.47 (IMO 2014/4).). Let P and Q be on segment BC of an acute triangle ABC such that $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Let M and N be the points on AP and AQ, respectively, such that P is the midpoint of AM and Q is the midpoint of AN. Prove that the intersection of BM and CN is on the circumference of triangle ABC.

Proof. Notice that $BB \parallel AP, CC \parallel AQ$, since $\angle QAC = \angle ABC = \angle(AC, CC)$. Therefore, Denote $X := AM \cap (ABC), X' := AN \cap (ABC)$, we have $ABXC, ABX'C$ both harmonic quadrilateral, then $X = X'$, hence the result. □

Problem 8.9 (Problem 9.48 (Shortlist 2004/G8).). Given a cyclic quadrilateral $ABCD$, let M be the midpoint of the side CD, and let N be a point on the circumcircle of triangle ABM. Assume that the point N is different from the point M and satisfies $\frac{BN}{AN} = \frac{BM}{AM}$. Prove that the points E, F, N are collinear, where $E = AC \cap BD$ and $F = BC \cap DA$.

Proof. Let $T := FE \cap CD,\ S := AB \cap CD$, we have $(D, C; T, S) = -1 \Longrightarrow SD \cdot SC = ST \cdot SM = SA \cdot SB$, which implies T lies on (ABM).

Notice that

$$-1 = (A, B; FT \cap AB) \overset{T}{=} (A, B, N, M)$$

hence the result. □

Problem 8.10 (Problem 9.49 (Sharygin 2013).). The incircle of triangle ABC touches BC, CA, and AB at points A', B', and C' respectively. The perpendicular from the incenter I to the C-median meets the line $A'B'$ in point K. Prove that $CK \parallel AB$.

Proof. Define $T := IK \cap CM,\ S := A'B' \cap CM$, where M is the midpoint of AB. Notice that it suffices to show $(A', B'; S, K) \overset{C}{=} (B, A; M, K) = -1$, since C, A', I, T, B' all lie on the circle with diameter CI, we have $ST \perp TK$, we also have CST bisects $\angle A'TB'$, which implies $(A', B'; S, K) = -1$, as result. □

Problem 8.11 (Problem 9.50 (Shortlist 2004/G2).). Let Γ be a circle and let d be a line such that Γ and d have no common points. Further, let AB be a diameter of the circle Γ; assume that this diameter AB is perpendicular to the line d, and the point B is nearer to the line d than the point A. Let C be an arbitrary point on the circle Γ , different from the points A and B. Let D be the point of intersection of the lines AC and d. One of the two tangents from the point D to the circle Γ touches this circle Γ at a point E; hereby, we assume that the points B and E lie in the same half-plane with respect to the line AC. Denote by F the point of intersection of the lines BE and d. Let the line AF intersect the circle Γ at a point G, different from A. Prove that the reflection of the point G in the line AB lies on the line CF.

Proof. Notice that applying Pascal's theorem on $CAGEEB$, we get $BC \cap GE$ is collinear with F, D, apply Pascal's theorem again on $CG'GEBB$, we have the result. □

Problem 8.12 (Problem 9.51 (USA January TST for IMO 2013).). Let ABC be an acute triangle. Circle ω_1, with diameter AC, intersects side BC at F (other than C). Circle ω_2, with diameter BC, intersects side AC at E (other than C). Ray AF intersects ω_2 at K and M with $AK < AM$. Ray BE intersects ω_1 at L and N with $BL < BN$. Prove that lines AB, ML, NK are concurrent.

Proof.

Claim 1 - N, K, L, M are concyclic.

Proof of Claim. Obviously, $H = KM \cap LN$, which is orthocenter. Define D lies on AB s.t. $CD \perp AB$. Notice that

$$HK \cdot HM = CH \cdot HD = HL \cdot HN$$

as result. ■

Notice that $AL^2 = AN^2 = AE \cdot AC = AK \cdot AM$, which means AL is tangent to $(NKLM)$, same also applies on BK. Applying Pascal's theorem on $MKNLLM$ solve the problem. □

Problem 8.13 (Problem 9.52 (Brazilian Olympiad 2011/5).). Let ABC be an acute triangle with orthocenter H and altitudes BD, CE. The circumcircle of ADE cuts the circumcircle of ABC at $F \neq A$. Prove that the angle bisectors of $\angle BFC$ and $\angle BHC$ concur at a point on BC.

Proof (Pure Geometry). Define M_1, M_2 lie on BC s.t. FM_1, HM_2 bisect $\angle BFC, \angle BHC$ respectively.

Computing...

$$\frac{BM_1}{CM_1} = \frac{BF}{CF} = \frac{BE}{CD}$$

$$\frac{BM_2}{CM_2} = \frac{BH}{CH} = \frac{BE}{CD}$$

as result. □

Proof (Projective Geometry). Denote H' as the reflection of the orthocenter H. It suffices to show $FBH'C$ is the harmonic quadrilateral. □

Problem 8.14 (Problem 9.53 (ELMO Shortlist 2013).). In $\triangle ABC$, a point D lies on line BC. The circumcircle of ABD meets AC at F (other than A), and the circumcircle of ADC meets AB at E (other than A). Prove that as D varies, the circumcircle of AEF always passes through a fixed point other than A, and that this point lies on the median from A to BC.

Proof. Notice that perform an Force-Overlaid Inversion at A. Denote Z^* as the inversion of Z, we have $C^* = B, B^* = C$

D^* become an interior point lies on (ABC), we have $E^* = AB \cap CD^*, F^* = AC \cap BD^*$. It suffices to show E^*F^* pass through a fixed point while lies on A-symmedian.

Notice that applying Pascal's theorem on CCD^*BBA, we have $X := CC \cap BB$ collinear w ith E^*, F^*, since AX is A-symmedian, as result. □

Problem 8.15 (Problem 9.54 (APMO 2008/3).). Let Γ be the circumcircle of a triangle ABC. A circle passing through points A and C meets the sides BC and BA at D and E, respectively. The lines AD and CE meet Γ again at G and H, respectively. The tangent lines to Γ at A and C meet the line DE at L and M, respectively. Prove that the lines LH and MG meet at Γ.

Proof. We define $T := DE \cap AC,\ X := BT \cap (ABC)$, we'll show that X is the point which satisfied this problem.

Notice that applying Pascal' theorem on

- $XGACCB \Longrightarrow DMT$ collinear.
- $XHCAAB \Longrightarrow ELT$ collinear.

Since DET already collinear, we have D, E, L, M, T collinear, as result. □

Problem 8.16 (Theorem 9.55 (Brianchon's Theorem).). Let $ABCDEF$ be a hexagon circumscribed about a circle ω. Prove that AD, BE, CF are concurrent.

Proof. We will use the convention in which the points will be denoted by capital latin letters, and their respective polars with the corresponding lowercase letters.

Denote by $M_i, i = 1, 2, ..., 6$, the points of tangency of A_iA_{i+1} with k. Obviously, $a_i = M_{i-1}M_i$.

Let $b_j = A_jA_{j+3}, j = 1, 2, 3$. Then $B_j = a_j \cap a_{j+3} = M_{j-1}M_j \cap M_{j+3}M_{j+4}$. We have to prove that there exists a point P such that $P \in b_1, b_2, b_3$, or analogously, that there is a line p such that $B_1, B_2, B_3 \in p$. In other words we have to prove that the points $B_1B_2B_3$ are collinear. However this immediately follows from the Pascal's theorem applied to $M_1M_3M_5M_4M_6M_2$. □

Problem 8.17 (Problem 9.56 (ELMO Shortlist 2014).). Suppose $ABCD$ is a cyclic quadrilateral inscribed in the circle ω. Let $E = AB \cap CD$ and $F = AD \cap BC$. Let ω_1, ω_2 be

the circumcircles of triangles AEF, CEF, respectively. Let G and H be the intersections of ω and ω_1, ω and ω_2, respectively, with $G \neq A$ and $H \neq C$. Show that $AC, BD,$ and GH are concurrent.

Proof. Denote K be the radical center of these three circle, O be the circumcenter of circle (ABC). Define $P := AC \cap BD,\ P' := AC \cap GH,\ X := AA \cap CC,\ S := AH \cap GC$

At first, applying Pascal's Theorem on

- $AABCCD \Longrightarrow EFX$ collinear
- $AAGCCH \Longrightarrow XSK$ collinear

Then we have E, F, K, X, S collinear

Applying Brocard's theorem on $ABCD$ and $AGCH$, we have $OP \perp EF, OP' \perp KS$. Since P, P' both $\in AC$. Implies that, $P = P'$, hence the result. □

Problem 8.18 (Problem 9.57 (ELMO Shortlist 2014).). Let $ABCD$ be a cyclic quadrilateral inscribed in circle ω. The tangent to ω at A intersects lines CD and BC at E and F. Lines BE and DF meet ω again G and I, and $H = BE \cap AD, J = DF \cap AB$. Prove that GI, HJ, and the B-symmedian of $\triangle ABC$ are concurrent.

Proof. It suffices to show AA, CC, GI, JH concurrent. Define $K := AA \cap GI,\ K' := CC \cap GI$

Applying Pascal's theorem on

- $ABGIDA \Longrightarrow JHK$ collinear
- $BGIDCC \Longrightarrow EFK'$ collinear

Therefore, we have AA, GI, JH concurrent, then CC, GI, AA concurrent, hence the result. □

Problem 8.19 (Problem 9.58 (Shortlist 2005/G6).). Let ABC be a triangle, and M the midpoint of its side BC. Let γ be the incircle of triangle ABC. The median AM of triangle ABC intersects the incircle γ at two points K and L. Let the lines passing through K and L, parallel to BC, intersect the incircle γ again in two points X and Y. Let the lines AX and AY intersect BC again at the points P and Q. Prove that $BP = CQ$.

Proof. Letting E and F denote the tangency points of the incircle. We have EF, XY, KL concurrent at Z.

Notice that

$$-1 = (A, Z; K, L) \overset{P_\infty}{=} (P_\infty, S; Y, X) \overset{A}{=} (P_\infty, M; Q, P)$$

hence the result. □

CHAPTER 9
Complete Quadrilateral

Problem 9.1 (Problem 10.17 (NIMO 2014).). Let ABC be an acute triangle with orthocenter H and let M be the midpoint of BC. Denote by ω_B the circle passing through $B, H,$ and M, and denote by ω_C the circle passing through $C, H,$ and M. Lines AB and AC meet ω_B and ω_C again at P and Q, respectively. Rays PH and QH meet ω_C and ω_B again at R and S, respectively. Show that $\triangle BRS$ and $\triangle CRS$ have the same area.

Proof. Notice that it suffices to show R, M, S collinear. Since $\angle HBC = \angle HSR$ by spiral similarity, obviously, $\angle HBM = \angle HSM$, done. □

Problem 9.2 (Problem 10.18 (USAMO 2013/1).). In triangle ABC, points P, Q, R lie on sides BC, CA, AB, respectively. Let $\omega_A, \omega_B, \omega_C$ denote the circumcircles of triangles AQR, BRP, CPQ, respectively. Given the fact that segment AP intersects $\omega_A, \omega_B, \omega_C$ again at X, Y, Z respectively, prove that $\frac{YX}{XZ} = \frac{BP}{PC}$.

Proof. Denote M be the Miquel Point of $\triangle ABC$. Notice that $\triangle MYZ \sim \triangle MBC$. It suffices to show $\angle ZMX = \angle CQP$. In fact,

$$\angle ZMX = \angle MXQ + \angle MQX + \angle MZQ + \angle MQZ = \angle QAX + \angle QPX = \angle CQP$$

□

Problem 9.3 (Problem 10.19 (Shortlist 1995/G8).). Suppose that $ABCD$ is a cyclic quadrilateral. Let $E = AC \cap BD$ and $F = AB \cap CD$. Prove that F lies on the line joining the orthocenters of triangles EAD and EBC.

Proof. Denote H_1, H_2 be the orthocenter of $\triangle EAD, \triangle EBC$, respectively. Notice that both H_1, H_2 lie on the radical axis of two circle with diameter AB, CD, obviously F lies on the radical axis also. □

Problem 9.4 (Problem 10.20 (USA TST 2007/1).). Circles ω_1 and ω_2 meet at P and Q. Segments AC and BD are chords of ω_1 and ω_2 respectively, such that segment AB and ray CD meet at P. Ray BD and segment AC meet at X. Point Y lies on ω_1 such that $PY \parallel BD$. Point Z lies on ω_2 such that $PZ \parallel AC$. Prove that points Q, X, Y, Z are collinear.

Proof. Denote $\ell \parallel BD$, s.t. $P \in \ell$. Redefine $Y := QX \cap \ell$. It suffices to show Y lies on $(APQC)$. In fact Q is the Miquel Point of the complete quadrilateral $AXDPCB$, we also have $CXDQ, ABQX$ concyclic. Therefore,

$$\measuredangle QYP = \measuredangle QXD = \measuredangle QCD = \measuredangle QCP$$

$$\measuredangle QZP = \measuredangle QZA = \measuredangle QBA = \measuredangle QBP$$

hence the result. □

Problem 9.5 (Problem 10.21 (USAMO 2013/6).). Let ABC be a triangle. Find all points P on segment BC satisfying the following property: If X and Y are the intersections of line PA with the common external tangent lines of the circumcircles of triangles PAB and PAC, then

$$\left(\frac{PA}{XY}\right)^2 + \frac{PB \cdot PC}{AB \cdot AC} = 1$$

Proof. Denote O_B, O_C as the circumcenter of circle $(APB), (APC)$. Assume X is closer to A than P. By the spiral similarity we have $\triangle AO_BO_C \sim \triangle ABC$, we will perform the calculation convenient by letting $AO_C = b', AO_B = c', O_BO_C = a', AM = h$, B', C' be the tangency points of X, also define $M := O_BO_C \cap AP$. Notice that $XB'^2 = XC'^2 = XM^2 - h^2$

Try to calculate $\left(\frac{PA}{XY}\right)^2$

$$\begin{aligned}
\left(\frac{PA}{XY}\right)^2 &= \left(\frac{AM}{XM}\right)^2 \\
&= \frac{h^2}{h^2 + XB'^2} \\
&= 1 - \frac{XB'^2}{h^2 + XB'^2} \\
&= 1 - \frac{B'C'^2}{(2h)^2 + B'C'^2} \\
&= 1 - \frac{a'^2 - (b'-c')^2}{\left(\frac{4}{a}[AO_BO_C]\right)^2 + a'^2 - (b'-c')^2} \\
&= 1 - \frac{a'^2\left(a'^2 - (b'-c')^2\right)}{(a'+b'+c')(a'+b'-c')(a'-b'+c')(-a'+b'+c') + a'^2\left(a'^2 - (b'-c')^2\right)} \\
&= 1 - \frac{a'^2}{(a'+b'+c')(-a'+b'+c') + a'^2} \\
&= 1 - \left(\frac{a'}{b'+c'}\right)^2 \\
&= 1 - \left(\frac{a}{b+c}\right)^2
\end{aligned}$$

WLOG Assume $PB = xa, PC = (1-x)a$

$$\begin{aligned}
1 - \left(\frac{a}{b+c}\right)^2 + \frac{x(1-x)a^2}{bc} &= 1 \\
\frac{x(1-x)a^2}{bc} &= \left(\frac{a}{b+c}\right)^2 \\
x(1-x) = \frac{bc}{(b+c)^2} &= \left(\frac{b}{b+c}\right)\left(\frac{c}{b+c}\right)
\end{aligned}$$

hence the result. □

Problem 9.6 (Problem 10.22 (USA TST 2007/5).). Acute triangle ABC is inscribed in circle ω. The tangent lines to ω at B and C meet at T. Point S lies on ray BC such that $AS \perp AT$. Points B_1 and C_1 lie on ray ST (with C_1 in between B_1 and S) such that $B_1T = BT = C_1T$. Prove that triangles ABC and AB_1C_1 are similar.

Proof.

Claim 1 - A is the Miquel point of BB_1C_1C.

Proof of Claim. Define $K := BB_1 \cap CC_1$. Notice that

$$\angle BKC = \angle B_1KC_1 = \pi - \angle CC_1T - \angle BB_1T$$
$$= \pi - \frac{\pi - \angle CTC_1}{2} - \frac{\pi - \angle BTB_1}{2} = \pi - \frac{\pi + \angle BTC}{2} = \pi - (\pi - \angle ABC) = \angle ABC$$

which mean $KBCA$ are concyclic. Since BB_1C_1C are also concyclic with circumcenter T, combine with $AS \perp AT$, determine that A is the Miquel point of BB_1C_1C. ■

The result is now obvious. □

Problem 9.7 (Problem 10.23 (IMO 2005/2).). Let $ABCD$ be a fixed convex quadrilateral with $BC = DA$ and $BC \nparallel DA$. Let two variable points E and F lie of the sides BC and DA, respectively, and satisfy $BE = DF$. The lines AC and BD meet at P, the lines BD and EF meet at Q, the lines EF and AC meet at R. Prove that the circumcircles of the triangles PQR, as E and F vary, have a common point other than P.

Proof. I claim that the desired point is M, which is the Miquel point of $ADBC$.

Notice that M is also the Miquel point of $AFEC$, which implies $FARM$ are concyclic. It reduces that M is also the Miquel point of $FAPQ$, solving this problem. □

Problem 9.8 (Problem 10.24 (USAMO 2006/6).). Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC, respectively, such that $\frac{AE}{ED} = \frac{BF}{FC}$. Ray FE meets rays BA and CD at S and T, respectively. Prove that the circumcircles of triangles SAE, SBF, TCF, and TDE pass through a common point.

Proof. Define $M := (SAE) \cap (SBF), M' = (TCF) \cap (TDE), X := AD \cap BC$. Notice that $M = (XAB) \cap (XEF), M' = (XEF) \cap (XDC)$, with condition $\frac{AE}{ED} = \frac{BF}{FC}$, implies that $M = M'$. □

Problem 9.9 (Problem 10.25 (Balkan Olympiad 2009/2).). Let MN be a line parallel to the side BC of a triangle ABC, with M on the side AB and N on the side AC. The lines BN and CM meet at point P. The circumcircles of triangles BMP and CNP intersect at a point $Q \neq P$. Prove that $\angle BAQ = \angle CAP$.

Proof. Notice that Q is the Miquel point of $AMPN$. It suffice to show

$$\begin{aligned}
\frac{\sin\angle BAQ}{\sin\angle CAQ} &= \frac{\sin\angle CAP}{\sin\angle BAP} \\
\frac{\sin\angle PNQ}{\sin\angle PMQ} &= \sqrt{\frac{CP}{MP}\frac{AM}{AC}\cdot\frac{PN}{PB}\frac{AB}{AN}} \\
\frac{PM\sin\angle NCP}{PN\sin\angle MBP} &= \frac{AB}{AC} \\
\frac{PM\cdot AM\frac{\sin\angle A}{MC}}{PN\cdot AN\frac{\sin\angle A}{NB}} &= \frac{AB}{AC} \\
\frac{AM}{AN} &= \frac{AB}{AC}
\end{aligned}$$

which is obvious. □

Problem 9.10 (Problem 10.26 (USA TSTST 2012/7).). Triangle ABC is inscribed in circle Ω . The interior angle bisector of angle A intersects side BC and Ω at D and L (other than A), respectively. Let M be the midpoint of BC. The circumcircle of triangle ADM intersects sides AB and AC again at Q and P (other than A), respectively. Let N be the midpoint of PQ, and let H be the foot of the perpendicular from L to line ND. Prove that line ML is tangent to the circumcircle of triangle HMN.

Proof. Define $S := (ADM) \cap (ABC)$. Obviously S, M, L are collinear. Notice that SD is the diameter of (ADM), therefore, S, D, N, H collinear, which also implies H lies on (ABC). Now, $NM \parallel DL$ is trivial, (according to the spiral similarity of the Miquel point S), hence the result. □

Problem 9.11 (Problem 10.27 (USA TSTST 2012/2).). Let $ABCD$ be a quadrilateral with $AC = BD$. Diagonals AC and BD meet at P. Let ω_1 and O_1 denote the circumcircle and the circumcenter of triangle ABP. Let ω_2 and O_2 denote the circumcircle and circumcenter of triangle CDP. Segment BC meets ω_1 and ω_2 again at S and T (other than B and C), respectively. Let M and N be the midpoints of minor arcs SP (not including B) and TP (not including C). Prove that $MN \parallel O_1O_2$.

Proof. Denote Q be the second intersection of ω_1, ω_2 Notice that $\triangle QAC \cong \triangle QBD$, which implies that both $\triangle QAB, \triangle QCD$ are isosceles triangle. Therefore

$$\angle BPQ = \angle DCQ = \angle CDQ = \angle CPQ$$

also $\angle PBM = \angle CBM, \angle PCN = \angle BCN$, which show that BM, CN, PQ are concurrent at I, the incenter of $\triangle PBC$, which implies that $BMNC$ concyclic. Well known $PQ \perp MN$, the result follows. □

Problem 9.12 (Problem 10.28 (USA TST 2009/2).). Let ABC be an acute triangle. Point D lies on side BC. Let O_B, O_C be the circumcenters of triangles ABD and ACD, respectively. Suppose that the points B, C, O_B, O_C lie on a circle centered at X. Let H be the orthocenter of triangle ABC. Prove that $\angle DAX = \angle DAH$.

Proof. Notice that A is the Miquel point of quadrilateral BO_BO_CC. Define $E := BO_B \cap CO_C, F := O_BO_C \cap BC$. Obviously, $AX \perp EF, AD \perp O_BO_C, AH \perp BC$. Since

$$\begin{aligned}
\angle DAX &= \angle AFO_B \\
\angle DAH &= \angle O_BFB
\end{aligned}$$

$\angle AFO_B = \angle O_BFB$ since $AO_B = BO_B$. □

Problem 9.13 (Problem 10.29 (Shortlist 2009/G4).). Given a cyclic quadrilateral $ABCD$, let the diagonals AC and BD meet at E and the lines AD and BC meet at F. The midpoints of AB and CD are G and H, respectively. Show that EF is tangent at E to the circle through the points $E, G,$ and H.

Proof. Define M as the midpoint of FE, $X := FE \cap AB, Y := FE \cap CD, I := AB \cap CD$, well known that M, G, H collinear, it suffices to show $ME^2 = MG \cdot MH$, since

$$(F, E; X, Y) = (B, A; X, G) = (C, D; Y, G) = -1$$

Therefore,

$$\begin{aligned} ME^2 = MG \cdot MH &\Leftrightarrow MX \cdot MY = MG \cdot MH \\ &\Leftrightarrow X, Y, H, G \text{ concyclic} \\ &\Leftrightarrow IX \cdot IG = IY \cdot IH \\ &\Leftrightarrow IA \cdot IB = IC \cdot ID \end{aligned}$$

which is trivial. □

Problem 9.14 (Problem 10.30 (Shortlist 2006/G9).). Points A_1, B_1, C_1 are chosen on the sides BC, CA, AB of a triangle ABC respectively. The circumcircles of triangles $AB_1C_1, BC_1A_1, CA_1B_1$ intersect the circumcircle of triangle ABC again at points A_2, B_2, C_2 respectively ($A_2 \neq A, B_2 \neq B, C_2 \neq C$). Points A_3, B_3, C_3 are symmetric to A_1, B_1, C_1 with respect to the midpoints of the sides BC, CA, AB respectively. Prove that the triangles $A_2B_2C_2$ and $A_3B_3C_3$ are similar.

Proof. Well known $\triangle A_2C_1B_1 \sim \triangle A_2BC$, we have

$$\frac{A_2C_1}{A_2B_1} = \frac{BC_1}{CB_1} = \frac{AC_3}{AB_3}$$

Therefore, $\triangle A_2C_1B_1 \sim \triangle AC_3B_3$, same applies on $\triangle B_2A_1C_1 \sim \triangle BA_3C_3, \triangle C_2B_1A_1 \sim \triangle CB_3A_3$. We have

$$\begin{aligned} \measuredangle B_3A_3C_3 &= \pi - \measuredangle CA_3B_3 - \measuredangle C_3A_3B \\ &= \pi - \measuredangle C_2A_1B_1 - \measuredangle C_1A_1B_2 \\ &= \pi - \measuredangle C_2CA - \measuredangle ABB_2 \\ &= \pi - \measuredangle C_2CB_2 \\ &= \pi + \measuredangle B_2A_2C_2 \\ &= \measuredangle B_2A_2C_2 \end{aligned}$$

same applies on other three angles, hence the result. □

Problem 9.15 (Problem 10.31 (Shortlist 2005/G5).). Let $\triangle ABC$ be an acute-angled triangle with $AB = AC$. Let H be the orthocenter of triangle ABC, and let M be the midpoint of the side BC. Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumcircles of $\triangle ABC$ and $\triangle ADE$.

Proof. Define $F :=$ ray $MH \cap (ABC), A_1 :=$ ray $HM \cap (ABC)$. It suffices to show $AFDE$ are concyclic $\Longleftrightarrow \triangle FBD \sim \triangle FCE$. In fact, we have.

$$\frac{FB}{FC} = \frac{\sin \angle FA_1B}{\sin \angle FA_1C} = \frac{A_1C}{A_1B}$$
$$\frac{BD}{EC} = \frac{HB}{HC}$$

well known HBA_1C is parallelogram, since $\triangle BDH \sim \triangle CEH$, hence the result. $\square$

CHAPTER 10
Evan Chen Personal favorite

Problem 10.1 (Problem 11.1 (Canada 2000/4).). Let $ABCD$ be a convex quadrilateral with $\angle CBD = 2\angle ADB, \angle ABD = 2\angle CDB$ and $AB = CB$. Prove that $AD = CD$.

Proof. Let $\angle CDB = \alpha, \angle ADB = \beta$ then $\angle ABD = 2\alpha, \angle CBD = 2\beta$, from the law of the sine:

$$\frac{\sin\beta}{AB} = \frac{\sin(\pi - (2\alpha + \beta))}{DB}$$
$$\frac{\sin\alpha}{CB} = \frac{\sin(\pi - (2\beta + \alpha))}{DB}$$

since $AB = CB$, we conclude that

$$\cos(3\alpha + \beta) = \cos(3\beta + \alpha)$$

with $\alpha + \beta < \frac{\pi}{2}$, we see that $\alpha = \beta$ is the only solution that satisfy the equation. Then, easily $\triangle ABD \cong \triangle CBD$, hence we are done. □

Problem 10.2 (Problem 11.2 (EGMO 2012/1).). Let ABC be a triangle with circumcenter O. The points D, E, F lie in the interiors of the sides BC, CA, AB respectively, such that DE is perpendicular to CO and DF is perpendicular to BO. Let K be the circumcenter of triangle AFE. Prove that the lines DK and BC are perpendicular.

Proof. Notice that

$$\angle CDE = \frac{\pi}{2} - \angle OCD = \frac{\pi}{2} - \left(\frac{\pi}{2} - A\right) = A$$
$$\angle BDF = \frac{\pi}{2} - \angle OBD = \frac{\pi}{2} - \left(\frac{\pi}{2} - A\right) = A$$

which shows $(AFDC), (AEDB)$ are concyclic. Therefore

$$\angle FKE = 2A$$

also since $\angle FDE = \pi - \angle CDE - \angle BDF = \pi - 2A$, we have $(DEKF)$ concyclic. we are done since $FK = EK$. □

Problem 10.3 (Problem 11.3 (ELMO 2013/4).). Triangle ABC is inscribed in circle ω. A circle with chord BC intersects segments AB and AC again at S and R, respectively. Segments BR and CS meet at L, and rays LR and LS intersect ω at D and E, respectively.

The internal angle bisector of $\angle BDE$ meets line ER at K. Prove that if $BE = BR$, then $\angle ELK = \frac{1}{2}\angle BCD$.

Proof. Well-known A is the midpoint of arc ED, $SR \parallel DE$.

Claim 1 - $BE = BR = BC$

Proof of Claim. Since $BE = BR$, we have AB is the perpendicular bisector of segmant ER, therefore,

$$\angle BCE = \angle BCS = \angle BRS = \angle BES$$

hence the result. ■

Claim 2 - K is the incenter of $\triangle DLE$

Proof of Claim. It suffices to show RE is the angle bisector of $\angle LED$. In fact

$$\angle RED = \angle ERS = \angle RES$$

as required. ■

Therefore $\angle ELK = \frac{1}{2}\angle ELD = \frac{1}{2}\angle BLC$, it suffices to show $\angle BLC = \angle BCD$. In fact

$$\angle BCL = \angle BCE = \angle BDC$$

the result is now obvious. □

Problem 10.4 (Problem 11.4 (Sharygin 2012).). Let BM be the median of right-angled triangle ABC with $\angle B = 90°$. The incircle of triangle ABM touches sides AB and AM in points A_1 and A_2; points C_1, C_2 are defined similarly. Prove that lines A_1A_2 and C_1C_2 meet on the bisector of angle ABC.

Proof. Define D, s.t. AD is the angle bisector of $\angle ABC$, $X_A := A_1A_2 \cap AD, X_C := C_1C_2 \cap AD$. Notice that A_1 is the midpoint of AB, C_1 is the midpoint of CB. Applying the Menelaus' theorem:

$$\frac{AD}{DA_2}\frac{A_2X_A}{X_AA_1}\frac{A_1B}{BA} = 1$$

$$\frac{A_2X_A}{X_AA_1} = 2\frac{\frac{cb}{c+a} - \frac{c}{2}}{\frac{cb}{c+a}}$$

$$\frac{A_2X_A}{X_AA_1} = \frac{(2cb) - c(a+c)}{cb}$$

$$\frac{A_2X_A}{X_AA_1} = \frac{2b - a - c}{b}$$

$$\frac{CD}{DC_2}\frac{C_2X_C}{X_CC_1}\frac{C_1B}{BC} = 1$$
$$\frac{C_2X_C}{X_CC_1} = 2\frac{\frac{ab}{c+a} - \frac{a}{2}}{\frac{ab}{c+a}}$$
$$\frac{C_2X_C}{X_CC_1} = \frac{(2ab) - a(a+c)}{ab}$$
$$\frac{C_2X_C}{X_CC_1} = \frac{2b - a - c}{b}$$

that completes the proof. □

Problem 10.5 (Problem 11.5 (USAMTS).). In quadrilateral $ABCD, \angle DAB = \angle ABC = 110°, \angle BCD = 35\circ, \angle CDA = 105°$, and AC bisects $\angle DAB$. Find $\angle ABD$.

Proof. Consider $T := (ABD) \cap AC$, we have $\angle ATB = 70° = 2\angle DCB$, also since AT bisects $\angle DAB \implies DT = BT$ which implies that T is the circumcenter of (CBD). Let $\angle ABD = \alpha$, notice that

$$\angle CTD = 2\angle CBD = 2(110° - \alpha)$$
$$\angle CTD = 180° - \angle ATD = 180° - \alpha$$

conclude that $\angle ABD = \alpha = 40°$ □

Problem 10.6 (Problem 11.6 (MOP 2012).). Let ABC be an acute triangle with circumcircle ω and altitudes AD, BE, CF. Circle γ is the image of ω when reflected across AB. Ray EF meets ω at P, and ray DF meets γ at Q. Prove that the points B, P, Q are collinear.

Proof. Well-known that H also lies on γ □

Problem 10.7 (Problem 11.7 (Sharygin 2013).). Chords BC and DE of circle ω meet at point A. The line through D parallel to BC meets ω again at F, and FA meets ω again at T. Let M denote the intersection of ET and BC, and let N be the reflection of A over M. Show that the circumcircle of $\triangle DEN$ passes through the midpoint of BC.

Proof. Redefine $K :=$ midpoint of BC, let G be the reflection of A over K, obviously, $DFGA$ is a isosceles trapezoid. Notice that

$$\measuredangle MED = \measuredangle TED = \measuredangle TFD = \measuredangle AFD = \measuredangle AGD = \measuredangle MGD$$

which implies M, D, G, E concyclic, since

$$AN \cdot AK = 2AM \cdot AK = AM \cdot AG = AD \cdot AE$$

hence the result. □

Proof. Perform an inversion around B with $r = \sqrt{BH \cdot BE} = \sqrt{BF \cdot BA} = \sqrt{BD \cdot BC}$, denote Z^* as the inversion of Z. Notice that $Q = P^*, P = Q^*$, hence the result. □

Problem 10.8 (Problem 11.8 (ELMO 2012/1).). In acute triangle ABC, let D, E, F denote the feet of the altitudes from A, B, C, respectively, and let ω be the circumcircle of $\triangle AEF$. Let ω_1 and ω_2 be the circles through D tangent to ω at E and F, respectively. Show that ω_1 and ω_2 meet at a point P on line BC other than D.

Proof. We claim that the second intersection of ω_1 and ω_2 is $EF \cap BC =: X$, well-known both the tangent line of (AEF) through E and F pass through M, which is the midpoint of BC. It suffices to show $\measuredangle MFD = \measuredangle FXD, \measuredangle MED = \measuredangle EXD$. In fact,

$$\begin{aligned}
\measuredangle MFD &= \measuredangle MFB + \measuredangle BFD \\
&= \measuredangle ABC + \measuredangle ACB \\
&= \measuredangle(EF, BC) \\
\measuredangle MED &= \measuredangle MEC + \measuredangle CED \\
&= \measuredangle ACB + \measuredangle ABC \\
&= \measuredangle(EF, BC)
\end{aligned}$$

that completes the proof. □

Problem 10.9 (Problem 11.9 (Sharygin 2013).). In trapezoid $ABCD$, $\angle A = \angle D = 90°$. Let M and N be the midpoints of diagonals AC and BD, respectively. Line BC meets (ABN) and (CDM) again at Q and R. If K is the midpoint of MN, show that $KQ = KR$.

Proof. Define $L := NM \cap BC$, let P as the midpoint of QR. Let $AB = 2a, CD = 2b, BC = 2c$. WLOG Assume $a < b$ Since

$$LR = \frac{LM^2}{LC} = \frac{a^2}{c},\ LQ = \frac{LN^2}{LM} = \frac{b^2}{c}$$

$$PL = \frac{1}{2}(LR + LQ) - LR = \frac{1}{2}(LQ - LR) = \frac{1}{2c}(b^2 - a^2)$$

$$KL = \frac{1}{2}(LM + LN) = \frac{1}{2}(a + b)$$

Notice that

$$\frac{KL}{PL} = \frac{(a+b)c}{b^2 - a^2} = \frac{c}{b-a} = \cos\angle BCD = \cos\angle PLK$$

which implies $KP \perp QR$, the result is now obvious. □

Problem 10.10 (Problem 11.10 (Bulgarian Olympiad 2012).). Let ABC be a triangle with circumcircle Ω and let P be a variable point in its interior. The rays PA, PB, PC meet Ω again at A_1, B_1, C_1. Let A_2 denote the reflection of A_1 over BC, and define B_2 and C_2 similarly. Prove that the circumcircle of triangle $A_2B_2C_2$ passes through a fixed point independent of P.

Proof. Construct parallelograms $XCAB, YABC,$ and $ZBCA$, obviously, (BHA_2CZ), $(AHC_2BY), (CHB_2AX)$. By Ceva's theorem in trigonometric form on triangle ABC and point P, we know that.

$$\frac{\sin\angle BAP}{\sin\angle PAC}\frac{\sin\angle CBP}{\sin\angle PBA}\frac{\sin\angle ACP}{\sin\angle PCB} = 1$$

But $\angle PAC = \angle A_1AC = \angle CXA_2$, since minor arcs A_1C and A_2C are identical. So the above rewrites as

$$\frac{\sin\angle BXA_2}{\sin\angle CXA_2}\frac{\sin\angle CYB_2}{\sin\angle AYB_2}\frac{\sin\angle AZC_2}{\sin\angle BZC_2} = 1$$

So rays XA_2, YB_2, ZC_2 concur at some point, say Q. Let H be the orthocenter of triangle ABC. We claim that H is the fixed point, and that in fact, the three points lie on a circle

with diameter HQ. Indeed, note that A_2 lies on the reflection of (ABC) over BC, which is a circle with diameter HQ, whence

$$\measuredangle HA_2X = \measuredangle HA_2Q = 90^\circ$$

as desired. □

Problem 10.11 (Problem 11.11 (Sharygin 2013).). Points $A_1, B_1, C_1, A_2, B_2, C_2$ lie inside a triangle ABC so that A_1 is on AB_1, B_1 is on BC_1, C_1 is on CA_1, A_2 is on AC_2, B_2 is on BA_2, C_2 is on CB_2. Suppose the angles $BAA_1, CBB_1, ACC_1, CAA_2, ABB_2, BCC_2$ are equal. Prove that $\triangle A_1B_1C_1$ and $\triangle A_2B_2C_2$ are congruent.

Proof. Obviously, $\triangle A_2B_2C_2 \sim \triangle A_1B_1C_1$, it suffices to show $A_1B_1 = A_2B_2$, let $\theta = \angle BAA_1 = \angle CBB_1 = \angle ACC_1 = \angle CAA_2 = \angle ABB_2 = \angle BCC_2$. WLOG Assume $R = 1$.

$$A_1B_1 = AB_1 - AA_1 = c\frac{\sin(B-\theta)}{\sin B} - b\frac{\sin\theta}{\sin A} = \frac{2\sin C\sin(B-\theta)}{\sin B} - \frac{2\sin B\sin\theta}{\sin A}$$
$$A_2B_2 = BB_2 - BA_2 = c\frac{\sin(A-\theta)}{\sin A} - a\frac{\sin\theta}{\sin B} = \frac{2\sin C\sin(A-\theta)}{\sin A} - \frac{2\sin A\sin\theta}{\sin B}$$

$$\begin{aligned}
A_1B_1 &= A_2B_2 \\
\Leftrightarrow \frac{2\sin C\sin(B-\theta)}{\sin B} - \frac{2\sin B\sin\theta}{\sin A} &= \frac{2\sin C\sin(A-\theta)}{\sin A} - \frac{2\sin A\sin\theta}{\sin B} \\
\Leftrightarrow \sin A\sin C\sin(B-\theta) - \sin^2 B\sin\theta &= \sin B\sin C\sin(A-\theta) - \sin^2 A\sin\theta \\
\Leftrightarrow \sin C(\sin A\sin(B-\theta) - \sin B\sin(A-\theta)) &= \sin\theta(\sin^2 B - \sin^2 A) \\
\Leftrightarrow \sin C(\sin B\cos A - \sin A\cos B) &= \sin^2 B - \sin^2 A \\
\Leftrightarrow \sin(A+B)\sin(B-A) &= \sin^2 B - \sin^2 A
\end{aligned}$$

which is trivial. □

Problem 10.12 (Problem 11.12 (Sharygin 2013).). Let ABC be a triangle, and let AD denote the bisector of $\angle A$ (with D on BC). Points M and N are the projections of B and C respectively to AD. The circle with diameter MN intersects BC at points X and Y . Prove that $\angle BAX = \angle CAY$.

Proof. It suffices to show

$$\frac{\sin\angle BAX}{\sin\angle CAX} = \frac{\sin\angle CAY}{\sin\angle BAY}$$

which is equivalent to

$$\frac{BX\cdot BY}{CX\cdot CY} = \left(\frac{AB}{AC}\right)^2$$

Since BM, CN is tangent to the circle mentioned in the question,

$$\text{LHS} = \left(\frac{BM}{CN}\right)^2 = \left(\frac{AB\sin\frac{A}{2}}{AC\sin\frac{A}{2}}\right)^2 = \text{RHS}$$

hence the result. □

CHAPTER 11
Note and configuration

11.1. Angle Chasing

Proposition 11.1 (Triangle Sum). The sum of the angles in a triangle is 180°.

Theorem 11.2 (Inscribed Angle Theorem). If $\angle ACB$ is inscribed in a circle, then it subtends an arc with measure $2\angle ACB$.

Theorem 11.3 (Cyclic Quadrilaterals). Let $ABCD$ be a convex quadrilateral. Then the following are equivalent.

i) $ABCD$ is cyclic.
ii) $\angle ABC + \angle CDA = 180°$.
iii) $\angle ABD = \angle ACD$.

Lemma 11.4 (The Orthic Triangle). Suppose $\triangle DEF$ is the orthic triangle of acute $\triangle ABC$ with orthocenter H. Then
i) Points A, E, F, H lie on a circle with diameter AH.
ii) Points B, E, F, C lie on a circle with diameter BC.
iii) H is the incenter of $\triangle DEF$.

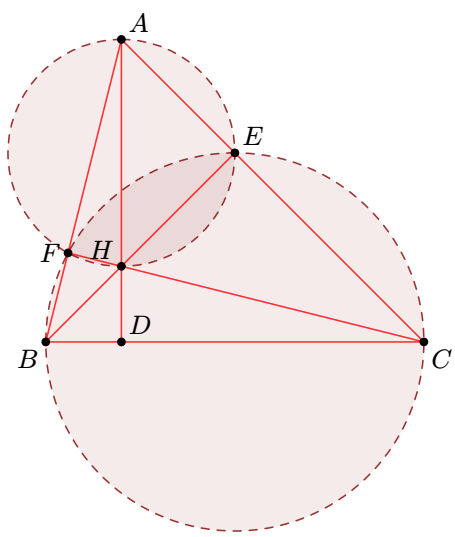

Figure 12.1A | The Orthic Triangle

Lemma 11.5 (Reflecting the Orthocenter). Let H be the orthocenter of $\triangle ABC$. Let X be the reflection of H over BC and Y the reflection over the midpoint of BC. We have

i) X and Y both lie on (ABC).
ii) AY is the diameter of (ABC).

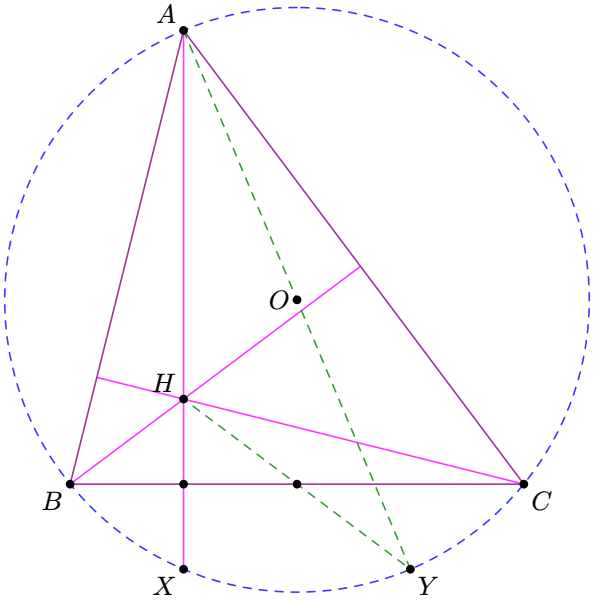

Figure 12.1B | Reflecting the Orthocenter

Lemma 11.6 (The Incenter/Excenter Lemma). Let ABC be a triangle with incenter I. Ray AI meets (ABC) again at L. Let IA be the reflection of I over S. Then,

i) The points $I, B, C,$ and IA lie on a circle with diameter II_A and center S. In particular, $SI = SB = SC = SI_A$.
ii) Rays BI_A and CI_A bisect the exterior angles of $\triangle ABC$.

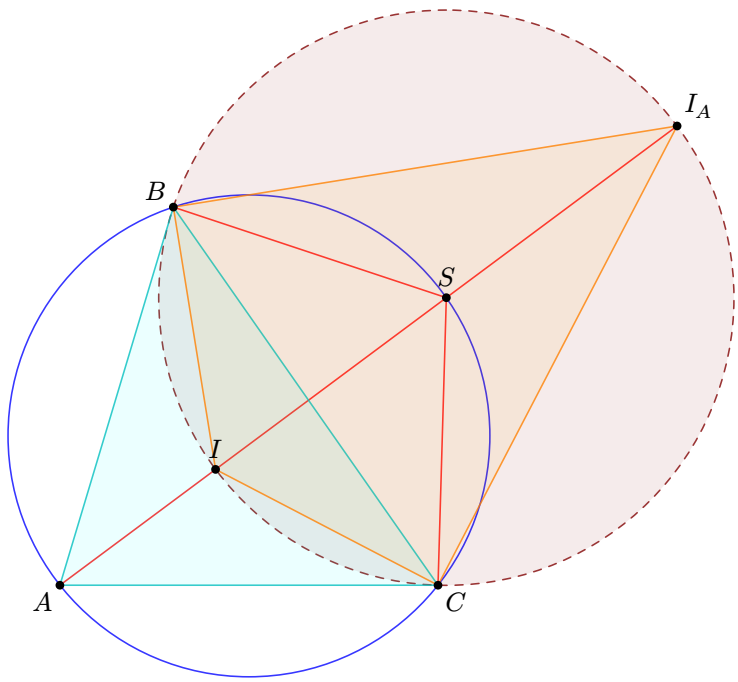

Figure 12.1C | The Incenter/Excenter Lemma

Theorem 11.7 (Cyclic Quadrilaterals with Directed Angles). Points A, B, X, Y lie on a circle if and only if $\measuredangle AXB = \measuredangle AYB$.

Proposition 11.8 (Directed Angles). For any distinct point A, B, C, P in the plane, we have the following rules:

Oblivion. $\measuredangle APA = 0$.

Anti-Reflexivity. $\measuredangle ABC = -\measuredangle CBA$.

Replacement. $\measuredangle PBA = \measuredangle PBC$ if and only if A, B, C are collinear. Equivalently, if C lies on line BA, then the A in $\measuredangle PBA$ may be replaced by C.

Right Angles. If $AP \perp BP$, then $\angle APB = \measuredangle BPA = 90°$.

Directed Angle Addition. $\measuredangle APB + \measuredangle BPC = \measuredangle APC$.

Triangle Sum. $\measuredangle ABC + \measuredangle BCA + \measuredangle CAB = 0$.

Isosceles Triangles. $AB = AC$ if and only if $\measuredangle ACB = \measuredangle CBA$.

Inscribed Angle Theorem. If (ABC) has center P, then $\measuredangle APB = 2\measuredangle ACB$.

Parallel Lines. If $AB \parallel CD$, then $\measuredangle ABC + \measuredangle BCD = 0$.

Lemma 11.9 (Miquel Point of a Triangle). Points D, E, F lie on lines $BC, CA,$ and AB of $\triangle ABC$, respectively. Then there exists a point lying on all three circles $(AEF), (BFD), (CDE)$.

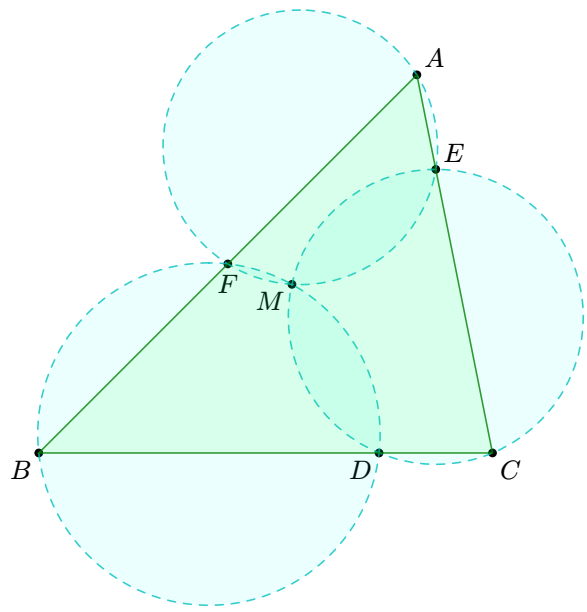

Figure 12.1D | Miquel Point of a Triangle

Proposition 11.10 (Tangent Criterion). Suppose $\triangle ABC$ is inscribed in a circle with center O. Let P be a point in the plane. Then the following are equivalent:

i) PA is tangent to (ABC).
ii) $OA \perp AP$.
iii) $\measuredangle PAB = \measuredangle ACB$.

Lemma 11.11. Let ABC be an acute triangle inscribed in circle Ω . Let X be the midpoint of the arc BC not containing A and define Y, Z similarly. Then the orthocenter of XYZ is the incenter I of ABC. (See Problem 1.7)

Lemma 11.12 (Three Tangents). Let ABC be an acute triangle. Let BE and CF be altitudes of triangle ABC, and denote by M the midpoint of BC. Then ME, MF, and the line through A parallel to BC are all tangents to (AEF). (See Problem 1.9)

Lemma 11.13 (Right Angles on Incircle Chord). The incircle of ABC is tangent to BC, CA, AB at D, E, F respectively. Let M and N be the midpoints of BC and AC, respectively. Than MN, BI, EF are concurrent, let K be the point of concurrency. Than $BK \perp CK$. (See Problem 1.10)

Theorem 11.14 (Simson Line). Let ABC be a triangle and P be any point on (ABC). Let X, Y, Z be the feet of the perpendiculars from P onto lines BC, CA, and AB. Than points X, Y, Z are collinear. (See Problem 1.13)

11.2. Circles

Proposition 11.15 (Similar Triangles). The following are equivalent for triangles ABC and XYZ.

i) $\triangle ABC \sim \triangle XYZ$.
ii) (AA) $\angle A = \angle X$ and $\angle B = \angle Y$.

iii) (SAS) $\angle B = \angle Y$, and $AB : XY = BC : YZ$.
iv) (SSS) $AB : XY = BC : YZ = CA : ZX$.

Theorem 11.16 (Power of a Point). Consider a circle ω and an arbitrary point P.

i) The quantity $Pow(P, \omega)$ is positive, zero, or negative according to whether P is outside, on, or inside ω, respectively.
ii) If ℓ is a line through P intersecting ω at two distinct points X and Y , then

$$PX \cdot PY = |\mathrm{Pow}_\omega(P)|$$

iii) If P is outside ω and PA is a tangent to ω at a point A on ω, then

$$PA^2 = \mathrm{Pow}_\omega(P)$$

Theorem 11.17 (Converse of the Power of a Point). Let A, B, X, Y be four distinct points in the plane and let lines AB and XY intersect at P. Suppose that either P lies in both of the segments AB and XY, or in neither segment. If $PA \cdot PB = PX \cdot PY$, then A, B, X, Y are concyclic.

Theorem 11.18 (Radical Axis). Let ω_1 and ω_2 be circles with distinct centers O_1 and O_2. The radical axis of ω_1 and ω_2 is a straight line perpendicular to O_1O_2. In particular, if ω_1 and ω_2 intersect at two points A and B, then the radical axis is line AB.

In general, consider three circles with distinct centers O_1, O_2, O_3. The pairwise radical axes will concur at a single point K, the radical canter, or the three radical axes will be pairwise parallel (or even the same line), this case can only occur if O_1, O_2, O_3 are collinear.

Theorem 11.19 (Radical Center of Intersecting Circles). Let ω_1 and ω_2 be two circles with centers O_1 and O_2. Select points A and B on ω_1 and points C and D on ω_2. Then the following are equivalent:

i) A, B, C, D lie on a circle with center O_3 not on line O_1O_2.
ii) Lines AB and CD intersect on the radical axis of ω_1 and ω_2.

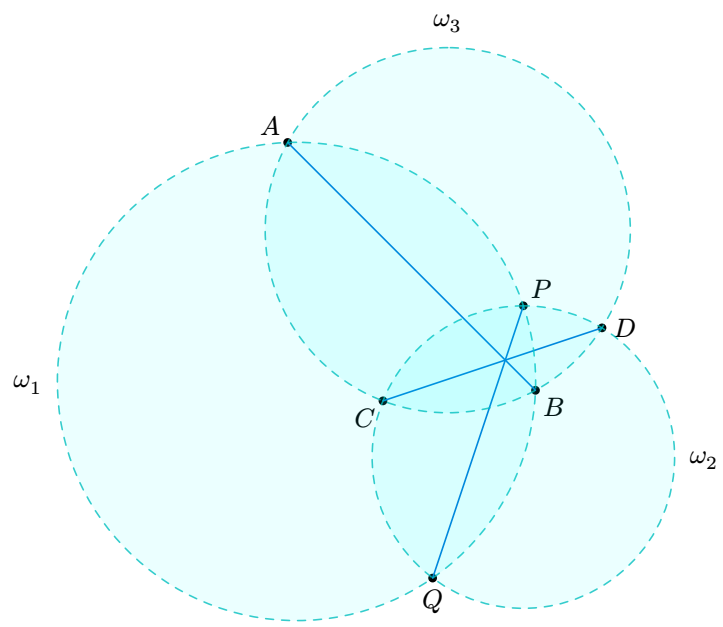

Figure 12.2A | Radical Center

Lemma 11.20. Let ABC be a triangle and consider a point P in its interior. Suppose that BC is tangent to the circumcircles of triangles ABP and ACP. Prove that ray AP bisects BC.

Lemma 11.21 (Finding Coaxial Circles). Three distinct circles $\Omega_1, \Omega_2, \Omega_3$ pass through a point X. Then their centers are collinear if and only if they share a second common point. Both conditions are equivalent to being coaxial.

Lemma 11.22 (Tangents to the Incircle). If DEF is the contact triangle of $\triangle ABC$, then $AE = AF = s - a$. Similarly, $BF = BD = s - b$ and $CD = CE = s - c$.

Lemma 11.23 (Tangents to the Excircle). If AB_1 and AC_1 are the tangents to the A -excircle, then $AB_1 = AC_1 = s$.

Lemma 11.24 (Length of Exradius). Prove that the A-exradius has length

$$r_a = \frac{s}{s-a} r$$

Lemma 11.25. Let ABC be a triangle. Suppose its incircle and A-excircle are tangent to BC at X and D, respectively. Show that $BX = CD$ and $BD = CX$.

Lemma 11.26 (Euler's Theorem). Let ABC be a triangle. Let R and r denote its circumradius and inradius, respectively. Let O and I denote its circumcenter and incenter. Then $OI^2 = R(R - 2r)$. In particular, $R \geq 2r$.

Lemma 11.27. Let ABC be a triangle with I_A, I_B, and I_C as excenters. Than triangle $I_A I_B I_C$ has orthocenter I and that triangle ABC is its orthic triangle. (See Problem 2.1)

Lemma 11.28 (The Pitot Theorem). Let $ABCD$ be a quadrilateral. If a circle can be inscribed in it, Than $AB + CD = BC + DA$. (See Problem 2.2)

11.3. Length and ratios

Theorem 11.29 (The Extended Law of Sines). In a triangle ABC with circumradius R, we have

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Theorem 11.30 (Angle Bisector Theorem). Let ABC be a triangle and D a point on BC so that AD is the internal angle bisector of $\angle BAC$. Show that

$$\frac{AB}{AC} = \frac{DB}{DC}$$

Theorem 11.31 (Ceva's Theorem). Let AX, BY, CZ be cevians of a triangle ABC. They concur if and only if

$$\frac{BX}{XC} \cdot \frac{CY}{YA} \cdot \frac{AZ}{ZB} = 1$$

Theorem 11.32 (Trigonometric Form of Ceva's Theorem). Let AX, BY, CZ be cevians of a triangle ABC. They concur if and only if

$$\frac{\sin\angle BAX \sin\angle CBY \sin\angle ACZ}{\sin\angle XAC \sin\angle YBA \sin\angle ZCB} = 1$$

Theorem 11.33 (Menelaus's Theorem). Let X, Y, Z be points on lines BC, CA, AB in a triangle ABC, distinct from its vertices. Then X, Y, Z are collinear if and only if

$$\frac{BX}{XC} \cdot \frac{CY}{YA} \cdot \frac{AZ}{ZB} = -1$$

where the lengths are directed.

Theorem 11.34 (Ceva's Theorem with Directed Lengths). Let ABC be a triangle and X, Y, Z be points on lines BC, CA, AB distinct from its vertices. Then lines AX, BY, CZ are concurrent if and only if

$$\frac{AZ}{ZB} \cdot \frac{BX}{XC} \cdot \frac{CY}{YA} = 1$$

where the ratios are directed.

Lemma 11.35 (Centroid Division). The centroid of a triangle divides the median into a 2:1 ratio.

Lemma 11.36 (Homothetic Triangles). Let ABC and XYZ be non-congruent triangles such that $AB \parallel XY, BC \parallel YZ,$ and $CA \parallel ZX$. Then lines AX, BY, CZ concur at some point O, and O is a center of a homothety mapping $\triangle ABC$ to $\triangle XYZ$.

Lemma 11.37 (Nine-Point Circle). Let ABC be a triangle with circumcenter O and orthocenter H, and denote by N_9 the midpoint of OH. Then the midpoints of AB, BC, CA, AH, BH, CH, as well as the feet of the altitudes of $\triangle ABC$, lie on a circle centered at N_9. Moreover, the radius of this circle is half the radius of (ABC).

We also have a homothety h mapping $(ABC) \mapsto (N_9)$ at H and with scale factor. Therefore, O get mapped to midpoint of OH, say N_9.

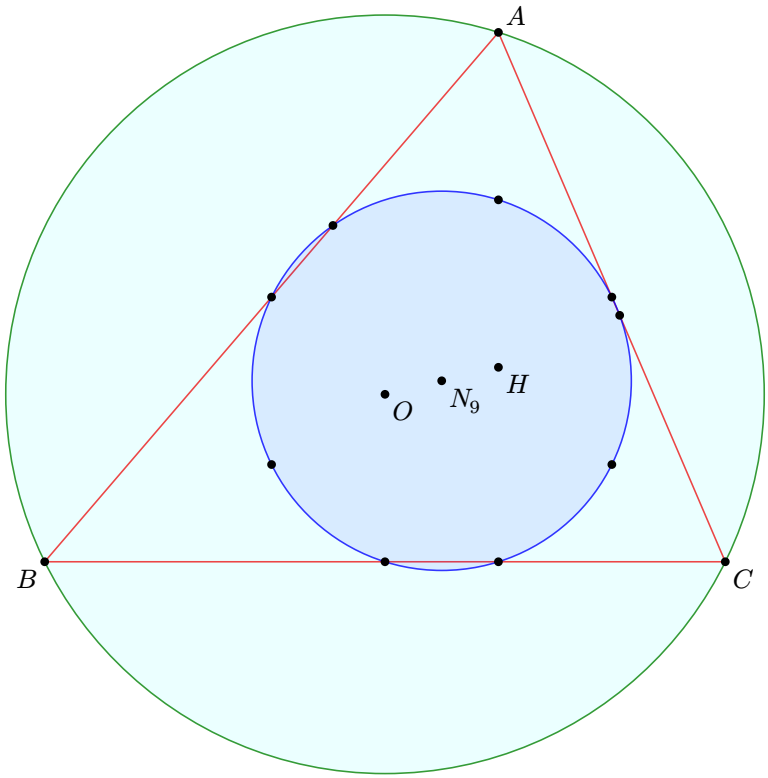

Figure 12.3A | Nine-Point Circle

Lemma 11.38 (Euler Line). In triangle ABC, prove that O, G, H (with their usual meanings) are collinear and that G divides OH in a $2:1$ ratio.

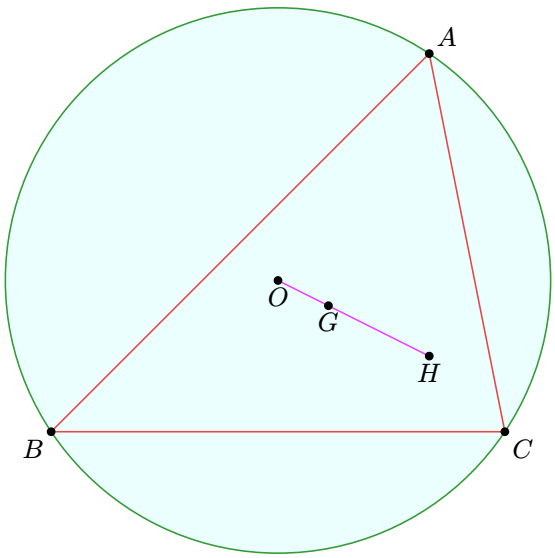

Figure 12.3B | Nine-Point Circle

Lemma 11.39. In cyclic quadrilateral $ABCD$, points X and Y are the orthocenters of $\triangle ABC$ and $\triangle BCD$. Than $AXYD$ is a parallelogram. (See Problem 3.2)

Theorem 11.40 (Monge's theorem). Consider disjoint circles $\omega_1, \omega_2, \omega_3$ in the plane, no two congruent. For each pair of circles, we construct the intersection of their common external tangents. Than these three intersections are collinear. (See Problem 3.7)

Theorem 11.41 (Cevian Nest). Let AX, BY, CZ be concurrent cevians of ABC. Let XD, YE, ZF be concurrent cevians in triangle XYZ. Than rays AD, BE, CF concur. (See Problem 3.8)

11.4. Assorted Configurations

Lemma 11.42 (Simson Line Bisection). Let ABC be a triangle with orthocenter H. If P is a point on (ABC) then its Simson line bisects PH.

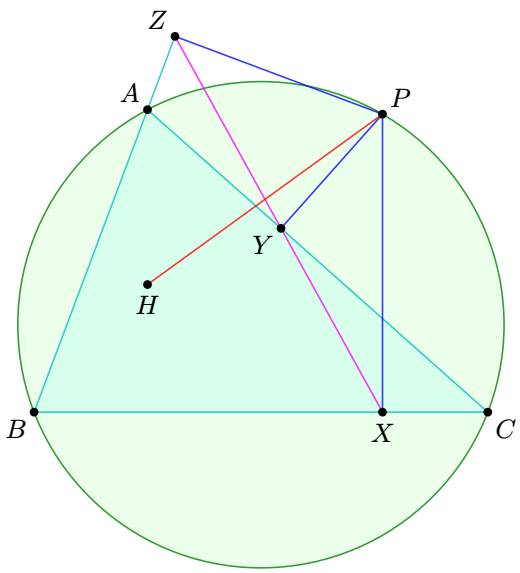

Figure 12.4A | Lemma 1.48 (Simson Line).

Lemma 11.43 (Duality of Orthocenters and Excenters). If I_A, I_B, I_C are the excenters of $\triangle ABC$, then triangle ABC is the orthic triangle of $\triangle I_A I_B I_C$, and the orthocenter is I.

Lemma 11.44 (The Diameter of the Incircle). Let ABC be a triangle whose incircle is tangent to BC at D. If DE is a diameter of the incircle and ray AE meets BC at X, then $BD = CX$ and X is the tangency point of the A-excircle to BC.

Lemma 11.45 (The Diameter of the Excircle). Let ABC be a triangle whose incircle is tangent to BC at D, A-excircle is tangent to BC at X. If XY is a diameter of the incircle, then D lies on AY.

Lemma 11.46 (Midpoint of Altitudes). Let ABC be a triangle with incenter I and A -excenter I_A, and let D and X be the associated tangency points on BC. Then lines DI_A and XI concur at the midpoint of the altitude from A.

Lemma 11.47 (An Incircle Concurrency). Let ABC be a triangle with incenter I and contact triangle DEF. If M is the midpoint of BC, then EF, AM and ray DI concur.

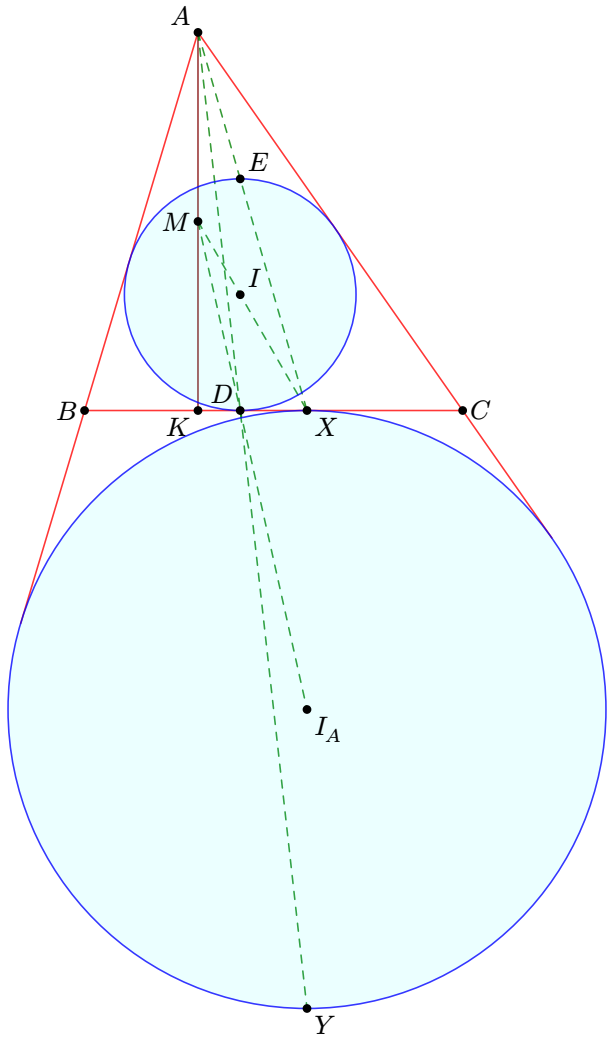

Figure 12.4B | Incenter and Excenter configuration

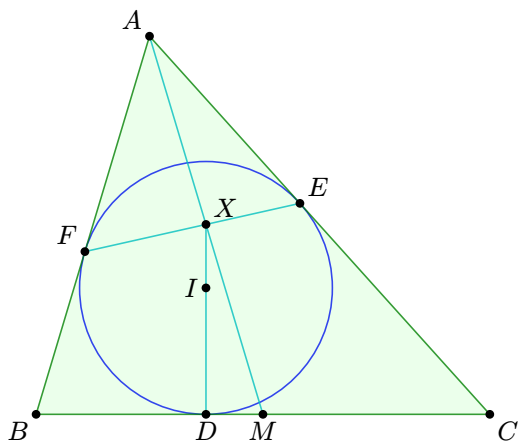

Figure 12.4C | An Incircle Concurrency

Lemma 11.48 (Isogonal Conjugates). Let ABC be a triangle and P any point not collinear with any of the sides. There exists a unique point P^* satisfying the relations

$$\measuredangle BAP = \measuredangle P^*AC, \quad \measuredangle CBP = \measuredangle P^*BA, \quad \measuredangle ACP = \measuredangle P^*CB.$$

Theorem 11.49 (Isogonal Ratios). Let D and E be points on BC so that AD and AE are isogonal. Then

$$\frac{BD}{DC} \cdot \frac{BE}{EC} = \left(\frac{AB}{AC}\right)^2.$$

Lemma 11.50 (Constructing the Symmedian). Let X be the intersection of the tangents to (ABC) at B and C. Then line AX is a symmedian.

Lemma 11.51 (Properties of the Symmedian). Let ABC be a triangle, and let the tangents to its circumcircle at B and C meet at point X. Let AX meet (ABC) again at K and BC at D. Then AD is the A-symmedian and

i) KA is a K-symmedian of $\triangle KBC$.
ii) $\triangle ABK$ and $\triangle AMC$ are directly similar.
iii) We have
$$\frac{BD}{DC} = \left(\frac{AC}{AB}\right)^2.$$
iv) We have
$$\frac{AB}{BK} = \frac{AC}{CK}.$$
v) (BCX) passes through the midpoint of AK.
vi) BC is the B-symmedian of $\triangle BAK$, and the C-symmedian of $\triangle CAK$.
vii) BC is the interior angle bisector of $\angle AMK$, and MX is the exterior angle bisector.

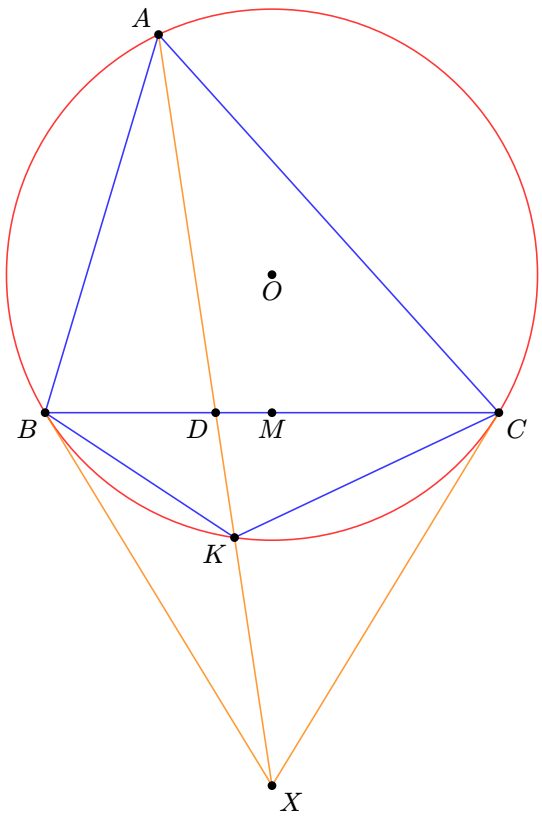

Figure 12.4D | Properties of the Symmedian

Lemma 11.52 (Symmedians in Cyclic Quadrilaterals). Let $ABCD$ be a cyclic quadrilateral. The following are equivalent.

i) $AB \cdot CD = BC \cdot DA$.
ii) AC is an A-symmedian of $\triangle DAB$.
iii) AC is a C-symmedian of $\triangle BCD$.
iv) BD is a B-symmedian of $\triangle ABC$.
v) BD is a D-symmedian of $\triangle CDA$.

Lemma 11.53 (Circles Inscribed in Segments). Let AB be a chord of a circle Ω. Let ω be a circle tangent to chord AB at K and internally tangent to ω at T. Then ray TK passes through the midpoint M of the arc AB not containing T . Moreover, $MA^2 = MB^2$ is the power of M with respect to ω.

Lemma 11.54 (Curvilinear Incircle Chords). Let ABC be a triangle and D be a point on AB. Suppose a circle ω is tangent to CD at L, AB at K, and also to (ABC). Then the incenter of ABC lies on line LK.

Lemma 11.55 (Mixtilinear Incircles). Let ABC be a triangle and let its A-mixtilinear circle be tangent to AB, AC, and (ABC) at K, L, and T, respectively. Denote by D and E the contact points of the incircle and A-excircle on BC, respectively.

i) The midpoint I of KL is the incenter of $\triangle ABC$.
ii) Lines TK and TL pass through the midpoints of arcs AB and AC not containing T .
iii) Line TI passes through the midpoint of arc BC containing A.
iv) The angles $\angle BAT$ and $\angle CAE$ are equal.
v) The angles $\angle BTA$ and $\angle CTD$ are equal.
vi) Quadrilaterals $BKIT$ and $CLIT$ are concyclic.

11.5. Computational Geometry

11.6. Complex Numbers

11.7. Barycentric Coordinates

I don't recommend people using the technique above to solve the problem, I will leave this for the reader who have interested in it.

11.8. Inversion

Lemma 11.56 (Inversion and Tangents). Let A be a point inside ω, other than O, and A^* be its inverse. Then the tangents from A^* to ω are collinear with A.

Theorem 11.57 (Inversion and Angles). If A^* and B^* are the inverses of A and B under inversion centered at O, then $\measuredangle OAB = -\measuredangle OB^*A^*$.

Proposition 11.58. A line passing through O inverts to itself.

Proposition 11.59. The inverse of a line ℓ not passing through O is a circle γ passing through O. Furthermore, the line through O perpendicular to ℓ passes through the center of γ.

Proposition 11.60. Let γ be a circle not passing through O. Then γ^* is also a circle and does not contain O.

Theorem 11.61 (Images of Clines). A cline inverts to a cline. Specifically, in an inversion through a circle with center O,

i) A line through O inverts to itself.
ii) A circle through O inverts to a line (not through O), and *vice versa*. The diameter of this circle containing O is perpendicular to the line.
iii) A circle not through O inverts to another circle not through O. The centers of these circles are collinear with O.

11.9. Projective Geometry

Theorem 11.62 (Cross-Ratio Under Perspectivity). Suppose that $P(A, B; X, Y)$ is a pencil of lines. If A, B, X, Y are collinear then

$$P(A, B; X, Y) = (A, B; X, Y).$$

Theorem 11.63 (Cross Ratios on Cyclic Quadrilaterals). Let A, B, X, Y be concyclic. If P is any point on its circumcircle, then $P(A, B; X, Y)$ does not depend on P. Moreover,

$$P(A, B; X, Y) = \pm\frac{XA}{XB} \div \frac{YA}{YB}$$

where the sign is positive if AB and XY do not intersect, and negative otherwise.

Lemma 11.64 (Midpoints and Parallel Lines). Given points A and B, let M be the midpoint of AB and P_∞ the point at infinity of line AB. Then $(A, B; M, P_\infty)$ is a harmonic bundle.

Lemma 11.65 (Harmonic Quadrilaterals). Let ω be a circle and let P be a point outside it. Let PX and PY be tangents to ω. Take a line through P intersecting ω again at A and B. Then

i) $AXBY$ is a harmonic quadrilateral.
ii) If $Q = AB \cap XY$, then $(A, B; Q, P)$ is a harmonic bundle.

Proposition 11.66 (Inversion Induces Harmonic Bundles). Let P be a point on line AB, and let P^* denote the image of P after inverting around the circle with diameter AB. Then $(A, B; P, P^*)$ is harmonic.

Lemma 11.67 (Cevians Induces Harmonic Bundles). Let ABC be a triangle with concurrent cevians AD, BE, CF (possibly on the extensions of the sides). Line EF meets BC at X (possibly at a point at infinity). Then $(X, D; B, C)$ is a harmonic bundle.

Lemma 11.68 (Complete Quadrilaterals Induces Harmonic Bundles). Let $ABCD$ be a quadrilateral whose diagonals meet at K. Lines AD and BC meet at L, and line KL meets AB and CD at M and N. Then $(K, L; M, N)$ is a harmonic bundle.

Lemma 11.69 (Midpoint Lengths). Points A, X, B, P lie on a line in that order, and $(A, B; X, P) = -1$. Let M be the midpoint of AB. Show that $MX \cdot MP = \left(\frac{1}{2}AB\right)^2$ and $PX \cdot PM = PA \cdot PB$.

Lemma 11.70 (Right Angles and Bisectors). Let X, A, Y, B be collinear points in that order and let C be any point not on this line. Then any two of the following conditions implies the third condition.

i) $(A, B; X, Y)$ is a harmonic bundle.
ii) $\angle XCY = 90°$.
iii) CY bisects $\angle ACB$.

Theorem 11.71 (Apollonian Circles). Let AB be a segment and $k \neq 1$ be a positive real. The locus of points C satisfying $\frac{CB}{CA} = k$ is a circle whose diameter lies on AB.

Lemma 11.72 (Harmonic Bundles on the Bisector). Let ABC be a triangle with incenter I and A-excenter I_A. Prove that

$$(I, I_A; A, AI \cap BC) = -1.$$

Theorem 11.73 (La Hire's Theorem). A point X lies on the polar of a point Y if and only if Y lies on the polar of X.

Proposition 11.74. Let AB be a chord of a circle ω and select points P and Q on line AB. Then $(A, B; P, Q) = -1$ if and only if P lies on the polar of Q.

Theorem 11.75 (Brocard's Theorem). Let $ABCD$ be an arbitrary cyclic quadrilateral inscribed in a circle with center O, and set $P = AB \cap CD, Q = BC \cap DA,$ and $R = AC \cap BD$. Then P, Q, R are the poles of QR, RP, PQ, respectively. In particular, O is the orthocenter of triangle PQR.

Lemma 11.76 (Self-Polar Orthogonality). Let ω be a circle and suppose P and Q are points such that P lies on the pole of Q (and hence Q lies on the pole of P). Prove that the circle γ with diameter PQ is orthogonal to ω.

Theorem 11.77 (Pascal's Theorem). Let $ABCDEF$ be a cyclic hexagon, possibly self-intersecting. Then the points $AB \cap DE, BC \cap EF,$ and $CD \cap FA$ are collinear.

Theorem 11.78 (Projective Transformations). Each of the following is achievable with a unique projective transformation.

i) Taking four points A, B, C, D (no three collinear) to any other four points W, X, Y, Z (no three collinear).
ii) Taking a circle to itself and a point P inside the circle to any other point Q inside the circle.
iii) Taking a circle to itself and any given line outside the circle into the line at infinity.

Furthermore, projective transformations preserve the cross ratio of any four collinear points. Moreover, if four concyclic points are sent to four concyclic points, then the cross ratio of the quadrilaterals are the same.

Theorem 11.79 (Butterfly Theorem). Let AB, CD, PQ be chords of a circle concurrent at M. Put $X = PQ \cap AD$ and $Y = PQ \cap BC$. If $MP = MQ$ then $MX = MY$.

Lemma 11.80 (Incircle Polars). Let ABC be a triangle with contact triangle DEF and incenter I . Lines EF and BC meet at K. Than $IK \perp AD$. (See Problem 8.1)

Theorem 11.81 (Desargues' Theorem). Let ABC and XYZ be triangles in the projective plane. We say that the two triangles are perspective from a point if lines AX, BY, and CZ concur (possibly at infinity), and we say they are perspective from a line if the points $AB \cap XY, BC \cap YZ, CA \cap ZX$ are collinear. Prove that these two conditions are equivalent.

11.10. Complete Quadrilateral

Lemma 11.82 (Spiral Centers). Let AB and CD be segments, and suppose $X = AC \cap BD$. If (ABX) and (CDX) intersect again at O, then O is the center of the unique spiral similarity taking AB into CD.

Proposition 11.83. The center of the spiral similarity taking AB to CD is also the center of the spiral similarity taking AC to BD.

Theorem 11.84 (Miquel's Theorem). The four circles $(PAB), (PDC), (QAD), (QBC)$ concur at the Miquel point M. Furthermore, M is the center of the spiral similarity sending AB to DC and BC to AD. (In particular, $\triangle MAB \sim \triangle MDC$ and $\triangle MBC \sim \triangle MAD$.)

Theorem 11.85 (Gauss-Bodenmiller Theorem). The circles with diameters AC, BD, PQ are coaxial. Their radical axis is a line passing through each of the four orthocenters of the triangles PAB, PCD, QAD, QBC.

Lemma 11.86 (Altitudes from the Miquel Point). The feet of the perpendiculars from M to lines AB, BC, CD, DA are collinear. Furthermore, the line though these four points is perpendicular to the Gauss line.

Theorem 11.87 (Miquel Point of a Cyclic Quadrilateral). Let $ABCD$ be a cyclic quadrilateral inscribed in circle ω with diagonals meeting at R. Then the Miquel point of M get the following property:

i) It is the common point of the six circles $(OAC), (OBD), (PAD), (PBC), (QAB), (QCD)$.
ii) It is the center of a spiral similarity taking AB to CD, as well as the spiral similarity taking BC to DA.
iii) It is the inverse of $R = AC \cap BD$ with respect to an inversion around $(ABCD)$. By Brocard's theorem, M is the foot of O onto PQ.